

REVIEW

In situ techniques for Li-rechargeable battery analysis

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Abstract

Reducing our carbon footprint is one of the most pressing issues facing humanity today. The technology of Li-rechargeable batteries is permeating every corner of our lives as a result of our efforts to reduce the use of carbon energy. Batteries can be seen metaphorically as “living cells”, and approaching the future of that technology requires observing and understanding the real-time phenomena that occur inside battery systems during (electro)chemical reactions. In this regard, in situ analysis techniques have made significant progress toward understanding the basic science of battery systems and finding better performance-improving factors. There are various analysis methods utilizing electromagnetic waves, electrons, and neutrons to perform multifaceted analyses of battery systems from the atomic to the macroscopic scale. Now is the opportune moment to construct a comprehensive guide that facilitates the design of advanced Li-rechargeable battery systems, adopting a highly discerning and all-encompassing approach toward these cutting-edge technologies. In this review article, we discuss and organize the key components such as capabilities, limitations, and practical tips with a comprehensive perspective on various in situ techniques. Moreover, this article covers a wide range of information from the nano to the micrometer scale, such as electronic, atomic, crystal, and morphological structures, from stereoscopic perspectives considering the probing depth.

KEYWORDS

in situ analyses, Li-rechargeable batteries, operando analyses, synchrotrons

1 | INTRODUCTION

Many countries are focusing on carbon-free energy as one of the most important global issues. Accordingly, the need for building a sustainable energy system and solving environmental problems related to carbon footprints has emerged, so the importance of the technology of Li-

rechargeable batteries has been highlighted. As a result, Li-rechargeable batteries are rapidly emerging as an imperative technology for various applications such as grid energy storage, electric vehicles, and portable devices.¹⁻³ Thanks to the tireless efforts toward achieving technological advancements in Li-rechargeable battery research, remarkable improvements have been achieved in energy density,

Seongeun Lee and Sangbin Park contributed equally to this study.

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cycle life, charging capabilities, and safety. However, despite the significant advances made so far, recent performances in Li-rechargeable batteries have begun to stagnate.^{4,5} To overcome this, a deeper understanding of the complicated reactions in battery systems is essential with multiple scale levels, from the atomic to the macroscopic level.⁶ This holistic perspective approach can provide insight into the challenges limiting battery performance and motivate the development of creative strategies to overcome related challenges. Numerous analytical methods have been developed to investigate the complex nature of battery reactions, but most of these were “ex situ” techniques that analyzed objects of interest outside their operating environment in the early days. However, these techniques have limitations in capturing real-time dynamics and changes in battery systems during cell operation. Therefore, it is necessary to develop a technology that can directly observe battery reactions under various operating conditions, and consequently, “in situ” technology is making remarkable progress.^{7,8}

Unlike ex situ methods, in situ analyses provide kinetic information, enabling a more profound comprehension of the underlying scientific principles governing dynamic phenomena. One of the key advantages of in situ techniques is to mitigate errors inherent in ex situ methods, such as sample preparation challenges, air or moisture exposure, and relaxation effects. Furthermore, in contrast to ex situ techniques measuring an equilibrium state, in situ methods enable the investigation of various transient states during battery operation, including phase transitions, surface reactions, and volume changes.^{9,10}

This review aims to build a comprehensive and well-organized library of in situ techniques for the constructive future of Li-rechargeable batteries. Accordingly, we will cover various in situ methodologies according to the analytical sources used, which cover a wide range of techniques utilizing electromagnetic waves, electrons, neutrons, and others (Figure 1). In X-ray-based in situ techniques, a comprehensive discussion encompasses X-ray diffraction (XRD), small-angle X-ray scattering

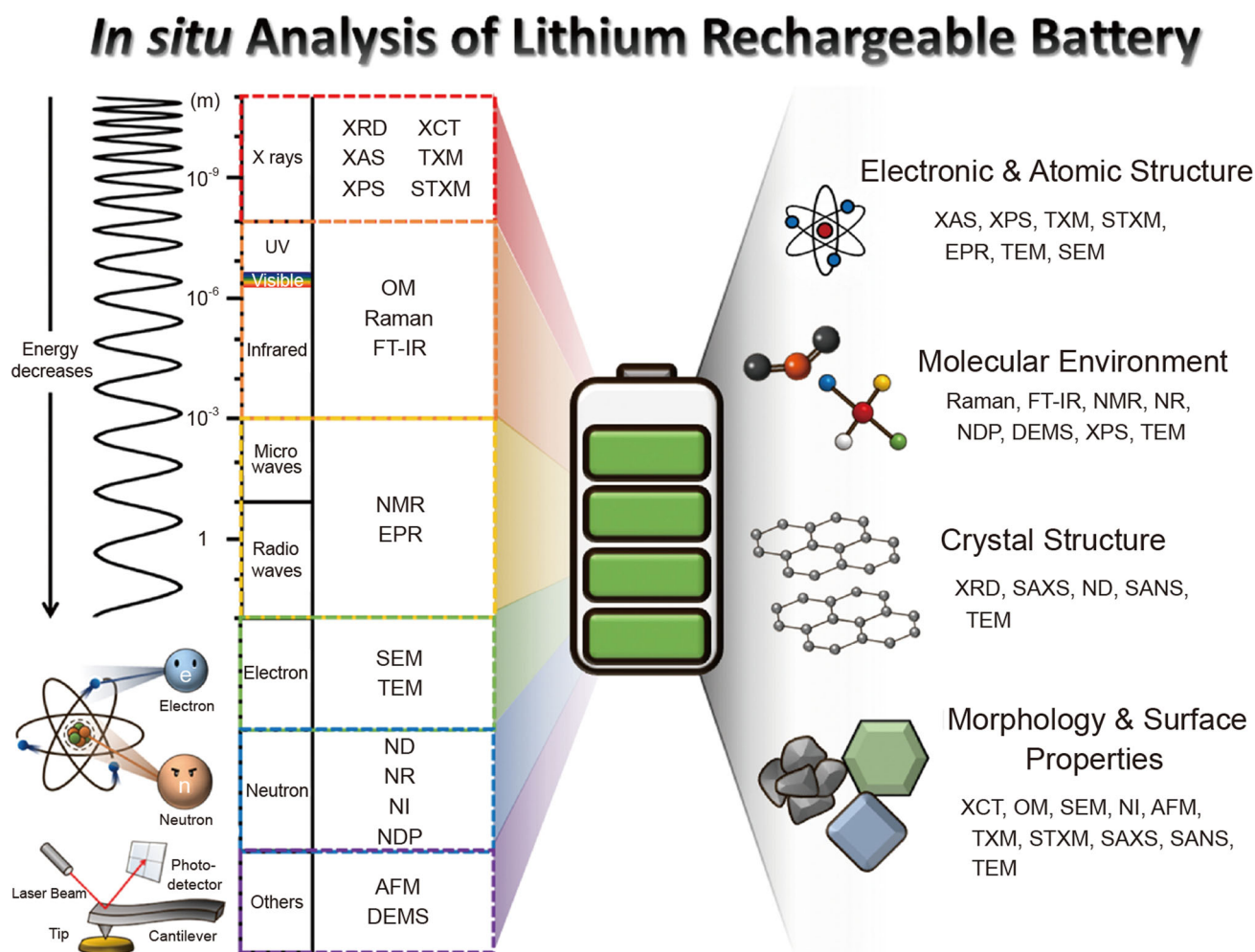


FIGURE 1 Schematic illustration for in situ analysis of a lithium rechargeable battery with analysis sources and classification of results from various techniques.

(SAXS), hard X-ray absorption spectroscopy (XAS), soft XAS, X-ray photoelectron spectroscopy (XPS), X-ray computed tomography (XCT), transmission X-ray microscopy (TXM), and scanning transmission X-ray microscopy (STXM). Subsequently, moving on to the higher wavelength region with ultraviolet (UV) radiation, visible light, and infrared (IR) radiation, the applications of optical microscopy (OM), Raman spectroscopy, and Fourier-transform infrared spectroscopy (FT-IR) are thoroughly explored. In the section on microwaves and radiowaves, the focus is on the applications of nuclear magnetic resonance (NMR) and electron paramagnetic resonance (EPR) techniques. In electron-based in situ techniques, the applications of scanning electron microscopy (SEM) and transmission electron microscopy (TEM) are examined. Neutron-based in situ techniques cover neutron diffraction (ND), small-angle neutron scattering (SANS), neutron reflectometry (NR), neutron imaging (NI), and neutron depth profiling (NDP). Furthermore, this review discusses differential electrochemical mass spectrometry (DEMS) and atomic force microscopy (AFM), which are widely used to gain deeper insights into the reaction mechanisms in battery systems. Overall, this review provides the general characteristics and practical applications of these in situ techniques to suggest feasible strategies to enhance the performances of current Li-rechargeable batteries.

2 | X-RAY-BASED ANALYTICAL TECHNIQUES

X-ray-based analytical techniques have been widely used to explore the structure of crystals and nanostructures, electronic structure, morphology, and chemical compositions. Advances in synchrotron radiation technology have greatly improved the quality of information obtained from X-ray-based methods owing to their high flux and brilliance. Furthermore, thanks to the high temporal resolution of synchrotron X-ray radiation, dynamic processes can be analyzed in real time with a high signal-to-noise ratio. In this section, X-ray-based in situ techniques mostly focus on synchrotron radiation, including XRD, SAXS, hard and soft XAS, XPS, XCT, TXM, and STXM.

2.1 | In situ XRD

After Laue et al.¹¹ discovered the diffraction of X-rays in crystalline substances in 1912, the field of X-ray crystallography rapidly advanced. XRD is a technique that takes advantage of the wave-like properties of X-rays. This technique can detect the organized arrangement of atoms

within a crystal to reveal crucial insights into the structure of electrode materials. In simpler terms, it helps us understand how the atoms are organized within a crystal. X-rays are scattered through the electrode material, and at specific angles, called Bragg angles, diffraction peaks occur due to constructive interference.¹² The diffraction angle and peak intensity are influenced by the crystal structure, including the lattice parameter and atom positions. Hence, quantitative and detailed information about the crystal structure can be deduced by refinement of the diffraction patterns.

The analytical technique using X-rays can be used under various conditions, that is, at high or low temperatures and in the air. The first in situ XRD for battery research was reported in the study of Chianelli et al.,¹³ who investigated the XRD patterns of Li/TiS₂ electrodes during discharge at an ambient temperature. In situ XRD techniques are frequently used to analyze the electrode material's electrochemical^{14–19} and thermal behaviors.^{9,20,21} Both can be measured using laboratory-based in situ XRD, but there are notable advantages to using synchrotron X-ray radiation.²² Synchrotron X-rays offer powerful and highly penetrating radiation, making them suitable for measurements in transmission mode using two-dimensional (2D) detectors.²³ As a result, in situ XRD utilizing synchrotron X-ray radiation enables simultaneous measurement of all angles in less than a few seconds. When conducting experiments during actual cell operation, synchrotron X-ray radiation permits measurements using the conventional coin cell or pouch cell. However, holes are drilled, and the cell is sealed with a Kapton or beryllium window because it is difficult for the X-ray to penetrate the stainless cap, case, and spacer of the coin cell, as shown in Figure 2A.^{14,24–26}

As mentioned above, information about the crystal structure can be obtained through the technique. For example, Lee et al.²⁷ investigated the structural behavior of a Co-fixed LiNi_{0.5+x}Co_{0.2}Mn_{0.3-x}O₂ series such as NCM523 ($x = 0.0$), NCM622 ($x = 0.1$), and NCM721 ($x = 0.2$) during the electrochemical reaction (Figure 2B). As the Ni content increases, the phase transition becomes a pseudo-one-phase reaction from a two-phase reaction. From the Rietveld refinement of the XRD patterns for each material at the initial state, it has been proven that the cause of this phenomenon is strongly correlated to the degree of cation disorder and the height of the Li-ion pathway. Also, Xu et al.¹⁴ have presented studies using laser-thinned coin cells, wherein holes were etched using lasers. They identified a fatigued phase in the (003) reflection of a cycled LiNi_{0.8}Co_{0.1}Mn_{0.1}O₂ cathode at a high state of charge (SoC) in in situ XRD, which can be attributed to the presence of significant lattice strain at the interface between the cathode material's surface and bulk. On the other hand,

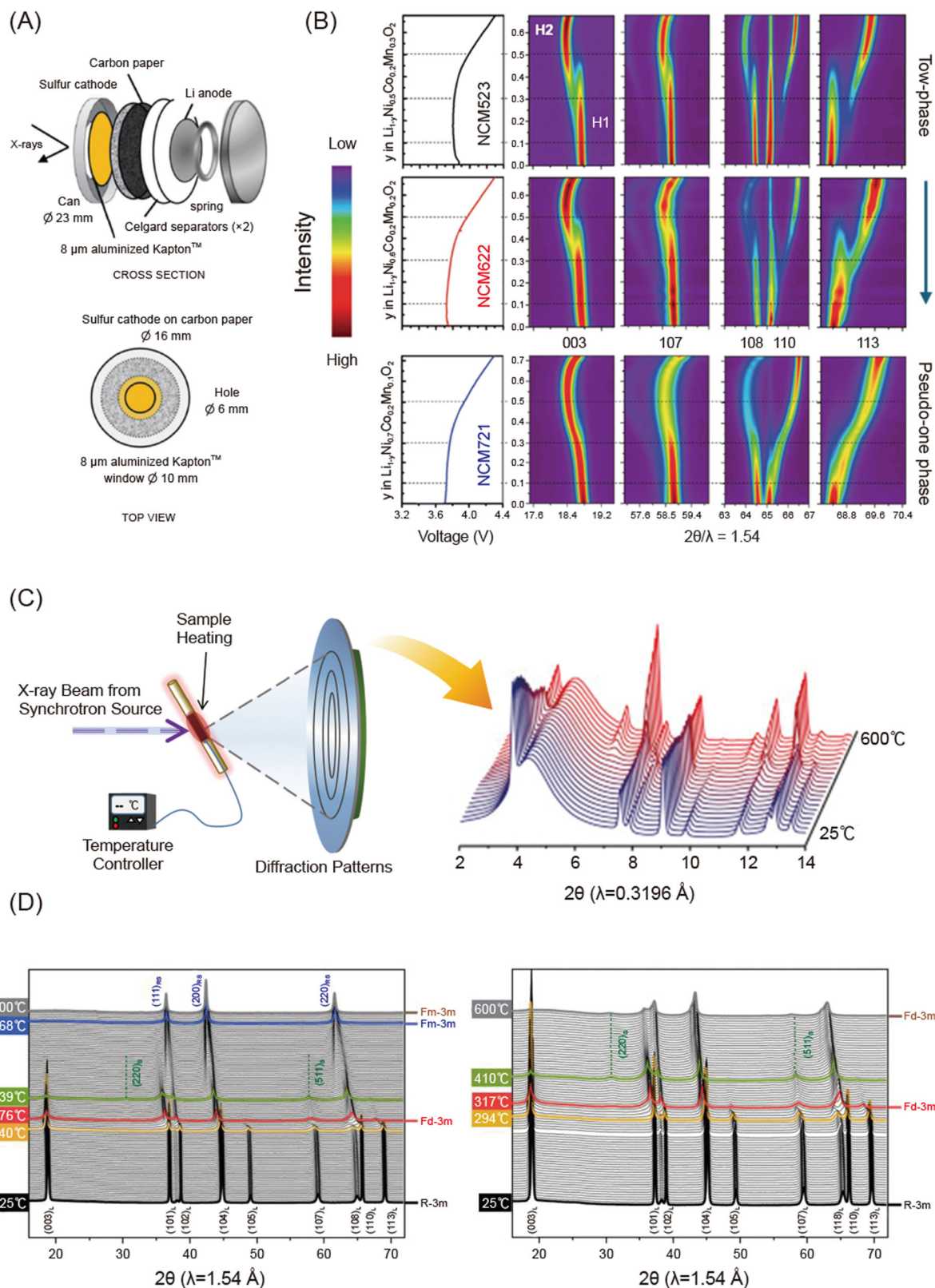


FIGURE 2 (A) Schematic illustration of an in situ XRD coin cell with a Kapton window. Reproduced with permission: Copyright 2019, Wiley.²⁶ (B) Contour plots for in situ XRD patterns of the $\text{LiNi}_{0.5+x}\text{Co}_{0.2}\text{Mn}_{0.3-x}\text{O}_2$ ($x = 0, 0.1, \text{ and } 0.2$) series during charge. Reproduced with permission: Copyright 2017, Wiley.²⁷ (C) Schematic illustration of the in situ XRD measurement process with quartz capillary during heating. Reproduced with permission: Copyright 2015, Elsevier.²⁸ (D) In situ XRD patterns of bare NCM and Al-doped NCM while heating it up to 600°C. Reproduced with permission: Copyright 2020, Elsevier.²⁹

Jeong et al.²⁹ obtained in situ XRD data of bare $\text{LiNi}_{0.80}\text{Co}_{0.15}\text{Mn}_{0.05}\text{O}_2$ (bare NCM) and Al-doped $\text{LiNi}_{0.80}\text{Co}_{0.15}\text{Mn}_{0.05}\text{O}_2$ (Al-doped NCM) in capillaries as the temperature increases up to 600°C with the experimental setup shown in Figure 2C.²⁸ The results depicted in Figure 2D demonstrated phase transitions from layered structures to disordered spinel structures and then to rock-salt structures, suggesting that Al doping could delay the onset temperatures of phase transitions and improve the thermal stability of Ni-rich layered oxide cathodes.

In situ XRD serves as a cornerstone for understanding structural changes in battery materials, offering valuable insights into phase transitions. However, this technique is mostly used for the investigation of crystalline materials and may not provide precise information about amorphous or liquid phases. This limits its capacity to describe structural behaviors in noncrystalline materials. Additionally, diffraction data obtained from in situ XRD can be difficult to interpret if there is an overlap of diffraction peaks among a target material and other components in the cell. This can complicate data processing and interpretation, necessitating careful peak separation and analysis procedures. While XRD excels in characterizing crystalline materials, complementary techniques such as XAS, Raman, NMR, and TEM should be used when analyzing amorphous or liquid phases in battery materials.

2.2 | In situ SAXS

SAXS allows the examination of the material morphology and nanostructure by measuring the scattered X-rays at small angles, typically 0.1°–10°, relative to the incident beam. It is especially useful for researching dispersed or isotropic systems such as macromolecules, colloidal suspensions, self-assembling materials, or nanoparticles in solution.^{30–32} This is because the X-rays strike the sample and scatter in various directions due to the diverse electron density within the sample. Grazing-incidence small-angle X-ray scattering (GISAXS) is an extension of SAXS specifically designed for examining thin films or surface characteristics.³³ By adjusting the grazing angle and sample-detector geometry, GISAXS can provide information regarding the lateral ordering, periodicity, and density variations of nanoscale structures at the surface.^{34,35} Because in situ SAXS techniques can be used to observe the scattering events of X-rays by the sample, similar experimental setups can be used for in situ SAXS measurements as for in situ XRD.^{36,37}

The first in situ SAXS study for batteries was conducted by Sandi et al.,³⁸ in which they monitored

the structure of carbons in lithium batteries and assessed their electrochemical performance using in situ SAXS. More recently, Xiong et al.³⁹ conducted in situ ultra-SAXS tests as shown in Figure 3A to investigate the structural variation of a polypropylene separator upon heating. They estimated various structural parameters, including porosity, specific interface, and pore size, and their findings revealed the melting temperature of the separator. In another case of in situ SAXS analysis by Prehal et al.,³⁶ shown in Figure 3B, the evolution of the Li_2O_2 phase in Li–O₂ cells was demonstrated. Through the combination of quantitative structural analysis and modeling for complex multiphase systems, they determined that the discharge capacities in Li–O₂ cells are predominantly dependent on the mobility of reactive O species rather than electronic transport through a Li_2O_2 film. In addition, in situ SAXS techniques can provide the ability to monitor mesoscopic structural changes. Park et al.^{40,41} observed the pore dynamics of ordered mesoporous electrode materials during electrochemical reactions, as illustrated in Figure 3C. They demonstrated that the in situ SAXS technique could provide direct evidence for the role of mesopores that effectively accommodates the volume expansion of a crystal structure by observing the changes in mesoscopic pore volumes during the (de)lithiation process.

In situ SAXS is used to study structural changes and particle size distributions in battery materials. However, it has the disadvantage of low scattering contrast, which makes it difficult to distinguish the untargeted components. Low scattering contrast can be caused by several factors, such as similar electron densities, similar atomic numbers, or similar molecular weights of the components. This limitation can hinder the precise identification of specific structural features in complex systems. Also, extracting meaningful structural information from the scattering patterns often requires advanced data processing methods and modeling techniques, considering factors such as particle size, shape, and distribution within the battery materials.

2.3 | In situ hard XAS

XAS is based on the interaction between X-rays and atoms, and de Broglie observed the first X-ray absorption spectrum in 1913.⁴² When X-rays are directed at a sample, they can cause one of the core electrons of an atom to jump to a higher energy level or even escape from the atom altogether, which is referred to as X-ray absorption.⁴³ The intensity of this absorption is not constant but varies with the energy of the incoming X-rays. The absorption coefficient is measured as a

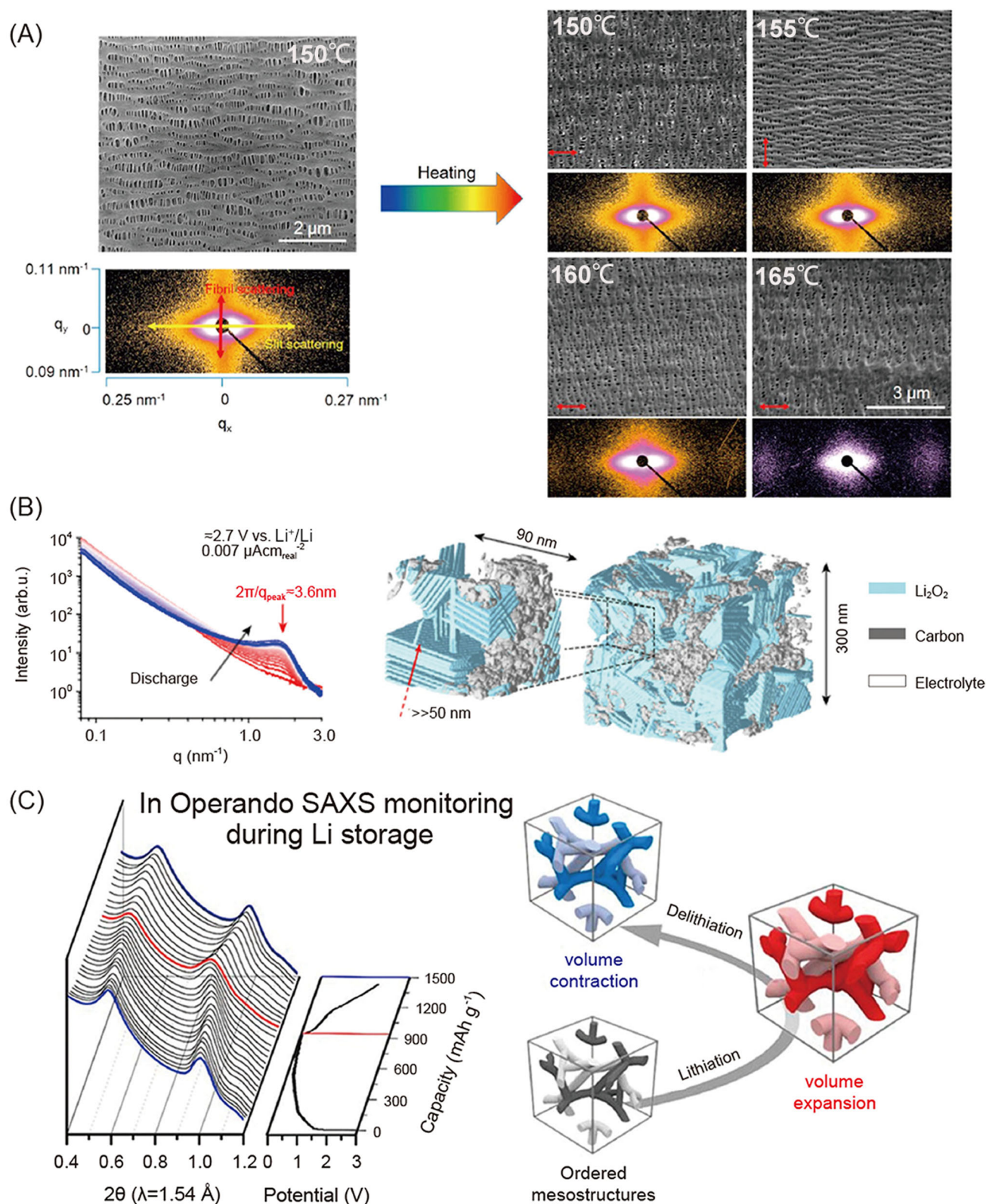


FIGURE 3 (A) Depiction of in situ ultra-small-angle X-ray scattering patterns along with a SEM micrograph showcasing the polypropylene separator. Reproduced with permission: Copyright 2018, Elsevier.³⁹ (B) Presentation of in situ SAXS data alongside modeling outcomes for electrolytes that dissociate LiO₂. Reproduced with permission: Copyright 2021, National Academy of Sciences.³⁶ (C) In situ SAXS spectra with corresponding voltage profiles (left) and schematic diagram of the pore dynamics in the ordered mesoporous metal oxide electrodes during the initial cycle (right). Reproduced with permission: Copyright 2015, American Chemical Society.⁴⁰

function of photon energy. The coefficient increases steeply near the binding energy, called the absorption edge. By analyzing the fine features near the absorption edge, it can investigate the electronic properties and local structures of specific elements.⁴⁴ More specifically, XAS spectra can be used to determine the element's valence state, which changes according to the chemical reactions.^{45–47} The local atomic structure encompasses coordination symmetry, bond length, and thermal/static disorder. According to the energy of the light source, XAS is divided into hard XAS (>5 keV) and soft XAS (<1 keV).

In battery research, hard XAS is primarily utilized to measure the transition-metal K-edge where the 1s electron is excited to the unfilled p orbital. The metal K-edge XAS spectrum involves two regions: X-ray absorption near-edge structure (XANES) and extended X-ray absorption fine structure (EXAFS).⁴⁸ XANES provides information about the site symmetry and valence states surrounding the absorbing element, covering an energy range from a few eV below to 50 eV above the absorption edge. EXAFS extends beyond the XANES region and reveals details about coordination numbers, bond lengths, bond angles, and disorder factors of the absorbing element and its neighboring atoms.^{49,50} It enables analysis of atomic-level properties and characterization of short- and long-range ordering in the electrode materials.

The experimental cell of in situ hard XAS can be fabricated using the same methodology as the in situ XRD cell. Openings for the inactive components must be included in order to allow X-rays to penetrate. Materials like a beryllium window, a high-density polyethylene foil, or Kapton can be used to keep a sealed system in place as shown in Figure 4A.^{9,51,52}

Shiraishi et al.⁵⁴ first applied in situ hard XAS to investigate the variations in the local structures of Mn atoms in LiMn_2O_4 in 1997, and numerous subsequent studies have utilized in situ hard XAS during various electrochemical processes in battery systems. Lee et al.⁵³ used in situ hard XAS to study the transition-metal redox chemistry and local environment of $\text{LiNi}_{0.88}\text{Co}_{0.1}\text{Al}_{0.02}\text{O}_2$ during the deintercalation process. As can be seen from the results of XANES and EXAFS spectra in Figure 4B, the correlation of the constituent elements in the Ni–Co–Ni configuration was confirmed, suggesting the possibility of an interplay between t_{2g} orbitals and e_g orbitals of Ni ions. This phenomenon indicates the importance of the local structural changes around not only Ni but also Co sites to achieve high-voltage operation batteries. Also, Nam et al.⁹ systematically studied and described the structural change during heating up to 600°C in overcharged $\text{Li}_{0.33}\text{Ni}_{0.80}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$ and $\text{Li}_{0.33}\text{Ni}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ using the in

situ hard XAS technique in combination with XRD and TEM. Based on the results presented in Figure 4C,D, a phase transition of each sample from a layered to a LiM_2O_4 -type and a rock-salt phase was found. Among them, the $\text{Li}_{0.33}\text{Ni}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ sample with a higher cobalt content compared to the $\text{Li}_{0.33}\text{Ni}_{0.80}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$ sample showed excellent thermal stability with contributions from a minor Co_3O_4 -type spinel phase.

In situ hard XAS plays a pivotal role in elucidating the electronic and local structures of specific elements within battery materials. However, caution must be exercised when using in situ hard XAS to obtain accurate and reliable data. First, the thickness of samples should be carefully controlled to achieve a proper edge step in hard XAS spectra. Therefore, prior knowledge of each sample and experimental procedures would be necessary and should be well understood to ensure meaningful data. Additionally, proficiency in complex software tools and data analysis techniques is essential for accurate XANES and EXAFS analyses. This involves expertise in data processing, refining, and interpretation to extract information about the electronic structure and chemical environment of the constituent elements in specific samples. These considerations, while demanding, pave the way for more precise and comprehensive investigations of battery performance.

2.4 | In situ soft XAS

Soft XAS was initially developed to measure surface X-ray absorption spectra, offering information about the electronic structure of specific elements.⁵⁵ Thereby, in comparison to bulk-sensitive hard XAS, soft XAS has the distinct feature of detecting substances at depths ranging from several to hundreds of nanometers from the surface. This technique also measures X-ray absorption resulting from the excitation of electrons in specific elements.⁴⁴ It effectively analyzes transition-metal L-edge spectra, involving the excitation of a 2p electron to an unfilled d orbital, as well as K-edge spectra of light elements like O. These spectra contain information about the bonding properties of metal or ligand ions within the battery materials.^{56–61}

Soft XAS often utilizes four detection modes: total electron yield (TEY), fluorescence yield (FY), partial electron yield (PEY), and partial fluorescence yield (PFY) mode. TEY mode analyzes the electron yield from the sample surface, providing information about the electronic structure of the surface or interfacial processes within approximately 10-nm depth. FY mode detects characteristic X-ray fluorescence emitted from the material. It provides information about the bulk and near-surface regions of the material, with a probing depth of more

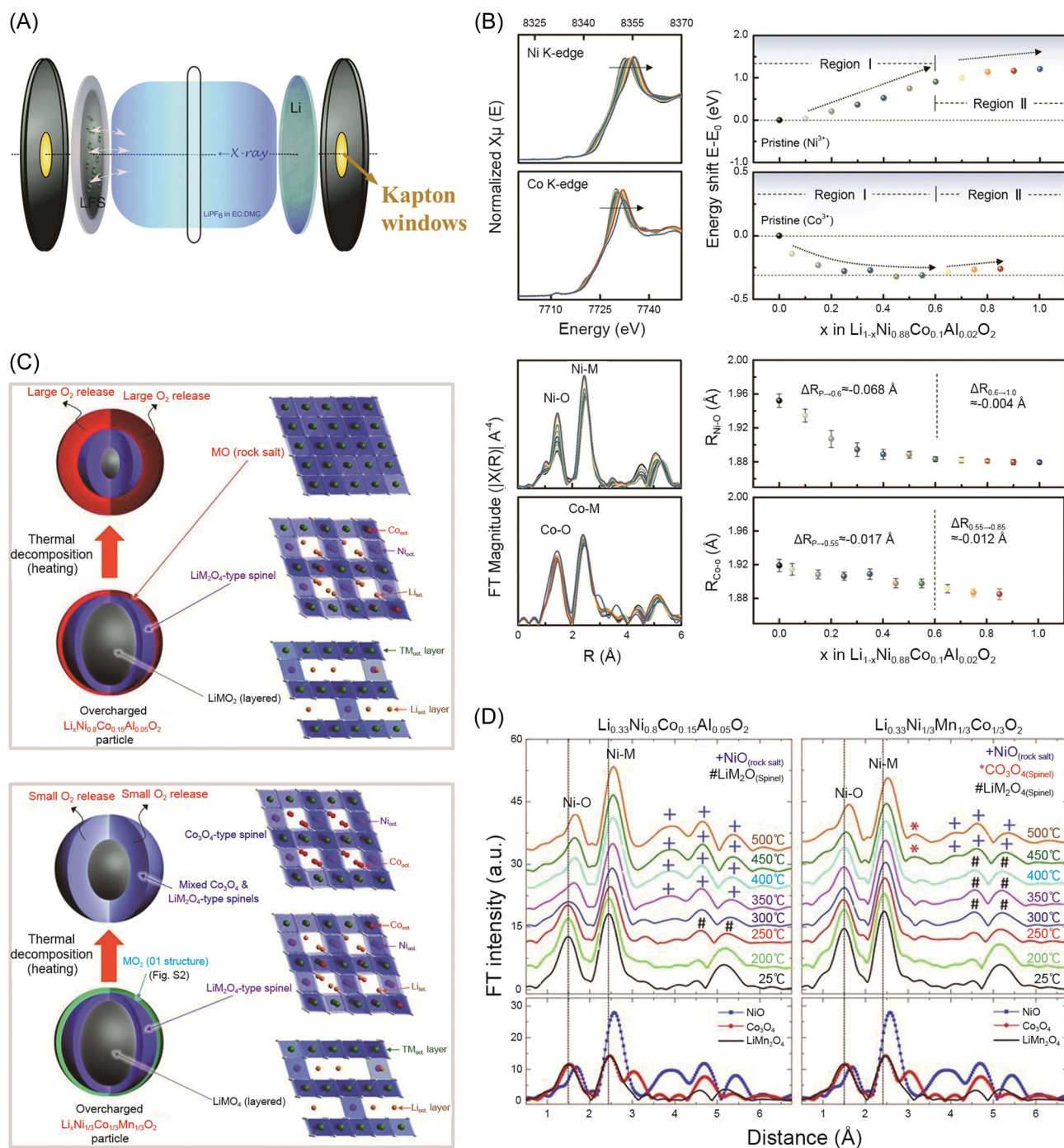


FIGURE 4 (A) Schematic illustration of a Kapton-sealed in situ XAS coin cell. Reproduced under the terms of the CC BY 3.0 Unported license: Copyright 2016, The Royal Society of Chemistry.⁵¹ (B) Ni K-edge and Co K-edge XANES spectra with an energy shift graph (top) and Ni K-edge and Co K-edge EXAFS spectra with the bond length of the first coordination shell (bottom) for $\text{Li}_{1-x}\text{Ni}_{0.88}\text{Co}_{0.1}\text{Al}_{0.02}\text{O}_2$ during the first charging. Reproduced with permission: Copyright 2020, Wiley.⁵³ (C) Schematic illustration of the thermal decomposition behavior of $\text{Li}_{0.33}\text{Ni}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$ (top) and $\text{Li}_{0.33}\text{Ni}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ (bottom). (D) Ni K-edge EXAFS spectra of delithiated $\text{Li}_{0.33}\text{Ni}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$ (left) and $\text{Li}_{0.33}\text{Ni}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ while heating it up to 500°C. Reproduced with permission: Copyright 2013, Wiley.⁹

than 100 nm.^{62,63} PEY mode (probing depth: ~5 nm) selectively detects the emitted Auger electrons, which makes soft XAS slightly more surface-sensitive than TEY mode.⁶⁴ In PFY mode, the energy range of emitted

photons is restricted to the measurement of only a fraction of FY signals. The electronic structure is obtained for bulkier regions compared to the regions detected by FY mode.

In situ soft XAS analysis faces difficulties owing to the shallow sampling depth and ultrahigh vacuum (UHV) condition requirement. UHV is essential for X-rays to avoid interference from unwanted gas-phase reactions that can affect measurement accuracy. Conventional coin cells may prevent X-ray penetration and hinder the detection of emitted electrons or photons. To overcome these challenges, specialized systems are required. For example, Liu et al.⁶⁵ constructed an extraordinary electrochemical cell for an in situ soft XAS experiment presented in Figure 5A. They designed a setup with detection windows drilled in an Al foil current collector to allow direct soft X-ray measurements on the electrode materials with the solid-state electrolyte. They discovered distinct charge dynamics between $\text{Li}(\text{Ni}_{1/3}\text{Mn}_{1/3}\text{Co}_{1/3})\text{O}_2$ (NMC) and LiFePO_4 (LFP) cathodes as shown in Figure 5B. In NMC electrodes, the transition metals immediately responded to electrochemical operations. In contrast, strong relaxation and SoC gradient effects are observed in LFP electrodes, with phase transformation starting near the current collectors and taking hours to reach a uniform charge distribution. Also, Yoon et al.⁶⁶ utilized the in situ soft XAS technique to analyze the thermal behavior of delithiated $\text{Li}_{0.33}\text{Ni}_{0.80}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$ cathode material as illustrated in Figure 5C–E. From the PEY data, it was found that the material showed O loss, resulting in a lower valence state of Ni and the formation of a NiO-like rock-salt structure at the surface. Interestingly, no surface reaction near the Co site was observed within the investigated temperature range, suggesting that the thermal behavior was mainly dependent on specific elements.

In situ soft XAS is useful for probing the chemical state of lightweight elements in battery materials, particularly essential in the study of surface reactions. Nevertheless, there are several important considerations to keep in mind. In situ soft XAS requires a high-intensity X-ray source, which may not always be accessible in a laboratory-based environment. The reliance on specialized facilities can limit its widespread adoption and accessibility. UHV requirements also hinder the use of liquid electrolyte systems as the cell needs to be open for soft X-rays to interact with the electrode. This limitation restricts the technique's applicability to a subset of battery configurations. Self-absorption effects can also complicate FY detection in soft XAS, where the sample absorbs a significant portion of the emitted X-rays, reducing the detected signal and distorting the absorption spectrum. To maximize the technique's utility in battery research, researchers must carefully weigh the benefits and limitations, considering the availability of facilities and their specific research objectives.

2.5 | In situ XPS

XPS, also known as electron spectroscopy for chemical analysis, is a surface-sensitive analytical technique frequently used in the examination of Li-rechargeable batteries. It provides only a few nanometers of surface information owing to the short mean free path of electrons.^{67,68} Furthermore, XPS has high sensitivity, capable of identifying tiny amounts of substances down to 0.1 atomic percent.⁶⁹ In principle, this technique uses X-ray sources with fixed photon energy and measures the kinetic energy of the released photoelectrons to calculate the binding energy. In the resultant electron spectrum, the chemical composition and electronic states of the sample surface can be determined by investigating the positions of characteristic peaks. Quantitative analysis of the integrated peak area reveals the relative amount or concentration of the chemical species.

Likewise, the in situ XPS technique provides a real-time examination of the interfaces between electrodes and electrolytes such as material surface degradation and the formation of the solid electrolyte interface layer during battery operation.^{70–75} The first in situ XPS study of battery systems was conducted by Hensley et al.⁷⁶, who detected compositional and structural changes in the interface region of a LiBO_2 film as a result of lithium deposition. Similar to soft XAS, in situ XPS requires the complicated fabrication of an electrochemical cell as depicted in Figure 6A, for which $\text{Li}_{6.6}\text{La}_3\text{Zr}_{1.6}\text{Ta}_{0.4}\text{O}_{12}$ was used as a solid electrolyte to accommodate the UHV environment.⁷⁷ This enabled the quantitative analysis of changes in the composition of the amorphous silicon layer and the formation of byproducts during the electrochemical reaction. As another example, Nandasiri et al.⁷⁸ created a setup that included Li and graphite electrodes mounted on a Teflon base as shown in Figure 6B. The in situ XPS results (Figure 6C–E) and computational modeling were combined to investigate the formation mechanism of the SEI layer on the Li anode of the Li–S battery. The results revealed the gradual formation of insoluble Li_2S compounds and Li–F species owing to the reduction processes of polysulfide species and decomposition of the electrolyte. Also, Yao et al.⁷⁹ utilized the in situ XPS technique in combination with the in situ XRD to study the thermal stability of Li_2O_2 and Li_2O powders in Li–air batteries. The surfaces of Li_2O_2 showed the appearance and growth of Li_2O and Li_2CO_3 starting at 250°C. These findings highlighted the challenge and importance of developing stable carbon-based electrodes for rechargeable Li–air batteries.

In situ XPS is useful for studying surface chemistry and SEI formation in battery materials during cell operation. However, it is essential to consider the following limitations. One significant limitation of in situ XPS is the requirement for UHV conditions during analysis. This can

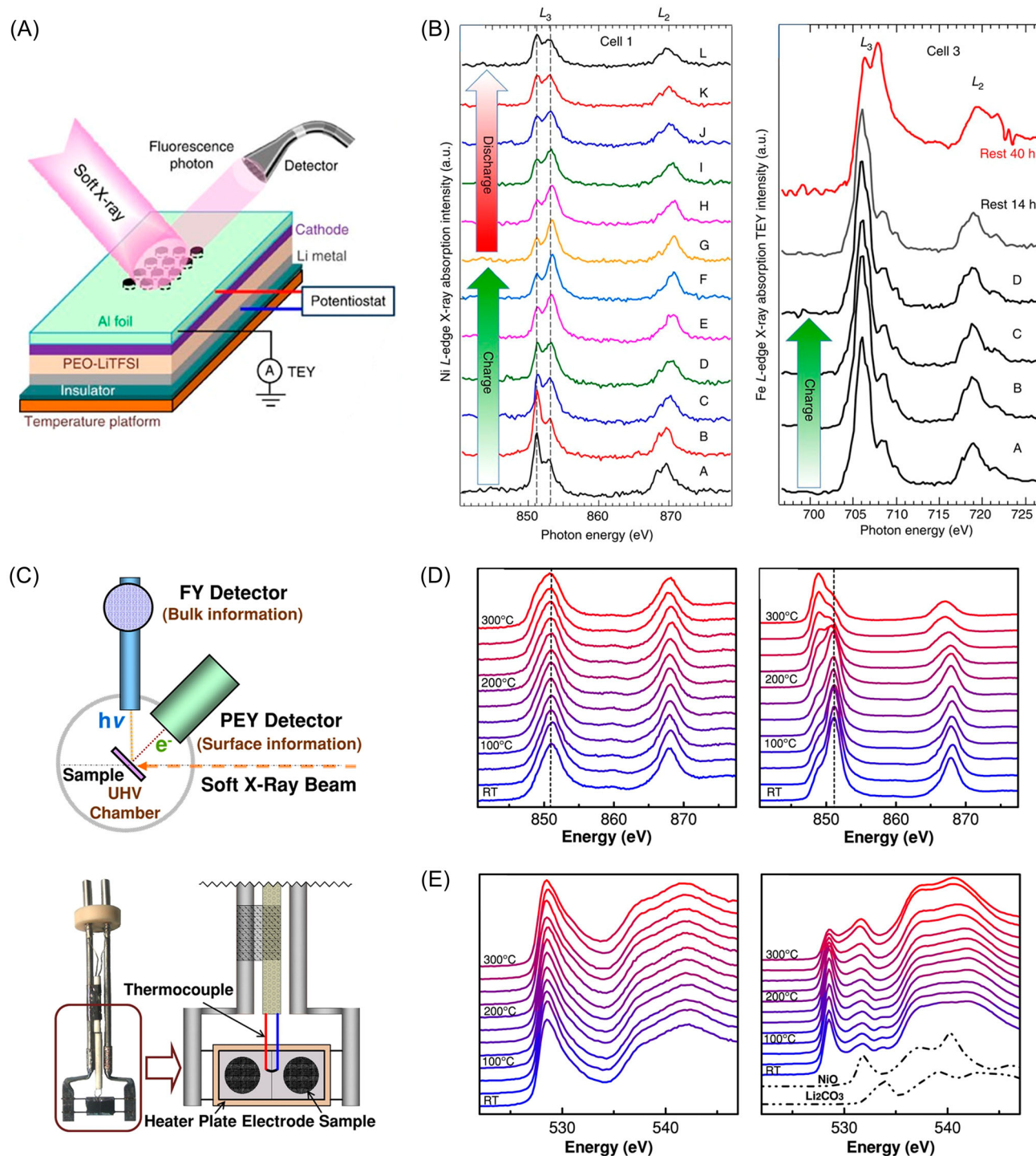


FIGURE 5 (A) Schematic illustration of an electrochemical cell for in situ soft XAS. (B) Ni L-edge spectra (left) of NMC obtained in TFM mode during a cycle at C/5 and Fe L-edge spectra (right) of LFP obtained in TEY mode during charge and subsequent relaxation. Reproduced under terms of the CC-BY license (<http://creativecommons.org/licenses/by/4.0/>). Copyright 2013, The Authors, published by Springer Nature.⁶⁵ (C) Schematic diagram of the experimental setup (top) and sample stage with a heating plate (bottom) for in situ soft XAS. (D) Ni L-edge spectra of NCA in FY mode (left) and PEY mode (right) during heating up to 300°C. (E) O K-edge spectra of NCA in FY mode (left) and PEY mode (right) during heating up to 300°C. Reproduced under terms of the CC-BY license (<http://creativecommons.org/licenses/by/4.0/>). Copyright 2014, The Authors, published by Springer Nature.⁶⁶

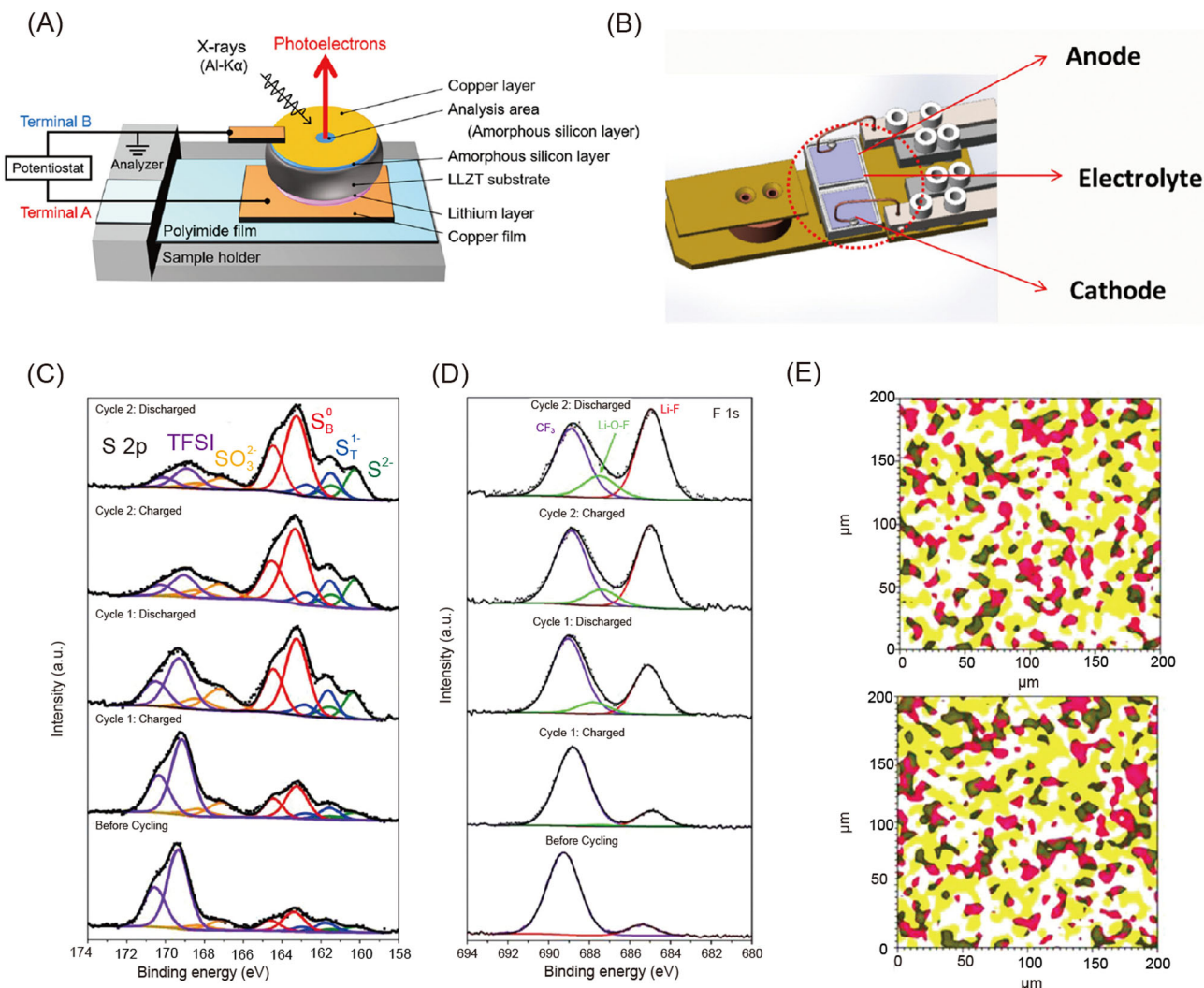


FIGURE 6 (A) Schematic illustration of the electrochemical cell for in situ XPS. Reproduced with permission: Copyright 2020, American Chemical Society.⁷⁷ (B) Schematic illustration of the in situ XPS cell. (C) Core level S 2p and (D) F 1s XPS spectra of the Li–electrolyte interface. (E) Chemical map of the Li–electrolyte interface for the first charge cycle (top) and discharge cycle (bottom). Reproduced with permission: Copyright 2017, American Chemical Society.⁷⁸

be particularly challenging when working with common liquid electrolytes, as they have the potential to evaporate and disrupt the vacuum condition. The existence of potential artifacts that may affect the accuracy of the results presents another difficulty for in situ XPS examination. Charging effects can occur when the sample becomes electrically charged, resulting in spectral data aberrations. Additionally, peak overlapping and shifting might make it challenging to distinguish and precisely quantify various chemical species, particularly in complex battery materials. Careful data analysis and deconvolution procedures are often necessary to extract meaningful information. In situ XPS may be challenging to use, but on balancing its undeniable advantages with the potential limitations, it proves valuable for observing surface-sensitive phenomena in the field of battery research.

2.6 | In situ XCT

The term tomography, which originates from ancient Greek and means slice, refers to the process of capturing 2D images of an object from multiple angles and reconstructing its internal structure in 3D. XCT allows imaging with a wide range of spatial resolutions, from micrometers to nanometers.^{80–82} This versatility enables the study of features at different length scales, ranging from large-scale electrode structures to small-scale particles and interfaces. The first 3D XCT images of battery electrodes were captured by Shearing et al.,⁸³ who successfully analyzed the 3D microstructure of a porous graphite electrode in an 18650 battery. By capturing fine-scale details, XCT facilitates a comprehensive understanding of the mechanical characteristics and

interactions within battery systems.^{84,85} The XCT technique relies on the differential attenuation of X-rays as they pass through materials.^{86,87} During an XCT scan, the sample is placed on a rotating platform, and an X-ray source emits a narrow beam that passes through the object. Detectors positioned opposite the X-ray source measure the intensity of the X-rays after they have passed through the object. The resulting X-ray intensities are affected by the object's internal structure, including density and composition variations.

In situ XCT enables the real-time analysis of various phenomena, including the investigation of electrode volume change,⁸⁸ crack formation,⁸⁹ and thermal behavior.⁹⁰ To conduct in situ XCT experiments, specially designed electrochemical cells are required for rotating the sample stage.^{91,92} For example, Figure 7A shows a

schematic illustration of a Swagelok-type cylindrical cell consisting of two plungers with stainless steel and straight unions with perfluoro alkoxy, which is commonly used to obtain unobstructed data during the 360° rotation.^{93,94} Also, it is possible to use a coin-type half-cell, which is shown in Figure 7B, if obtaining 2D images is not a concern.⁹⁵

Nelson et al.⁹⁷ investigated the structural evolution in Li-S batteries during electrochemical reactions using in situ XCT for the first time. They visualized the morphological changes of the sulfur/Super P composite and observed the dissolution of polysulfides through a slightly smaller particle. In 2014, Wang et al.⁹⁶ utilized synchrotron in situ X-ray nanotomography to track the microstructural changes and phase distribution of Sn and Li_xSn phases in Sn anodes. Figure 7C shows the chemical-phase maps of Sn particles from pristine states

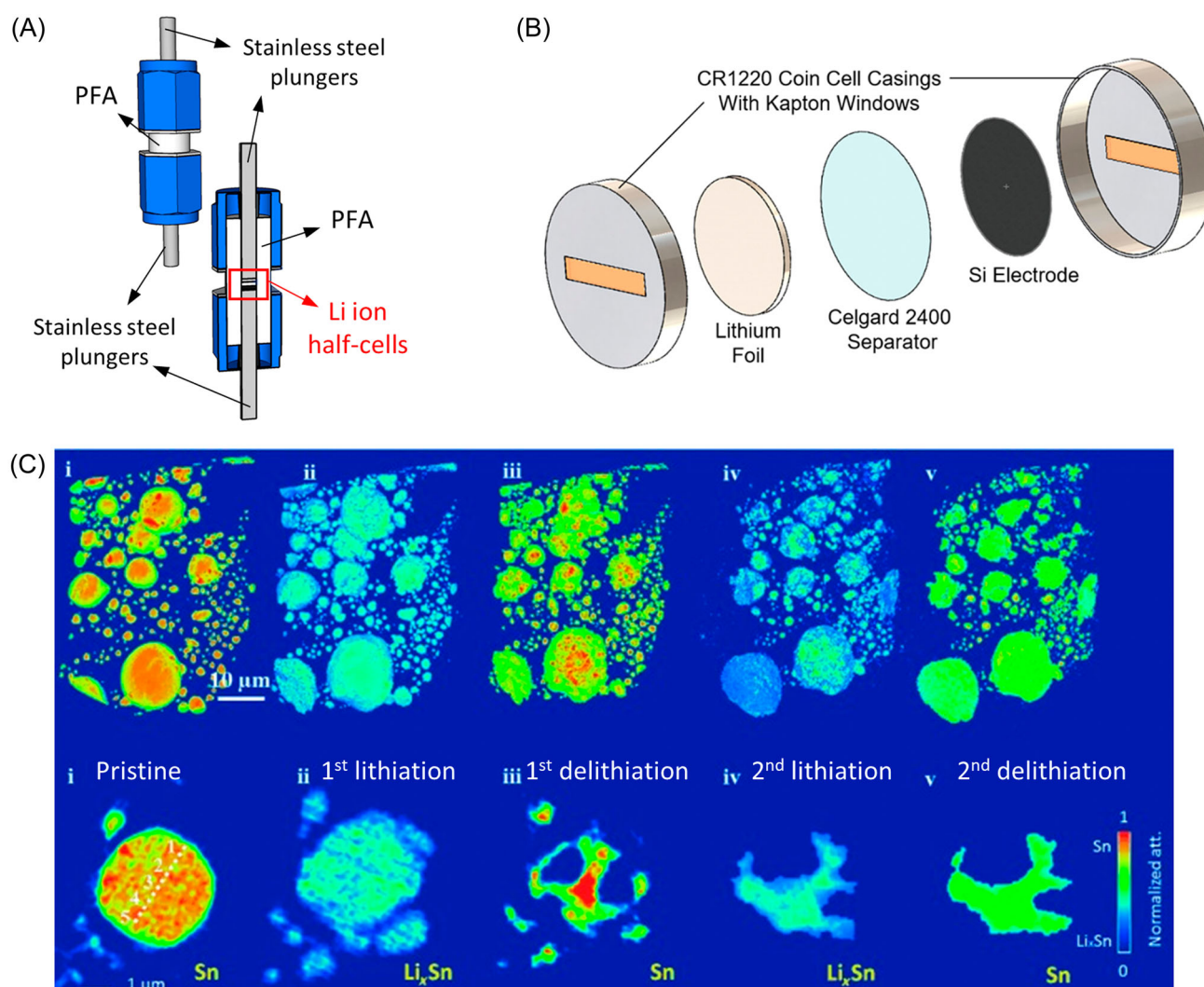


FIGURE 7 (A) Schematic illustration of a modified Swagelok cell for in situ XCT. Reproduced under terms of the CC-BY license (<http://creativecommons.org/licenses/by/4.0/>). Copyright 2018, The Authors, published by MDPI.⁹³ (B) Schematic of a coin cell setup for in situ XCT. Reproduced with permission: Copyright 2021, American Chemical Society.⁹⁵ (C) Microstructural changes of Sn particles during the initial two electrochemical cycles. Reproduced with permission: Copyright 2014, Wiley.⁹⁶

to the second delithiation process. The images below show the selected particles. During the first delithiation (ii–iii), severe fracture and pulverization were observed, which were caused by the extraction of Li ions and volume shrinkage. Additionally, they found that this deformation leads to inhomogeneous phase distribution in the particle, resulting in a higher attenuation value in the core after the first delithiation process.

In situ XCT nondestructively visualizes the internal structures and dynamic processes in battery materials. However, the penetration depth of X-rays is limited, which hampers the visualization of internal structures in thick or high-density objects. Because of this restriction, detailed information from deeper regions cannot be obtained. Additionally, there is a trade-off between spatial and temporal resolution, as achieving high spatial resolution requires longer acquisition times, limiting the observation of fast dynamic processes. To best suit research objectives, the spatial and temporal resolution should be balanced. Finally, differentiating materials with similar X-ray attenuation coefficients is challenging, leading to ambiguities especially when distinguishing phases, interfaces, or reactions involving such materials.

2.7 | In situ TXM

TXM is an advanced imaging technique based on X-ray absorption that can generate high-resolution images. It may use either hard or soft X-rays, and yet, the focus of this section will be on TXM using hard X-rays. Hard X-rays are advantageous for assessing the internal structures within the battery owing to their higher energy and superior capacity to penetrate thicker materials. In addition, TXM can produce spatial maps that show element-specific details on the distribution and localization of different phases, chemical elements, and oxidation states inside the electrode.^{98–100} Analysis of these spatial distributions allows for an in-depth evaluation of the electrode's homogeneity and aids in detecting possible degradation processes.^{101,102} TXM detects the transmitted intensity of X-rays directed onto a sample using a circular grating known as a Fresnel zone plate (FZP).^{103,104} FZPs intricately focus X-rays via diffraction, producing images with extraordinary spatial resolutions of 100 nm or even finer. Thus, TXM can produce high-resolution contrast images and XANES spectra by accurately altering the photon energy to align with the absorption edge of a particular element.

In situ TXM is helpful in identifying active and inactive regions within the electrode, which can be correlated with electrochemical performance.^{105–107} Furthermore, they allow for continuous monitoring of crack and void development, which can impact the mechanical

and electrical integrity of the electrode.^{108,109} By measuring X-ray absorption contrast and applying image processing algorithms, in situ TXM can offer quantitative information about the SoC, state of health, and state of lithiation of the electrodes.¹¹⁰ To efficiently use in situ TXM for battery research, the cell should include thin and transparent windows that allow for X-ray transmission, as well as electrical connectors to apply voltage and current. Various cell types can be used, including coin cell, pouch cell, and plastic cell, as shown in Figure 8A,B.^{111–113}

Since Wang et al.¹⁰⁷ first demonstrated the potential of in situ TXM for nondestructive and precise chemical mapping of CuO battery materials, numerous in situ TXM studies have been conducted to analyze complex battery systems during electrochemical processes. Li et al.¹¹¹ used in situ TXM to visualize the solid-state phase transformations. They compared two samples of iron fluoride with different microstructures and porosity and observed how they affect phase evolution during cycling. As shown in the chemical phase maps of Figure 8C, the porous polycrystalline iron fluoride showed faster and more complete Li storage than that of dense single-crystal iron fluoride. Another study used in situ TXM to compare chemical phase mappings of single-crystal Ni-rich cathodes between the 1st and 201st cycles at high voltage. Zhang et al.¹¹⁴ visualized the surface chemistry and phase distribution changes from homogeneity to heterogeneity during cycling, which induced heterogeneous internal strain and performance degradation. Moreover, Lou et al.¹⁰⁸ found interfacial issues for a solid-state battery system, as shown in Figure 8D. They compared the changes of the chemical state between solid–solid interfaces and solid–liquid interfaces in polycrystalline $\text{LiNi}_{0.6}\text{Co}_{0.2}\text{Mn}_{0.2}\text{O}_2$. Through chemical phase mapping of single cathode particles in the early SoC, they inferred physical contact loss at solid–solid interfaces, which also impacted the heterogeneity of ion transport kinetics within the interior local environment of secondary particles.

In situ TXM offers the real-time visualization of microscale structural changes simultaneously with electronic states in battery materials. However, the spatial and temporal resolutions of an in situ TXM are subject to a trade-off, necessitating a compromise between the two. Moreover, in situ TXM relies on a synchrotron radiation source, which can be expensive and often limited in availability.¹¹⁵ Additionally, the technique's restricted field of view and shallow depth of field may pose challenges in capturing the entire electrode or resolving intricate details accurately. Therefore, researchers must carefully balance these parameters to suit their specific research objectives, potentially limiting their ability to capture fine details during rapid dynamic processes.

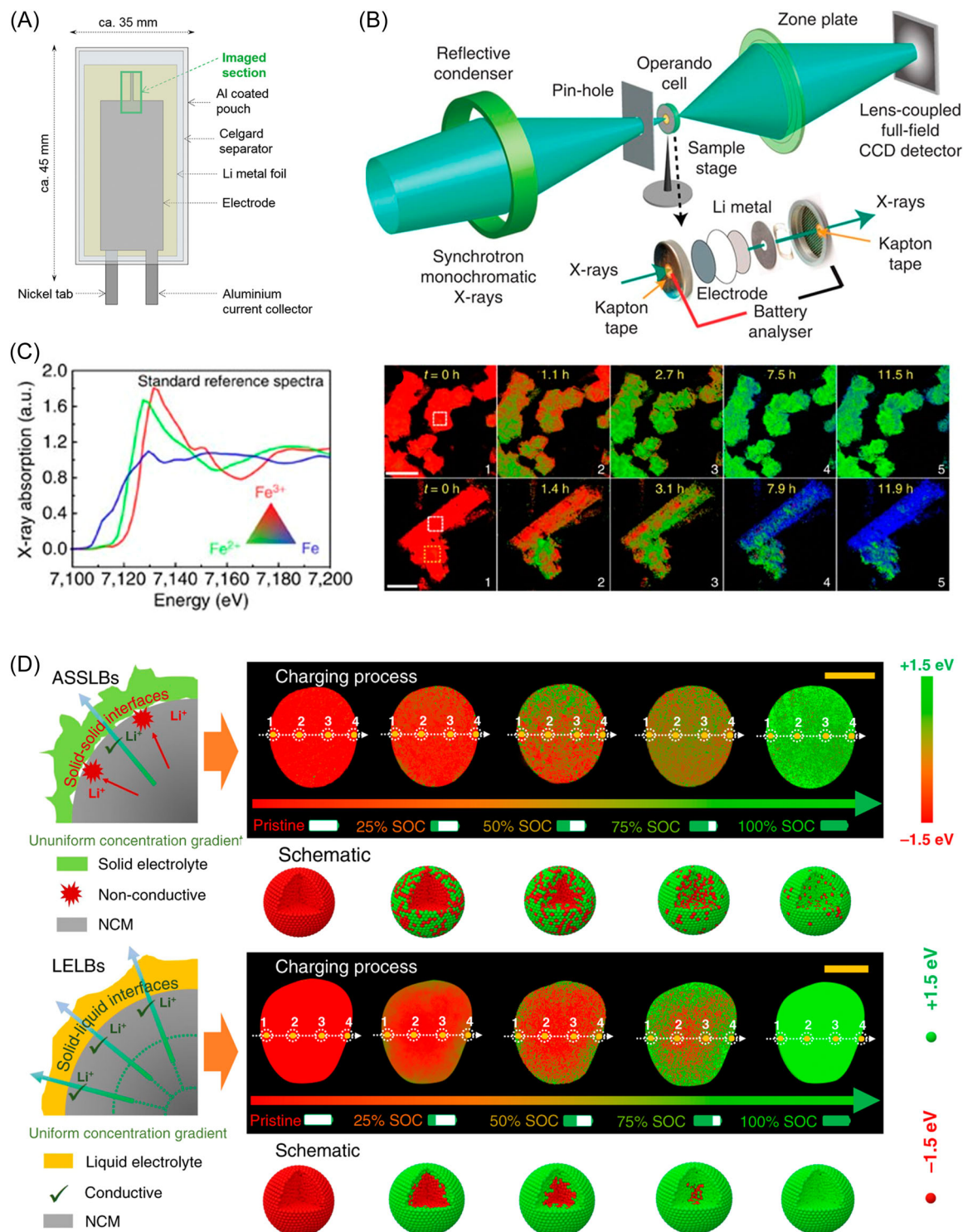


FIGURE 8 (A) Pouch cell configuration for an in situ TXM experiment. Reproduced under the terms of the CC BY-NC-ND license: Copyright 2021, Elsevier.¹¹³ (B) Perforated coin cell configuration for in situ TXM measurements. (C) Reference spectra of standard compounds (left) and chemical phase maps displaying the FeF₃ polyhedral structure and multiwire structure during the discharge processes (right). Reproduced under terms of the CC-BY license (<http://creativecommons.org/licenses/by/4.0/>). Copyright 2015, The Authors, published by Springer Nature.¹¹¹ (D) In situ 2D TXM-XANES mapping of a commercial LiNi_{0.6}Co_{0.2}Mn_{0.2}O₂ material during the initial charging process using both all-solid-state lithium batteries and liquid electrolyte lithium-ion batteries. Reproduced under terms of the CC-BY license (<http://creativecommons.org/licenses/by/4.0/>). Copyright 2020, The Authors, published by Springer Nature.¹⁰⁸

2.8 | In situ STXM

STXM uses soft X-rays for investigating the chemical and structural properties of materials at the nanoscale. Similar to TXM, STXM facilitates the generation of high-resolution images and element-specific spectra, enabling the identification of phase transitions, electronic structures, and degradation processes within the battery electrode.^{116,117} STXM entails the detection of X-ray absorption by comparing initial and transmitted intensities. By tuning the X-ray energy to correspond with specific absorption edges, STXM achieves element-specific contrast and chemical sensitivity. Each pixel in the resultant image corresponds to a soft XAS spectrum, capturing transitions at the transition-metal L-edge and light element K-edge. A zone plate is used to focus the X-ray beam, concentrating it to a spot size on the order of tens of nanometers. Thanks to the ability to obtain nanoscale chemical mapping images, the first STXM for battery research was used to observe the different states of charge in individual nanosized LFP particles.¹¹⁸

The application of in situ STXM enables the direct observation of variations in the morphology of active materials and dynamic chemical and structural changes.^{119,120} Also, by utilizing soft X-rays, it becomes possible to measure the K-edge of ligands such as O.^{121,122} This information aids in understanding anionic

redox mechanisms, such as the O evolution reaction and potential O loss.

To utilize in situ STXM, the sample must be placed in a specially designed electrochemical cell that allows simultaneous cycling and imaging under controlled environmental conditions. In the first study on in situ STXM, Ohmer et al.¹¹⁹ produced an all-solid-state micro-sized battery with a single LFP crystal along the (010) orientation to investigate the real-time phase transformation of LFP during electrochemical (de)lithiation. They scanned the particles for energies that are indicative of Fe^{2+} and Fe^{3+} . Through the comparison of the optical densities at each pixel, they were able to visualize and track the two oxidation states of iron. On the other hand, to study the reaction dynamics of LFP in a liquid electrolyte system, Lim et al.¹²⁰ used a multimodal microfluidic cell, depicted in Figure 9A, comprising two sandwiched Si membrane chips with 75-nm-thick SiN_x windows. Using in situ STXM, they measured the relative concentrations of Fe^{2+} and Fe^{3+} in LFP electrode material. Figure 9B shows line cuts of Li composition under different cycling conditions. At fast cycling rates, solid solutions formed as Li ions are removed and inserted. However, at slow rates, nanoscale phase separation occurs, which eventually shortens the battery life.

In situ STXM investigates the nanostructures within battery materials, allowing for detailed chemical and structural insights. However, it is essential to consider the

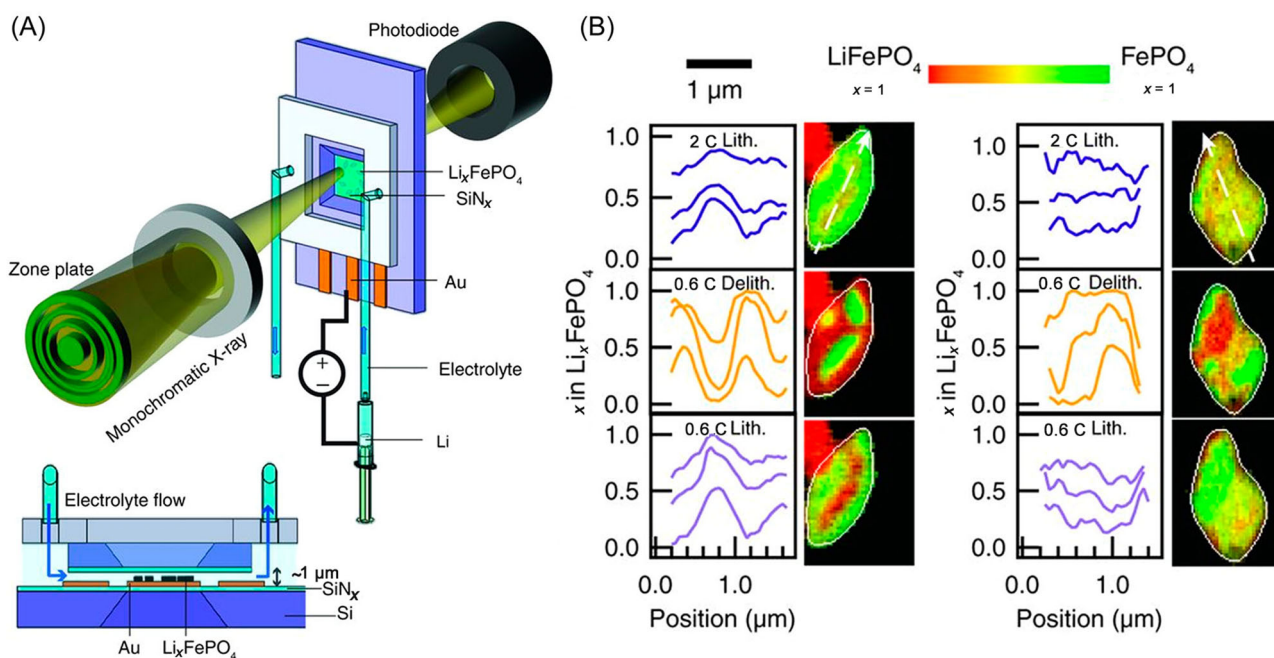


FIGURE 9 (A) Diagram illustrating the operational setup of in situ STXM for liquid electrochemical experiments involving LiFePO_4 . (B) Graphs showing line cuts of Li composition (x) within identical particles subjected to different cycling conditions. Reproduced with permission: Copyright 2016, The American Association for the Advancement of Science.¹²⁰

various limitations associated with this technique. In situ STXM necessitates sophisticated instrumentation and specialized facilities, including synchrotron radiation sources. The requirement for UHV conditions should also be considered for cell construction, making it necessary to have the appropriate expertise and access to specialized facilities. Furthermore, it shows limited penetration depth and field of view, thereby constraining observations to thin electrodes and smaller-scale particles. This requires the careful selection of their region of interest, which may result in the potential sacrifice of study of larger regions within the battery systems. Additionally, it presents challenges in terms of low temporal resolution, impeding the capture of fast dynamics and transient phenomena. This limitation makes it less suitable for studying rapid processes and necessitates a trade-off between spatial and temporal resolution.

2.9 | Recent studies

X-ray-based techniques, including XRD, XAS, XPS, XCT, TXM, STXM, and related methods, are instrumental in comprehending diverse aspects such as crystal structure, electronic configuration, morphology, and composition, all of which are pivotal for gaining understanding in battery research. Recently, these techniques have been evolving toward use for observation of fine structures by integrating with imaging technologies. Since 2006, the introduction of Bragg coherent diffraction imaging has facilitated simultaneous measurements of morphology and diffraction patterns, primarily used to assess lattice strain within microcrystals.^{123–125} In the case of XAS, approaches like TXM and STXM, which combine XAS spectra with X-ray imaging, have become prevalent in rechargeable battery research, as described in the preceding sections.¹²⁶ To overcome the limitations of vacuum conditions, researchers have developed ambient pressure XPS,¹²⁷ with the inaugural application of in situ ambient pressure XPS in rechargeable batteries dating back to 2012.¹²⁸ Table 1 provides additional recent instances of in situ techniques. Hence, it is evident that X-ray-based methods continue to be actively researched, and these efforts are expected to contribute to the significant development of Li-rechargeable batteries.

3 | UV AND VISIBLE LIGHT AND IR-BASED

UV-visible light induces electronic transitions, while IR stimulates vibrational transitions. By observing the electronic excitation-relaxation process and vibrational

transitions, the vibrational states of constituent chemical species can be investigated. On the other hand, visible light is perceptible to the naked eye and thereby provides an intuitive means of analysis. Consequently, in situ techniques utilizing UV-visible-IR can effectively track variations in morphological and vibrational characteristics. In this section, three in situ techniques based on UV-visible-IR radiation are introduced: OM, Raman spectroscopy, and FT-IR.

3.1 | In situ OM

The optical microscope, with its initial creation attributed to Hans and Zacharias Janssen around 1590,¹⁴⁸ is the first type of microscope. They designed the microscope with lenses placed inside a tube. Similarly, most modern OMs continue to use visible light and lenses to optically capture enlarged images of objects, maintaining their simplicity compared to more advanced microscopy techniques. By directly observing the electrode material and interface, OM allows for exploring the morphological characteristics of materials. When visible light passes through an object, it can be absorbed, reflected, or refracted. The resulting image is formed based on the differences in light intensity or color between the object and its background. OMs use lenses to magnify the image and focus it on a detector. The magnification depends on the focal length of the lenses, while the resolution is limited by the wavelength of light and the numerical aperture of the lenses.

In situ OM can detect phenomena such as Li dendrite growth^{149–155} and the propagation of phase boundary at various length scales.^{156–160} For instance, He et al.¹⁵⁵ utilized an electrochemical cell for in situ OM, as depicted in Figure 10A, to observe the formation and progression of Li dendrites at the interface between Li metal and gel polymer electrolyte. By applying polyphosphazene-based polymer electrolytes with various Li salts, they found that gel polymers with a higher Li transference number and reduced interface resistance effectively suppressed dendrite formation. In a more recent study, Gao et al.¹⁵⁹ used a custom-designed electrochemical cell, as depicted in Figure 10B, to observe Li plating on the basal surface of a single graphite particle in real time. Figure 10C shows the voltage profile for the graphite cell and microscopic images for the lithiation process into a graphite particle. Their observations revealed that the onset of plating is closely linked to phase separation in graphite and occurs only on the fully lithiated edges of the particles. Merryweather et al.¹⁶⁰ developed a custom-made lid of a half-cell to accommodate the microscope objective and glass observation windows. They used optical

TABLE 1 Recent studies of X-ray-based in situ techniques.

Techniques	Material @ cell	Main description/results
XRD	ZnMn ₂ O ₄ @ in situ cell with Be window	Tang et al. ¹²⁹ revealed the electrochemical conversion reaction mechanism of the carbon nanotube-coated ZnMn ₂ O ₄ anode.
	LiNi _{0.84} Co _{0.07} Mn _{0.09} O ₂	Deng et al. ¹³⁰ conducted a comparative investigation for the magnitude of lattice change between poly- and single-crystal LiNi _{0.84} Co _{0.07} Mn _{0.09} O ₂ .
	LiNi _{0.83} Co _{0.10} Mn _{0.07} O ₂ @ modified 2430 cell with Be window	Pang et al. ¹³¹ demonstrated the structural reversibility of single-crystalline LiNi _{0.83} Co _{0.10} Mn _{0.07} O ₂ during cycling.
	LiNi _{0.8} Mn _x Al _y O ₂ @ coin cell with Be window	Wang et al. ¹³² confirmed the enhanced reversibility of a Co-free cathode through the Al and cation vacancy effect.
SAXS	Cu @ customized sputter chamber	Schaper et al. ¹³³ revealed the growth of Cu on polystyrene-block-poly(ethylene oxide) diblock copolymer thin films.
	High-molecular-weight polyethylene-liquid paraffin gel film	Wu et al. ¹³⁴ examined how ultra-high-molecular-weight polyethylene behaves in terms of cavitation during stretching in the wet and dry process.
Hard XAS	LiCoO ₂ @ Al-laminated cell	Mukai et al. ⁴⁷ elucidated the reaction mechanisms of overstoichiometric LiCoO ₂ at deep charge with quick-scanning XAS.
	LiMn _{0.5} Fe _{0.5} PO ₄ @ in situ coin cell	Pleuksachat et al. ¹³⁵ evaluated the redox reactions of Fe ^{2+/3+} and Mn ^{2+/3+} concurrently, demonstrating the nonequilibrium phase transition behavior.
Soft XAS	Si @ SiN _x window-based liquid cell	Schellenberger et al. ¹³⁶ studied the formation process and constituents of the SEI layer on the Si anodes.
XPS	Li metal @ specially designed gas dosing cell	Ettxebarria et al. ¹³⁷ investigated the interaction between Li metal and CO ₂ gas using in situ ambient pressure XPS to detect the formation of oxalate formation during the reaction.
	Li _{1.5} Al _{0.5} Ge _{1.5} (PO ₄) ₃ @ photoelectron-transparent solid electrolyte	Inozemtseva et al. ¹³⁸ examined the electrochemical reduction of a Li _{1.5} Al _{0.5} Ge _{1.5} (PO ₄) ₃ glass-ceramic electrolyte with a two-layer graphene electrode, which is transparent for photoelectrons.
	Li-CO ₂ @ Ionic liquid-based air cells	Wang et al. ¹³⁹ developed in situ ambient pressure XPS to investigate the chemical reactions of Li-CO ₂ batteries in an ionic liquid electrolyte.
	Li/liquid electrolyte/graphite	Li et al. ¹⁴⁰ elucidated the chemical compositions of intercalants on the graphite surface.
XCT	Li/Li ₁₀ SnP ₂ S ₁₂ @ customized solid-state cell	Lewis et al. ¹⁴¹ studied the chemo-mechanical changes in Li/solid-state electrolyte interfaces during battery operation.
	Li/β-Li ₃ PS ₄ /Li cell @ Torlon casing	Seitzman et al. ¹⁴² observed how Li behaves in response to device operations, such as stack pressure, particle sizes, and temperature.
	LiCoO ₂ /graphite @ pouch cell	Du et al. ¹⁴³ investigated the gas evolution and structural changes in a pouch full-cell during cycling.
	CuS, β-Li ₃ PS ₄ , carbon black @ custom solid-state cell	Zhang et al. ¹⁴⁴ visualized the prevented crack formation of solid-state batteries by increasing the stack pressure.
TXM	LiNi _{0.8} Co _{0.1} Mn _{0.1} O ₂ @ quartz capillary tubes	Alvarado et al. ¹⁴⁵ explored the thermally induced reactions taking place within the delithiated LiNi _{0.8} Co _{0.1} Mn _{0.1} O ₂ particles.
	LiNi _{0.6} Co _{0.2} Mn _{0.2} O ₂	Lou et al. ¹⁰⁸ investigated the relationship between the local interface conditions and the initial electrochemical characteristics in solid-state batteries.
	LiCoO ₂ /Fe ₂ (MoO ₄) ₃ @ thin-film all-solid-state battery	Ishiguro et al. ¹⁴⁶ examined the chemical state changes within the cathode and anode layers before and after cycling.
STXM	LiFePO ₄ @ microfluidic cell	Koo et al. ¹⁴⁷ analyzed both the surface and internal Li compositions within submicron particles of LFP oriented along the [100] direction.

Abbreviations: SAXS, small-angle X-ray scattering; STXM, scanning transmission X-ray microscopy; TXM, transmission X-ray microscopy; XAS, X-ray absorption spectroscopy; XCT, X-ray computed tomography; XPS, X-ray photoelectron spectroscopy; XRD, X-ray diffraction.

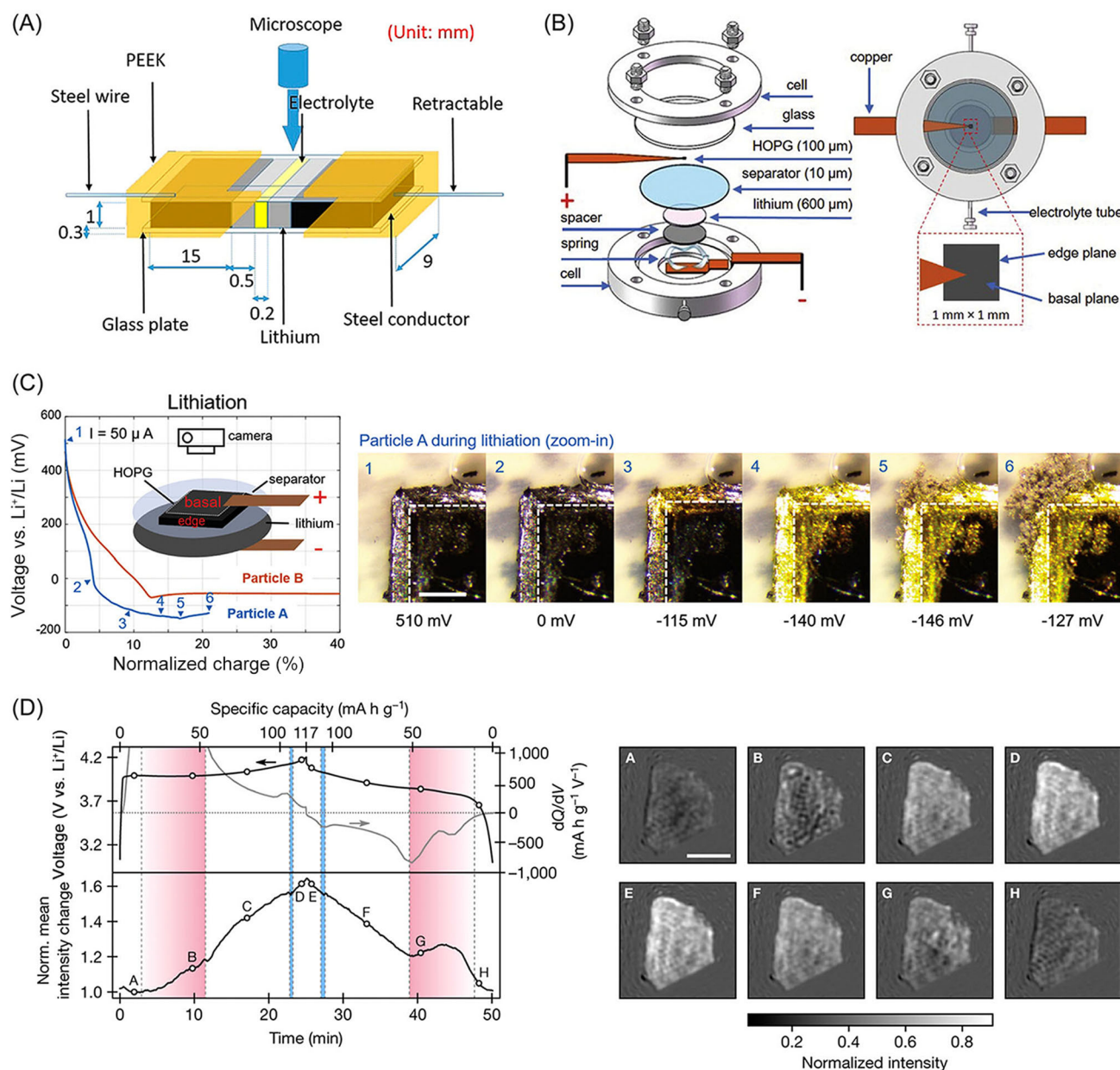


FIGURE 10 (A) Schematic representing the electrochemical cell for in situ OM to examine lithium dendrites. Reproduced with permission: Copyright 2019, Wiley.¹⁵⁵ (B) Schematic illustration of the electrochemical cell incorporating micro-sized graphite. (C) Plot displaying the voltage profile of a highly oriented pyrolytic graphite cell, accompanied by optical microscopic images capturing the lithiation process into a graphite particle. Reproduced with permission: Copyright 2021, Elsevier.¹⁵⁹ (D) Galvanostatic and differential capacity plots, along with iSCAT images of the active particle, at specific time points marked as open circles. Reproduced with permission: Copyright 2021, Springer Nature.¹⁶⁰ OM, optical microscopy.

interferometric scattering microscopy (iSCAT) to investigate the cycling behavior of individual particles of Li_xCoO_2 as shown in Figure 10D. Using this approach, they determined the rates of Li diffusion at the single-particle level and identified different mechanisms during the charge and discharge processes.

In situ OM observes the morphological changes in battery systems with high temporal resolution. However, there are several limitations of in situ OM, including the

challenge of improving spatial resolution owing to the diffraction limit of light. This limitation can hinder the ability to visualize fine details at the nanoscale. Enhancing contrast and signal-to-noise ratio is another challenge, as these factors are influenced by the optical properties of the electrodes and electrolytes, such as reflectance, transmittance, absorption, scattering, and fluorescence. Therefore, it is essential to address these factors to obtain clear and meaningful data, potentially

requiring specialized sample preparation and data processing depending on the area of interest.

3.2 | In situ Raman spectroscopy

In 1928, C. V. Raman discovered the phenomenon of Raman scattering, which laid the groundwork for the development of Raman spectroscopy.¹⁶¹ Raman spectroscopy is widely used to study the structural and chemical properties of materials based on molecular vibrations. This technique relies on the interaction between a laser beam and the material being investigated. The laser beam has a wavelength that can range from UV to near-IR, depending on the desired resolution and penetration depth. When the laser beam interacts with the material, most of the photons are elastically scattered, retaining their original energy and wavelength. However, a small fraction of photons undergoes inelastic scattering, resulting in photons with different energy. This phenomenon, known as Raman scattering, provides information about vibrational frequencies, which can be utilized to identify bond characteristics.^{162,163}

Through in situ Raman spectroscopy, researchers can investigate reaction kinetics,¹⁶⁴ changes in chemical composition,^{165,166} phase transitions,¹⁶⁷ interfacial phenomena,¹⁶⁸ and stress-strain change.^{164–166,168–172} This technique requires a specialized cell setup to enable the interaction between the laser beam and the electrode materials, as depicted in Figure 11A,B. An extensively used setup is the coin cell configuration, which comprises a transparent window positioned on one side of the cell, opposite to a metal disk on the other side.^{173–175} Alternatively, specially designed prefabricated cells or pouch cells with a window can also be used for Raman spectroscopy investigations.^{167,170,176}

Venkateswara et al.¹⁷⁷ investigated the electrochemical behavior and structural stability of $\text{Li}_{1.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}\text{O}_2$ using in situ Raman spectroscopy as shown in Figure 11C. Their findings revealed the extraction of Li and oxide ions from the lattice during the charging process, followed by cation rearrangement. During discharging, Li ions were inserted back into the structure without significant changes. In a study by Ghanty et al.,¹⁶⁷ pouch cells were used for in situ Raman spectroscopy to investigate the (de)lithiation processes of Ni-rich layered cathode material. They observed that the peak wavenumbers and widths showed reversible behavior during discharge, with slight deviations from the charging process. These findings indicated that the cathode material underwent reversible structural changes in the initial charge/discharge cycles. Moreover, Li et al.¹⁷⁸ explored the thermal decomposition process of

the near-surface structure of fully charged high-Ni-layered cathode materials using in situ heating Raman spectroscopy as shown in Figure 11D. They discovered that the intensities of A_{1g} (Ni, Co) and E_g (Ni, Co) peaks began to decrease at 90°C and the reduced peak intensities were closely related to the local metal–O coordination structural changes by the reduction of Ni ions and the release of O species. This implied that there was a structural change from an ordered layered structure to a disordered spinel structure at that temperature.

In situ Raman spectroscopy probes the chemical bonding and molecular changes within battery materials, providing information about structural changes in accordance with the interpretation. This technique, however, requires the material to have polarizability. Polarizability refers to the ability of molecules or materials to form temporary dipoles. This restricts the use of Raman when analyzing materials with low polarizability. Additionally, interference or background noise from electrolytes, binders, current collectors, or other battery components may also affect the Raman spectra. These unwanted signals can complicate data interpretation and require careful processing to extract meaningful information. Lastly, laser irradiation during Raman spectroscopy may induce undesirable effects on the sample, such as heating, degradation, or polarization. Therefore, in situ Raman spectroscopy must be applied judiciously, assessing whether the sample is suitable for Raman measurement and considering cell design for the specific area of interest.

3.3 | In situ IR spectroscopy

FT-IR spectroscopy is utilized in the analysis of molecular interaction with IR radiation, which has wavelengths longer than visible light but shorter than microwaves. IR radiation corresponds to the energy required to excite molecular vibrations, including stretching, bending, and twisting of chemical bonds. By examining the vibrational modes of functional groups like carbonates and sulfates, FT-IR facilitates the investigation of SEI layer formation and the oxidation states, coordination environments, and molecular interactions involved in electrochemical reactions.^{179–182} Fundamentally, FT-IR involves exposing the sample to an IR beam and measuring the transmitted or reflected light. The interaction of IR light with the molecular bonds induces characteristic vibrations, and these vibrations are like unique fingerprints, revealing details about the type, number, arrangement, and bonding characteristics of atoms within the molecules or materials.

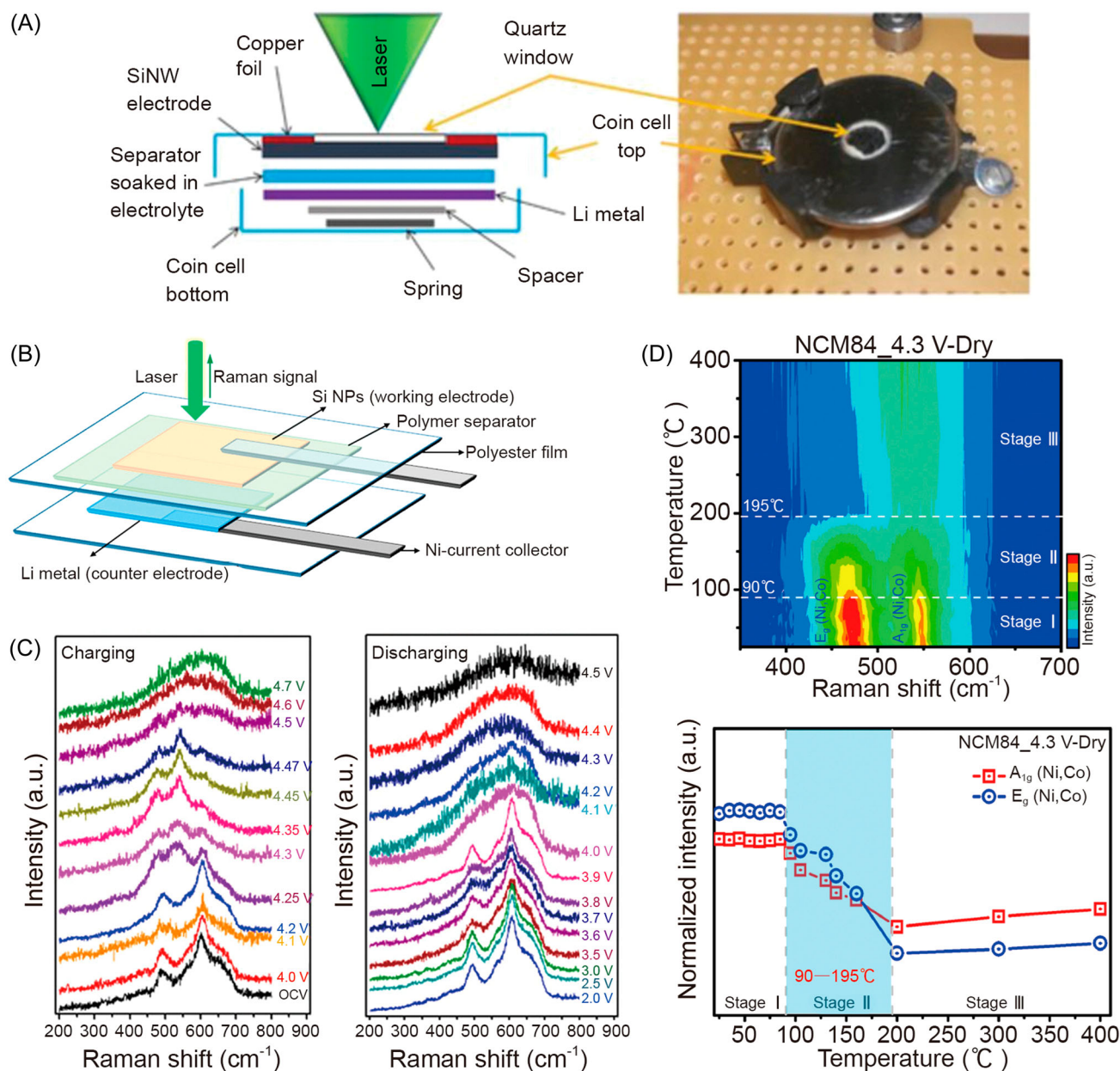


FIGURE 11 (A) Configuration depicting the silicon nanowire electrode setup for Raman measurements, including the presence of a quartz window within the coin cell. Reproduced under terms of the CC-BY license (<http://creativecommons.org/licenses/by/4.0/>). Copyright 2019, The Authors, published by IOP Publishing.¹⁷⁵ (B) Schematic diagram of pouch cell used for in situ Raman spectroscopy studies. Reproduced with permission: Copyright 2016, Elsevier.¹⁷⁰ (C) In situ Raman spectra measured during the first cycle of the $\text{LiNi}_{0.84}\text{Co}_{0.10}\text{Mn}_{0.06}\text{O}_2$ cathode coated with AlPO_4 . Reproduced with permission: Copyright 2014, American Chemical Society.¹⁷⁷ (D) Contour map of in situ Raman spectroscopy for charged $\text{LiNi}_{0.84}\text{Co}_{0.10}\text{Mn}_{0.06}\text{O}_2$ at 4.3 V (top) and intensity changes of A_{1g} (Ni, Co) and E_g (Ni, Co) peaks (bottom) during heating. Reproduced with permission: Copyright 2022, Elsevier.¹⁷⁸

In situ FT-IR is used to explore real-time interfacial interactions, including electrolyte adsorption/desorption, and interphase formation.^{183–186} This analysis can also investigate the redox reactions of electrolytes¹⁸⁷ and electrode materials.¹⁸⁸ Depending on the configuration of the IR beam and detector, different types of techniques and setups are available. For instance, attenuated total reflection FT-IR (ATR

FT-IR), shown in Figure 12A, uses a prism or waveguide to reflect the IR beam at the interface between the electrode and the electrolyte, enhancing sensitivity for detecting surface species and interfacial reactions.^{179,189} Transmission FT-IR, on the other hand, utilizes a transparent window or a thin-film electrode to allow the IR beam to pass through the entire battery cell, as shown in Figure 12B.¹⁹⁰ This

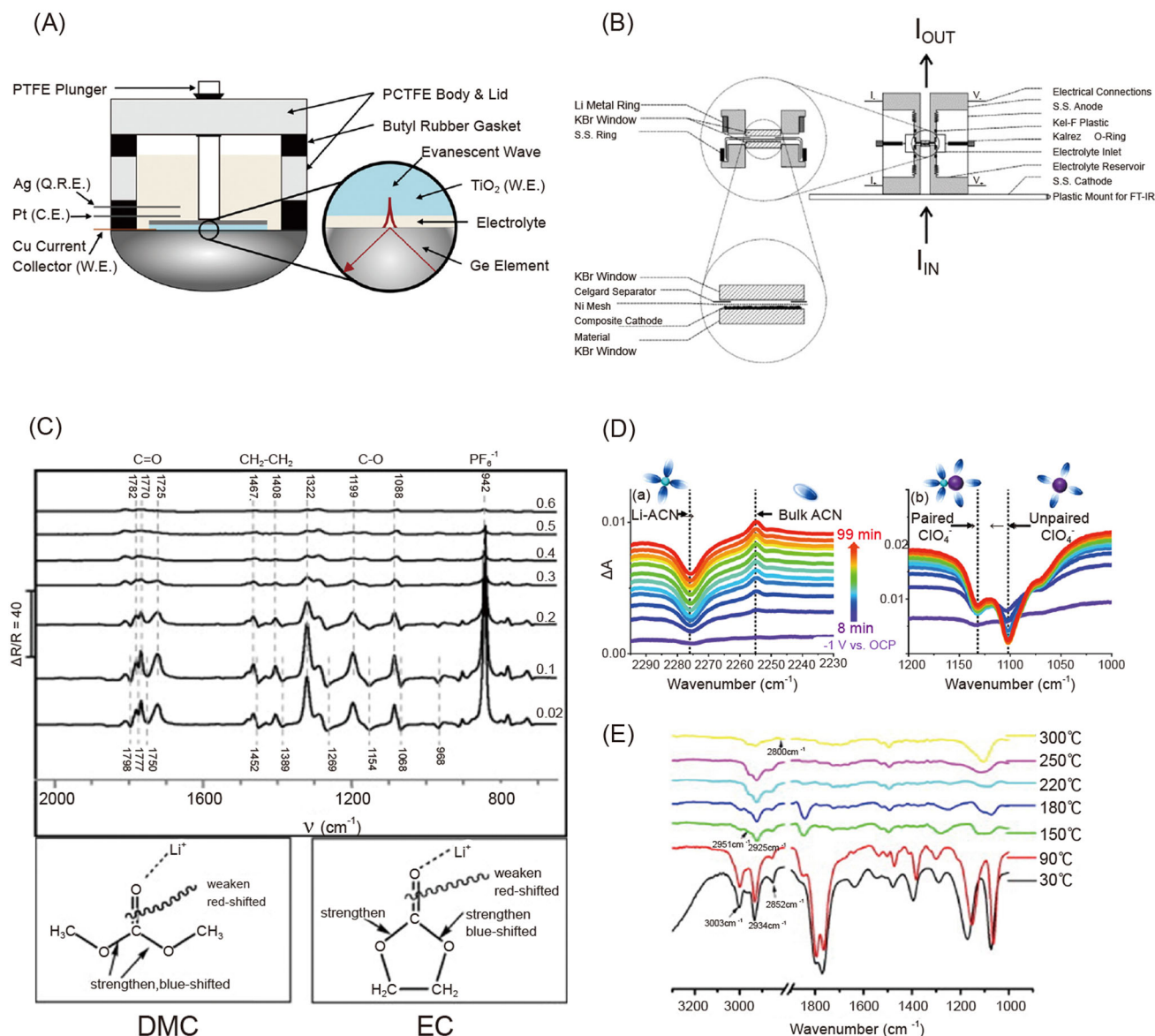


FIGURE 12 (A) Cross-sectional diagram outlining the air-free in situ vibrational spectroelectrochemical cell specifically constructed for potential-dependent ATR-FT-IR measurements. Reproduced with permission: Copyright 2021, American Chemical Society.¹⁷⁹ (B) Diagram representing the design of an in situ transmission FT-IR spectroelectrochemical cell. Reproduced with permission: Copyright 2006, Elsevier.¹⁹⁰ (C) In situ MFTIR spectra of a Sn-Co alloy thin-film anode immersed in a 1 M LiPF₆/EC + DMC (1:1, vol%) electrolyte during the lithiation process, and a schematic illustration showcasing the molecular structures of solvated DMC and EC with lithium ions. Reproduced with permission: Copyright 2010, Elsevier.¹⁹¹ (D) In situ ATR-FT-IR spectroelectrochemistry data demonstrating the signatures of solvation of lithium-coordinated acetonitrile (Li-ACN) and ion-pairing (paired ClO₄⁻). Reproduced with permission: Copyright 2021, American Chemical Society.¹⁷⁹ (E) In situ FT-IR spectra of the LiPF₆ electrolyte in a 1 M LiPF₆/EC + DMC (1:1, wt%) composition during thermal heating up to 300°C. Reproduced with permission: Copyright 2009, Elsevier.¹⁹²

technique provides information on both bulk and surface phenomena and enables spatially resolved measurements.

For example, Istone et al.¹⁹³, who applied in situ FT-IR for the first time in battery research, tracked the reaction intermediate in the electrochemical process of the Li-SOCl₂ battery system. More recently, Li et al.¹⁹¹ used in situ microscope FT-IR (MFTIR) spectroscopy to study

the interfacial properties of Sn-Co alloy thin-film anodes in a 1 M LiPF₆/ethylene carbonate (EC) + dimethyl carbonate (DMC) (1:1, vol%) electrolyte during discharge/charge processes. The MFTIR spectra in the lithiation process and a schematic illustration for the molecular structure of solvated DMC and EC are shown in Figure 12C. They found shifts in the IR bands associated with C=O, C-O, and C-H vibrations, which are attributed to the change in

the degree of solvation of Li ions during the (de)lithiation processes. Also, Cave et al.¹⁷⁹ developed an air-free in situ vibrational cell with the ATR FT-IR setup. Through the changes in the vibrational bands shown in the IR spectra in Figure 12D, they investigated the dissociation of contact ion pairs, desolvation of Li, and electron accumulation in a TiO₂/acetonitrile model system. Furthermore, Xiang et al.¹⁹² conducted in situ FT-IR experiments to investigate the thermal stability of commercial LiPF₆-based electrolytes. Figure 12E shows the in situ FT-IR spectra of an electrolyte containing 1 M LiPF₆ in EC/DMC (1:1, wt%) under thermal heating up to 300°C. They observed strong exothermic reactions occurring below 225°C, primarily attributed to the ring-opening reactions of EC and the decomposition of polymeric products.

In situ FT-IR provides insights into chemical processes within battery systems, including electrode–electrolyte interface, electrolyte, and even active materials in electrodes. Nevertheless, it is crucial to note that FT-IR signals appear when there is a change in the dipole moment during vibration. Consequently, molecules or materials with bonds that do not show changes in the dipole moment while vibrating are limited in their analysis using

FT-IR. Additionally, in situ FT-IR may face challenges in capturing all the details of SEI layer formation and evolution, particularly at extremely small scales. The resolution of nanoscale heterogeneities and dynamics through in situ FT-IR analysis may pose difficulties, which could potentially impact the accuracy and comprehensiveness of the findings. Therefore, researchers should be aware of the potential challenges in the restricted range of materials and capturing nanoscale dynamics, to properly apply the technique in battery research.

3.4 | Recent studies

In this section, we have introduced three techniques that use UV, visible light, and IR as sources: OM, Raman spectroscopy, and FT-IR. Recent studies for each in situ technique are summarized in Table 2. These techniques have all been evolving toward increasing their spatial resolution capabilities. A prominent microscope, OM, is currently under development with super-resolution optical microscopy techniques to overcome its lower spatial resolution compared to electron

TABLE 2 Recent studies of ultraviolet and visible light and infrared-based in situ techniques.

Techniques	Material @ cell	Main description/results
OM	LiCoO ₂ @ OM half-cell	Merryweather et al. ¹⁶⁰ explored the Li-diffusion kinetics within a single LiCoO ₂ particle.
	LiFePO ₄ @ ECC-Opto-Std cell	Rittwegert et al. ²⁰⁶ investigated the charge carrier dynamics in positive electrodes.
	Graphite @ specially designed OM cell	Gao et al. ¹⁵⁹ observed the lithiation and Li-deposition process at the graphite particle surface.
	Graphite @ cross-sectional setup	Hogrefe et al. ²⁰⁷ estimated the lithiation fronts for LiC ₁₂ and LiC ₆ using their distinct colors.
Raman	Au	Gogoi et al. ¹⁹⁸ studied the SEI formation mechanisms on a Au electrode utilizing in situ surface-enhanced Raman spectroscopy.
	Natural graphite	Zhou et al. ²⁰⁸ demonstrated exfoliation and continuous SEI formation on graphite and verified the interfacial stabilization by introducing an artificial SEI layer.
	β-Li ₃ PS ₄ , Li ₁₀ GeP ₂ S ₁₂ /Au	Sang et al. ²⁰⁹ investigated the reactions in which the ionic framework in the solid electrolyte requires breaking and rebuilding during Li deposition and stripping.
FT-IR	Graphene/Li/solid polymer electrolyte	He et al. ²¹⁰ found that nanoscale structural heterogeneities of the solid polymer electrolyte induce additional heterogeneous interfacial reactions.
	Covalent organic framework (COF)/carbon nanotube (CNT)	Yang et al. ²¹¹ reported COF@CNT composite electrode as a high-capacity anode, which shows enhanced capacity with a higher amount of CNT.
	Air electrode @ 3D printed Li–O ₂ cell	Nepel et al. ²¹² conducted in situ synchrotron IR nanospectroscopy and demonstrated the dependence of the atmosphere on Li product formation.

Abbreviations: FT-IR, Fourier-transform infrared; OM, optical microscopy; SEI, solid electrolyte interface.

microscopy.^{194–197} In the case of Raman spectroscopy, it has advanced beyond conventional Raman microscopy to include super-resolution Raman techniques such as surface-enhanced Raman spectroscopy,¹⁹⁸ tip-enhanced Raman spectroscopy,^{199,200} and stimulated Raman scattering microscopy.^{201,202} Attempts have been made to achieve improvements in FT-IR as well, to improve spatial resolution, using FT-IR imaging techniques to obtain chemical images.^{203–205} While certain advanced techniques within these categories are yet to be fully tailored for battery research or in situ applications, we anticipate that the technical constraints of these methods will be successfully overcome in the future. Consequently, they are poised to significantly contribute to the analysis of Li-rechargeable batteries, broadening the analytical horizons within this field.

4 | MICROWAVE AND RADIOWAVE BASED

Microwaves and radiowaves are used for studying the magnetic properties of atoms and ions. Using these waves in magnetic fields, the magnetic properties of chemical species with unpaired electrons can be probed. All battery systems discussed in this literature involve Li (⁷Li) that has paramagnetic and magnetic field-responsive characteristics. Consequently, the transport or deposition process of Li during the operation of batteries could be probed by analyzing the changes in magnetic responses. In this section, we will introduce NMR and EPR, which are two prominent in situ analysis tools that utilize microwave and radiowave radiation sources.

4.1 | In situ NMR

The first experiment of NMR is known from the research carried out in 1950 by Hahn,²¹³ who described the principles of spin echoes with ¹H and ¹⁹F signals in liquids. NMR utilizes radiowaves to investigate the interaction between atomic nuclei and a magnetic field. In 1994, Tanaka et al.²¹⁴ conducted a pioneering study of ⁷Li solid-state NMR, marking the initial utilization of NMR in Li-rechargeable battery research. Due to continuous advancements, NMR study now enables the estimation of structural evolution of electrode material²¹⁵ and the growth patterns of Li dendrites.^{216,217} Moreover, NMR-based techniques enable the quantification of mass transport properties and ion-specific behavior in batteries.²¹⁸ They can be used to determine diffusion coefficients and concentration changes of NMR-active

species, thus allowing investigations of ion transport. In particular, NMR imaging, commonly referred to as magnetic resonance imaging (MRI), offers spatial information about concentration gradients, diffusion properties, and electrolyte dynamics.^{219,220}

NMR analysis is based on the magnetic response of nuclei possessing spins like ¹H, ⁷Li, ¹³C, ¹⁹F, ²⁹Si, and ³¹P, which makes them behave like tiny magnets. When these nuclei are placed in a strong external magnetic field, the nuclear spins align parallel or antiparallel to the field direction, resulting in a slight energy difference between the spin states. During the NMR experiment, a radio-frequency (RF) pulse at a specific frequency matching the energy difference of the spin states excites the nuclei, causing a resonance transition. The frequency of the signal depends on the strength of the magnetic field and the type of nucleus, which allows us to identify different elements and isotopes. Upon relaxation, the excited nuclei revert to their original spin state, leading to the emission of an electromagnetic signal known as free induction decay (FID). FID carries information about the local chemical environment and molecular interactions. Liquid-state NMR techniques are limited because battery materials are mostly composed of solids. Therefore, solid-state NMR techniques like magic angle spinning NMR are often used for battery electrode analysis.²²¹

During charge–discharge cycling, in situ NMR provides real-time information about structural changes,^{222–225} variations in Li concentration,^{226,227} and the Li-deposition process.^{228–230} Integration of in situ MRI with NMR offers diagnostic information about battery chemistry,^{231–235} including chemical reactions at electrode–electrolyte interfaces.²³⁶ As shown in Figure 13A, specially designed electrochemical cells are used for in situ NMR experiments.^{224,225,237} The setup uses a plastic bag battery to minimize magnetic field interference and ensure efficient RF excitation and signal detection. Then, the battery is positioned in the coil of an NMR magnet to measure in situ data in the magnetic field.

For the first time in the battery system, Tsubouchi et al.²³⁸ utilized in situ ⁷Li NMR spectroscopy to investigate the reversible electrochemical reaction of Li ions in graphitic carbon electrodes. In addition to anode materials, Shimoda et al.²²³ conducted in situ ⁷Li NMR measurements on a LiCoO₂ cathode in a plastic cell battery. The NMR signal of Li_xCoO₂ (0 ≤ x ≤ 1) was analyzed to investigate structural changes corresponding to different Li extraction/insertion stages as depicted in Figure 13B. A narrow solid solution region of Li_xCoO₂ (~0.99 ≤ x < 1) was observed, associated with the localized electronic spin of paramagnetic Co⁴⁺ ions during early delithiation. Also, Zhou et al.²²² investigated the first cycle of Li_{1.08}Mn_{1.92}O₄ using in

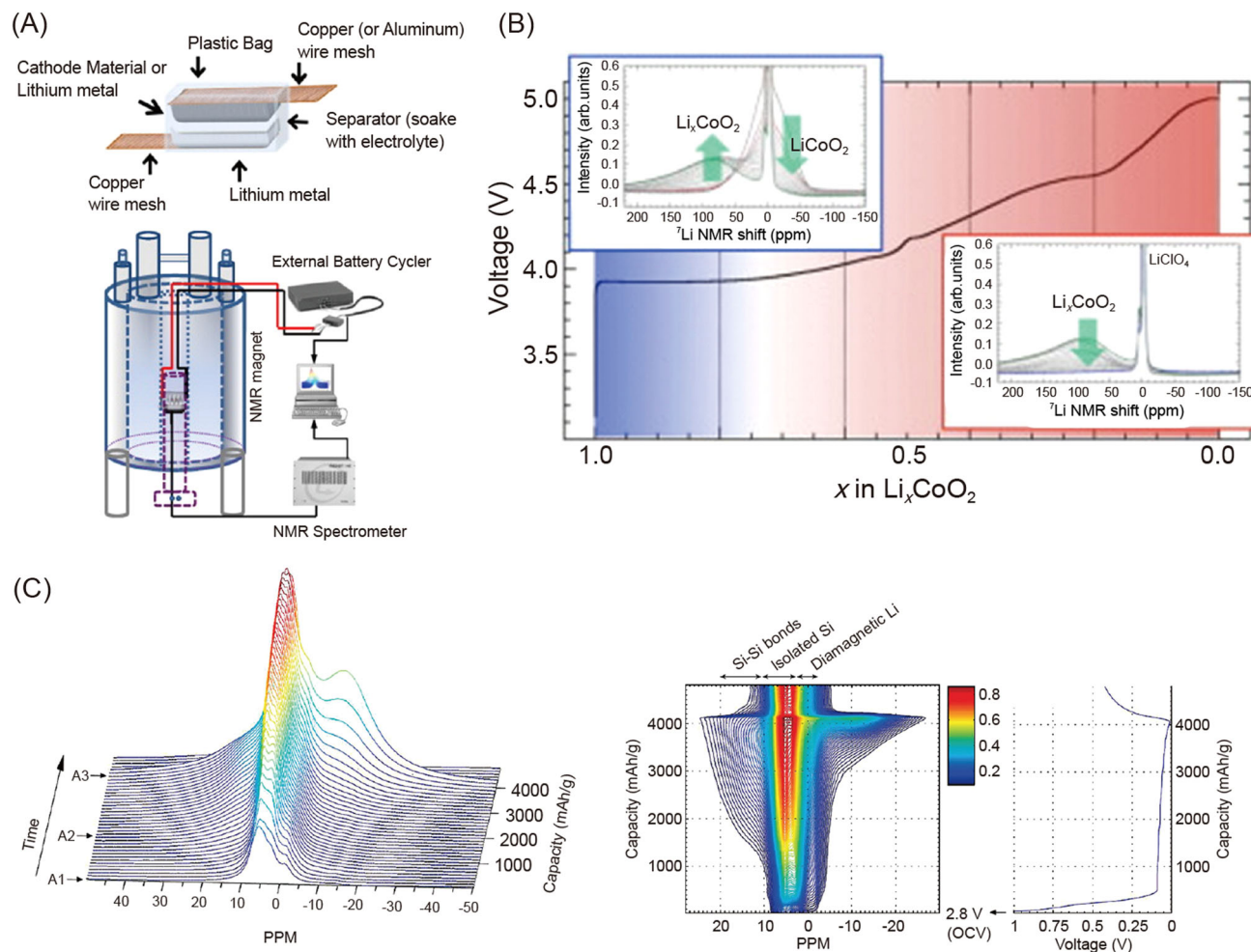


FIGURE 13 (A) Diagram illustrating the setup of an in situ NMR configuration within a bag cell battery. The bag cell is positioned inside the coil of an NMR probe and undergoes cycling within the magnetic field. Reproduced with permission: Copyright 2012, Elsevier.²³⁷ (B) Electrochemical profile of the plastic cell battery during the charging process, reaching up to 5.0 V, and in situ ^7Li NMR spectra acquired with a horizontal orientation for the Li_xCoO_2 cathode material ($0 \leq x \leq 1$) throughout the charging process. Reproduced with permission: Copyright 2013, Elsevier.²²³ (C) Stacked and contour plots displaying the in situ ^7Li static NMR spectra alongside the electrochemical profile for the first discharge of an actual crystalline Si versus Li^+/Li battery. Reproduced with permission: Copyright 2009, American Chemical Society.²²⁴

situ NMR spectroscopy to study the Li-ion mobility during cycling. They found that the NMR signal intensity did not show a continued decrease or increase during charging or discharging. They ascribed such deviation to Li motion changes during the refocusing times used in NMR data acquisition. Key et al.²²⁴ revealed the formation of isolated Si and Si-Si clusters in the amorphous phase during the discharge of a Si-Li battery. From in situ NMR spectra, which are shown in Figure 13C, they found that the excess Li phase formed at the end of the first discharge of crystalline silicon was very reactive in the electrolyte, causing a self-discharge process with capacity loss.

In situ NMR offers a perspective into the chemical and structural changes by the magnetic response of nuclei, allowing for direct monitoring of Li species in Li-rechargeable batteries. However, this analysis has several

limitations. Only nuclei with spins can be analyzed and high concentrations of nuclei are required for detection due to limited sensitivity. This limitation affects the study of low-concentration species and minority components. Additionally, the strong magnetic field of NMR instruments may interfere with battery operation, altering diffusion behavior and inducing unwanted side reactions. For this reason, researchers must consider the potential impact of the magnetic field when designing in situ NMR experiments.

4.2 | In situ EPR

After E. K. Zavoisky first detected the EPR peak of a cupric chloride crystal, EPR was developed to utilize a

magnetic field and microwave radiation to focus on the spin properties of electrons rather than nuclei.²³⁹ Since Euler²⁴⁰ first proposed that EPR would be a useful tool to investigate the electrochemical electrode in the battery field, it has been applied for the study of battery systems, particularly in the breakdown mechanisms of electrolyte salts and solvent,^{241,242} as well as the redox mechanism of electrode materials.²⁴³ One noteworthy application is the identification of Li deposition, as EPR can discern metallic Li from Li ions with high specificity.^{244–247} On comparing EPR to NMR, it is found that EPR offers enhanced sensitivity and accuracy in the detection of materials with unpaired electron spins. However, the detection depth of EPR for Li is limited compared to the depth achieved with radio frequencies in NMR. As a result, EPR analysis primarily provides surface area estimation of structural growth rather than bulk

estimation, as observed in NMR studies.²⁴⁷ The principle of EPR analysis revolves around the behavior of unpaired electrons in paramagnetic species, which gives rise to a net magnetic moment. When subjected to an external magnetic field, the unpaired electrons align either parallel or antiparallel to the field. By applying microwave radiation, the energy difference between the spin states of the unpaired electrons can be probed, corresponding to a specific frequency. Resonance occurs when the applied microwave frequency matches this energy difference, leading to a change in the energy state of the paramagnetic species, with the excess energy emitted as an electromagnetic signal.²⁴⁴

Continuous monitoring of in situ EPR facilitates the analysis of the kinetics of electrolyte decomposition and Li-deposition behavior,^{245–248} as well as the anionic redox mechanism of electrode materials.^{249–251} Figure 14A

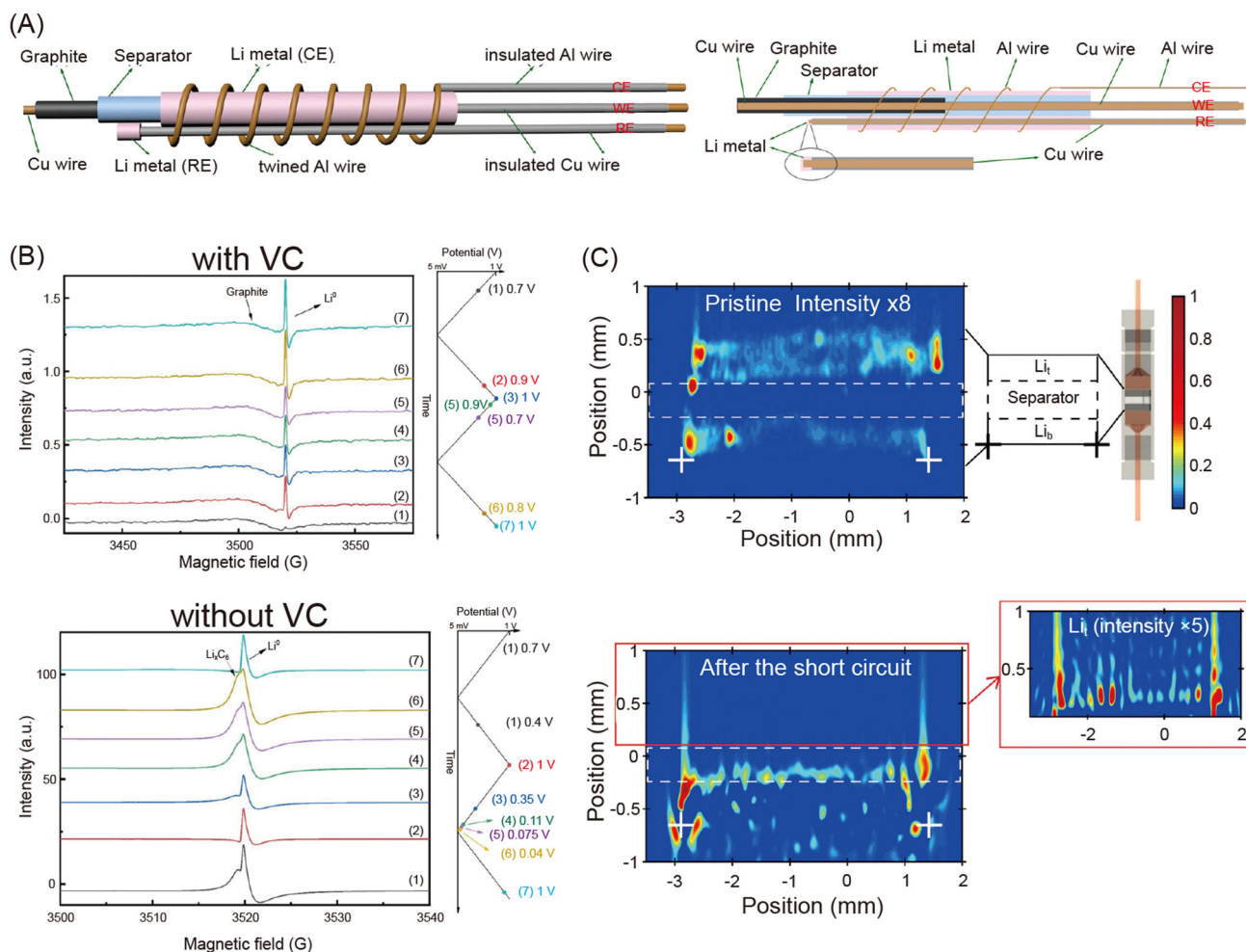


FIGURE 14 (A) Diagram depicting the three-electrode in situ EPR cell, including a cross-section showcasing the graphite anode and Li metal as the counter and reference electrodes. (B) EPR spectra of metallic lithium observed at graphite surfaces with (top) and without (bottom) VC during the charging process at a scan rate of 0.1 mV s^{-1} . Reproduced under terms of the CC-BY license (<http://creativecommons.org/licenses/by/4.0/>). Copyright 2021, The Authors, published by Wiley.²⁴⁵ (C) In situ spatial-spatial EPR images obtained in the pristine state, with a peak-to-peak amplitude multiplied by eight and after a short circuit, respectively. Reproduced under terms of the CC-BY license (<http://creativecommons.org/licenses/by/4.0/>). Copyright 2021, The Authors, published by Springer Nature.²⁴⁸

shows an exemplary electrochemical cell configuration of in situ EPR with a three-electrode system in a quartz tube. The battery cell is designed to accommodate the EPR setup while minimizing magnetic field interference and is positioned carefully within the resonator to ensure that the species of interest fall within the region of the resonant magnetic field.^{245,248,251}

Sathiya et al.²⁵¹ conducted EPR in situ during cycling to explore the anionic redox mechanism of $\text{Li}_2\text{Ru}_{0.75}\text{Sn}_{0.25}\text{O}_3$, a Li-rich layered oxide. By tracking the signals of paramagnetic O species and Ru^{5+} , they revealed that the redox reaction of O anions contributes to its high capacity. Wang et al.²⁴⁵ utilized in situ EPR to explore the lithiation/deposition behavior on the graphite anode with a three-electrode system. They highlighted that Li deposition on the graphite electrode can occur at potentials as low as 0.04 V (vs. Li^+/Li) when the scan rate is reduced to 0.04 mV s^{-1} . They also investigated the inhibitory effect of a vinylene carbonate (VC) additive on Li deposition, as observed from the EPR spectra in Figure 14B. As another example, Dutoit et al.²⁴⁸ combined in situ EPR spectroscopy and imaging to investigate the morphology and location of metallic Li in a symmetric Li/ LiPF_6/Li electrochemical cell during polarization. Through variations in the shape, resonance field, and amplitude of the EPR spectra, they monitored the nucleation of submicrometric Li particles and the fragmentation of bulk Li. Additionally, they demonstrated the visualization of dendrites formed through the separator in the entire cell as shown in Figure 14C.

In situ EPR is useful to investigate the electronic properties of paramagnetic species such as radicals. Conversely, it is important to note that EPR analysis is

only applicable when paramagnetic species are present in the electrode materials since EPR signals arise solely from unpaired electrons. This limitation restricts the use of in situ EPR to systems containing paramagnetic species and may not be suitable for certain battery materials. Similar to NMR, the strong magnetic field used in EPR can influence diffusion behavior and trigger side reactions, potentially altering the dynamics of the battery material during analysis. Therefore, when conducting in situ EPR experiments, it is essential to take these considerations into account when designing the experiments.

4.3 | Recent studies

Microwave- and radiowave-based techniques have been proven to be useful for understanding Li deposition and redox mechanisms. In the 2020s, in situ NMR and in situ EPR techniques have been introduced within the domain of Li-rechargeable battery research, as documented in Table 3. Recent applications of NMR and EPR have extended to encompass a diverse array of in situ cells and materials. Moreover, research into time-resolved techniques is underway to observe the rapid lithium diffusion and redox mechanisms. In the case of NMR, time-resolved NMR methodologies have been used to monitor phenomena such as self-discharge and electrolyte decomposition within redox flow batteries.²⁵² Similarly, time-resolved EPR techniques have been applied to scrutinize the intricate process of lithium deposition.²⁵³ These time-resolved techniques are crucial for the real-time monitoring of dynamic processes, notably in fast-charging technologies.

TABLE 3 Recent studies of microwave- and radiowave-based in situ techniques.

Techniques	Material @ cell	Main description/results
NMR	$\text{LiFePO}_4/\text{Cu}$	Gunnarsdóttir et al. ²⁵⁴ demonstrated the high-rate corrosion caused by SEI formations on Li and Cu and suggested polymer coatings and Cu surface modification as solutions.
	Graphite @ parallel-plate NMR cell	Krachkovskiy et al. ²⁵⁵ suggested a parallel-plate resonator concept and classified two Li-deposition processes.
	Li-S @ homemade in situ MRI cell	Dorai et al. ²⁵⁶ suggested ^1H MRI as an effective tool to study the intermediate species during Li-S battery operation.
EPR	Graphite	Wang et al. ²⁴⁵ highlighted the suppression of Li deposition by using a VC additive.
	$\text{LiCoO}_2/\text{graphite}$ @ pouch cell	Geng et al. ²⁴⁶ used the EPR imaging technique to trace the various Li deposits during battery operation.
	LiCoO_2 @ 2032-type coin cells	Hu et al. ²⁵⁷ utilized dual-mode EPR, which is valuable for measuring various electron spin states to study redox mechanisms in LiCoO_2 .

Abbreviations: EPR, electron paramagnetic resonance; MRI, magnetic resonance imaging; NMR, nuclear magnetic resonance; SEI, solid electrolyte interface; VC, vinylene carbonate.

5 | ELECTRON-BASED

Electron-based microscopy techniques are universally used to study multiscale features ranging from atomic to particle scale. The electron-atom interaction arises in various ways, including elastic-inelastic scattering, transmission, and X-ray emission. Using different imaging modes, a wide range of information can be obtained, such as crystalline structures, particle morphology, and chemical composition. Taking advantage of the abundant information, in situ analyses with electron microscopy techniques allow for the investigation of structural and compositional changes at the nanometer to micrometer scale. In this section, two key electron-based techniques are addressed: SEM and TEM.

5.1 | In situ SEM

SEM utilizes a focused electron beam to achieve high-resolution images by scanning across a sample surface. In the context of the battery system, Holleck et al.²⁵⁸ first observed the SEM images of titanium disulfide ($\text{Ti}_{1+x}\text{S}_2$) and trisulfide (TiS_3) cathodes. By obtaining surface images, SEM investigates various features, such as crack formation,²⁵⁹ electrode-electrolyte interfaces,²⁶⁰ and morphology or distribution of electrode materials.^{261,262} Secondary electrons (SE) and backscattered electrons (BSE) are typically detected and used to construct an image of the electrode surface. Scanning the electron beam across the sample allows for the generation of detailed images, and their brightness depends on sample topography, atomic composition, and conductivity.²⁶³

In situ SEM offers the real-time observation of electrode-electrolyte interface degradation,²⁶⁴ Li dendrite,²⁶⁵⁻²⁶⁷ and electrode volume change.^{268,269} To conduct in situ SEM experiments, an electrochemical cell should be placed into the SEM chamber. The cell design must be compatible with high vacuum conditions and ensure proper electrical contact between the electrodes and the external equipment. Figure 15A shows an example of an in situ SEM cell in which a monolayer atomic sheet of MoS_2 on a carbon film is supported by a Cu grid to interact with the electrolyte in the coin cell.²⁷⁰ Figure 15B depicts another form of electrochemical cell where electrode material and a solid electrolyte are coated on a current collector mesh to operate inside the SEM chamber itself. Generally, in situ SEM cell design should include a thin and transparent window that allows electron beam transmission for observation of the electrode surface.²⁶⁸

The first in situ SEM for battery research was identified by Baudry et al.²⁷¹ They compared the morphological changes of FeS , TiS_2 , and V_6O_{13} cathode

materials in polymer lithium batteries. In another study, Orsini et al.²⁶⁵ conducted SEM to study the cross-sections of plastic cells in a quasi in situ mode. Figure 15C shows SEM images after the first and the 14th charge. After cycling, a mossy layer was found at the Li-separator interface, primarily consisting of Li deposits accumulated during the charging process. Chen et al.²⁶⁸ performed an in situ SEM study on silicon anodes, simulating the configuration of actual batteries with an ionic liquid electrolyte. As can be seen in the SE images in Figure 15D, they observed a significant correlation between the morphological changes in silicon materials and their size and shape during electrochemical cycling. Additionally, the BSE images were utilized to visualize the diffusion of atomic Li into the silicon materials. Golozar et al.²⁶⁶ used in situ SEM to investigate the cycling behavior of all-solid Li metal batteries featuring LFP cathodes. As shown in Figure 15E, needle-like Li dendrites were observed, which tended to penetrate the polymer material.

In situ SEM observes the microstructural surface morphology of battery systems during cell operation. However, it is important to recognize certain considerations associated with this technique. SEM requires high vacuum conditions, which restricts the range of samples suitable for analysis. Materials with low vapor pressure may not be compatible with in situ SEM analysis. Moreover, SEM provides surface-sensitive information, limited to the outer surface of the sample. To address these limitations, additional techniques like TEM or sample preparation methods such as cross-section polisher or focused ion beam (FIB) may be necessary. Another limitation is the potential damage caused by the electron beam. Prolonged or intense exposure to the electron beam can lead to structural modifications or even the destruction of beam-sensitive materials. Therefore, careful control of beam parameters and exposure time is crucial to prevent sample damage.

5.2 | In situ TEM

By transmitting an electron beam through a thin specimen and capturing the resulting electron interactions, TEM provides information about the morphology, crystal structure, composition, and defects on the atomic scale. In the field of battery research, TEM plays a pivotal role in investigating the electrochemical behavior of electrode materials, including changes in morphology, interphase evolution, phase transitions, and degradation processes.^{272,273} Figure 16A shows schematic diagrams of the two primary modes of TEM: image mode and diffraction mode. In the image mode, electron images

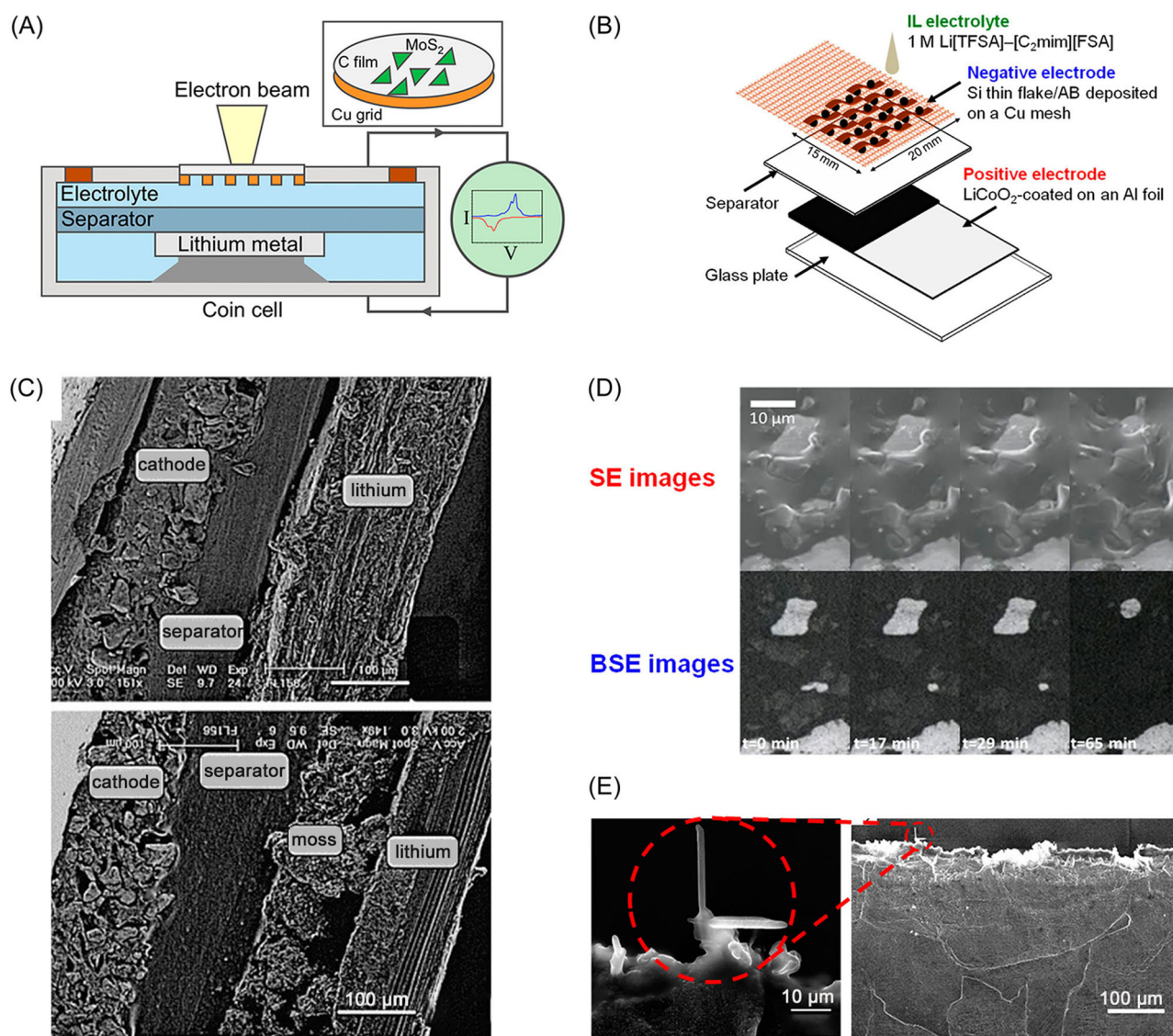


FIGURE 15 (A) Schematic illustration of the Li–MoS₂ cell, demonstrating the setup for in situ SEM of the electrochemical reactions occurring on the MoS₂ anode. Reproduced with permission: Copyright 2020, American Chemical Society.²⁷⁰ (B) Depiction of a lithium-ion battery system featuring a binder-free Si thin flake anode and an ionic liquid, designed for in situ SEM analysis. Reproduced under terms of the CC-BY license (<http://creativecommons.org/licenses/by/4.0/>). Copyright 2016, The Authors, published by Springer Nature.²⁶⁸ (C) Images displaying the section of a lithium battery after the first charge (top) and after the 14th charge (bottom), where a charging rate of C/5 (0.45 mA cm^{−2}) was applied. Reproduced with permission: Copyright 1998, Elsevier.²⁶⁵ (D) In situ secondary and backscattered electron images showcasing the evolution of a Si thin flake anode during the third charge process. Reproduced under terms of the CC-BY license (<http://creativecommons.org/licenses/by/4.0/>). Copyright 2016, The Authors, published by Springer Nature.²⁶⁸ (E) Morphology of needle-like structures observed in the plane view of SEM images during cycling. Reproduced with permission: Copyright 2018, American Chemical Society.²⁶⁶

are recorded, offering information about material morphology, size, shape, and distribution. The diffraction mode involves capturing electron diffraction patterns, which unveil the crystal structure, lattice parameters, orientation, and strain of the materials. Of the two modes, in situ analyses in battery research commonly utilize the imaging mode. The imaging mode provides three different signals depending on the position of the detector: bright field (BF), annular dark field (ADF), and

high-angle annular dark field (HAADF).²⁷⁴ The BF detector captures unscattered electrons on the optic axis, providing high-resolution and high-contrast imaging for thin specimens. The ADF detector collects scattered electrons from a small inner angle ($10 < \theta < 50$ mrad), enhancing contrast for heavy elements while reducing diffraction effects. HAADF mode collects scattered electrons from a larger inner angle ($\theta > 50$ mrad), enabling high-contrast imaging based on atomic number

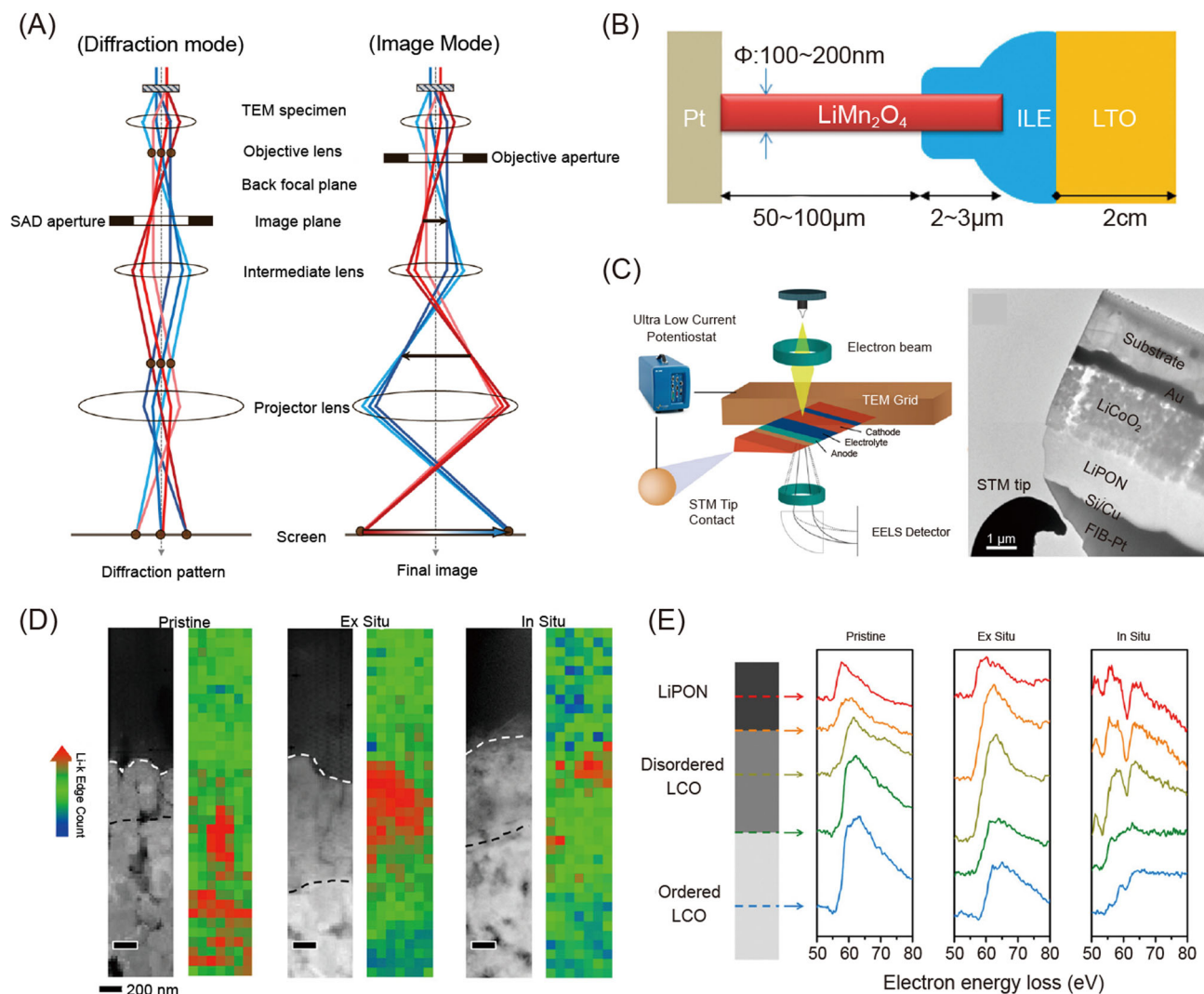


FIGURE 16 (A) Simplified schematic diagram illustrating the two fundamental operations of a TEM: diffraction mode and imaging mode. (B) Schematic illustration depicting a nanowire battery setup. A bundle of LiMn₂O₄ nanowires connects the Pt current collector to an ionic liquid electrolyte. Reproduced with permission: Copyright 2013, American Chemical Society.²⁷⁷ (C) Schematic diagram and TEM bright-field image displaying in situ TEM biasing of a nanobattery. (D) HAADF images and (E) Li K-edge EELS analyses spectra conducted on the nanobattery stack in its pristine state, ex situ, and in situ samples. The scale bar in the images corresponds to a length of 200 nm. Reproduced with permission: Copyright 2016, American Chemical Society.²⁷⁸

(Z-contrast) while minimizing multiple scattering effects. The choice of mode depends on the specific information required for the sample under investigation.²⁷⁵ Concurrently with imaging, TEM analysis uses spectroscopic tools, such as energy-dispersive X-ray spectroscopy, to analyze elemental distribution. Furthermore, TEM offers an additional spectroscopic tool called electron energy-loss spectroscopy (EELS). EELS measures the energy loss of electrons passing through a sample, providing insights into electronic structure, oxidation states, and bonding configurations.²⁷⁶

In situ TEM provides the capability for real-time observation of morphology and composition changes in the electrode–electrolyte interphase.^{279–281} Furthermore,

it enables the investigation of kinetics related to crystallographic changes, such as phase transitions of electrode materials.^{282–285} By utilizing spectroscopic techniques in situ, spatially resolved information on the chemical and electronic states of battery materials can be obtained at nanometer resolution.^{278,286} In situ TEM requires special experimental setups to allow the battery system to operate under UHV conditions, as illustrated in Figure 16B,C. The selection of an appropriate electrolyte is crucial, ensuring compatibility with the vacuum environment and electrode materials, as well as stability under electron irradiation.^{277,278,287,288}

The first in situ TEM for a battery system was utilized by Kim et al.,²⁸⁹ who investigated the electrochemical

behavior of non-graphitized and graphitized electrodes using $\text{LiN}(\text{CF}_3\text{SO}_2)_2/\text{PC}$ and $\text{LiN}(\text{CF}_3\text{SO}_2)_2/(\text{EC}:\text{DMC})$ electrolytes in microcells with electron-transparent walls. In a study carried out by Liu and Huang,²⁹⁰ they first developed different types of the electrochemical nanobatteries for in situ TEM experiments to investigate the lithiation process of various anode materials. This included a liquid cell with an ionic liquid electrolyte, a solid cell with a naturally grown Li_2O layer as the solid-state electrolyte, and a solid cell with a solid electrolyte coating on nanowires. The results revealed distinct lithiation behaviors for different materials, with SnO_2 nanowires demonstrating high-stress tolerance and Si nanowires showing a core-shell lithiation mechanism. In another study by Wang et al.,²⁸⁵ in situ TEM was used to investigate the structural evolution and phase transition mechanisms during charge-discharge processes. Crystallization of $\text{Li}_{15}\text{Si}_4$ from amorphous Li_xSi occurred spontaneously and congruently, deviating from expected nucleation and growth processes. Moreover, Wang et al.²⁷⁸ used a $\text{LiCoO}_2/\text{LiPON}/\text{Si}$ thin-film battery for in situ TEM experiments. They examined the cathode-electrolyte interfacial layer using scanning TEM and EELS, and the HAADF images and Li K-edge EELS spectra, which are shown in Figure 16D,E. They observed a disordered layer that accumulated Li and transformed into rock-salt CoO after cycling. Increasing the layer thickness resulted in rapid capacity decay, rendering a larger portion of the cathode inactive.

In situ TEM observes dynamic processes and nanoscale structural changes with high spatial resolution. However, one of the significant challenges in in situ TEM analysis lies in designing liquid cells that accurately

replicate real batteries while enabling electron beam transmission and maintaining vacuum conditions. Strategies to minimize electron irradiation damage to the liquid electrolyte are crucial. Moreover, the sample preparation process can pose challenges, often requiring meticulous dispersion methods for obtaining small particles or utilizing FIB techniques. These steps can be time-consuming and may require specialized expertise in addition to in situ TEM analysis itself. Therefore, considering careful experimental design and meticulous sample preparation can help address these challenges and maximize the technique's potential in battery research.

5.3 | Recent studies

Electron microscopes provide the distinct advantage of directly observing the invisible microworld. However, the requirement of high vacuum conditions for the use of electron microscopes poses substantial challenges when designing in situ electrochemical cells. Notably, recent research has explored environmental electron microscopy as a promising solution to avoid these demanding vacuum conditions. In this context, “environmental” implies the capacity to capture the images and perform analyses under precisely controlled gas atmospheres and varying pressures. Thanks to this feature, many cases of applying environmental SEM^{291–294} and environmental transmission electron microscopy (ETEM)^{5,295,296} to in situ techniques have been reported. Additionally, Table 4 provides a comprehensive overview of various applications of in situ SEM and in situ TEM. Particularly in the

TABLE 4 Recent studies of electron-based in situ techniques.

Techniques	Material @ cell	Main description/results
SEM	MoS_2 @ modified coin cell	Wei et al. ²⁷⁰ studied the phase transformations of monolayer MoS_2 anode sheets.
	$\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$ @ heating stage	Tang et al. ²⁹⁷ monitored the morphological change of $\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Al}_{0.1}\text{O}_2$ during the solid-state synthesis process.
TEM	Si @ open cell	Zhang et al. ²⁹⁸ simultaneously measured the mechanical stresses and observed real-time dimensional changes in yolk-shelled Si/C structural units during cycling.
	Li metal @ heating holder with a Cu grid	Li et al. ²⁹⁹ examined the impact of temperature on the oxidation mechanisms of lithium dendrites.
	$\text{LiNi}_{0.6}\text{Co}_{0.2}\text{Mn}_{0.2}\text{O}_2$ @ gas cell with a Si_3N_4 membrane chip	Park et al. ³⁰⁰ explored phase evolution during the solid-state synthesis of $\text{LiNi}_{0.6}\text{Co}_{0.2}\text{Mn}_{0.2}\text{O}_2$ by in situ environmental TEM.
	$\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$ @ heating holder	Zhang et al. ³⁰¹ utilized in situ environmental TEM, revealing that internal stress plays a significant role in shaping the morphology during the solid-state synthesis of $\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$.

Abbreviations: SEM, scanning electron microscopy; TEM, transmission electron microscopy.

case of ETEM, there have been endeavors to couple it with a heating holder to enable real-time assessment of thermal stability and synthesis processes. Electron microscopes continue to serve as indispensable tools for the examination of microstructures and nanostructures. Their widespread utilization persists across various battery-related fields with significant technological advancements.

6 | NEUTRON-BASED

Neutrons, as uncharged particles, primarily interact with atomic nuclei. Neutron-based methods differ in their sensitivity to specific elements compared to X-ray-based methods. X-rays, which are scattered by electron clouds, show an overall increase in scattering power with increasing atomic number. The scattering power of neutrons, on the other hand, does not vary consistently, and there are differences even between isotopes. Furthermore, because the attenuation rate of neutrons is less sensitive to the density, neutrons show superior transmittance. Because of these features, neutron-based in situ techniques have great advantages when analyzing light elements, or when analyzing whole battery cells. This section covers the fundamental principles and practical uses of neutron-based in situ techniques, including ND, SANS, NR, NI, and NDP.

6.1 | In situ ND

ND, also known as neutron powder diffraction, is used to study crystal structures such as phases, and lattice parameters of the unit cell.^{302,303} The working principle of ND is based on Bragg's law, similar to XRD. However, because this method utilizes neutrons, ND is more sensitive to light elements such as Li, so that it can be used for the investigation of crystal structure depending on the Li content. In battery research, Rossouw et al.³⁰⁴ first applied ND to observe the possibility of whether high-crystallinity α -MnO₂ can intercalate the lithium or not.

In situ ND provides insights into the structural change of electrode materials during charge and discharge cycles.^{305–310} The first in situ ND was conducted by Berg et al.,³¹¹ who used a tube-type electrochemical cell to detect the phase transition of $\text{Li}_x\text{Mn}_2\text{O}_4$ to λ -MnO₂ during the lithium extraction. Presently, setting up an in situ ND experiment requires custom-designed cells to minimize interferences and optimize neutron scattering, involving cylindrical and pouch cell geometry, which are shown in Figure 17A,B or other configurations suitable for neutron penetration.^{307,312–315}

Zinth et al.³⁰⁵ conducted in situ ND analysis to investigate the effect of plated Li on the electrochemical performance of the cell in commercial $\text{LiNi}_{1/3}\text{Mn}_{1/3}\text{Co}_{1/3}\text{O}_2$ /graphite cells at -20°C . In this study, they estimated the amount of deposited Li by measuring the intensity ratio of diffraction peaks for LiC_6 and LiC_{12} phases. After fast charging (C/5), more Li was deposited on the graphite anode than the slower-charging case (C/30). On the other hand, Liu et al.³⁰⁷ used in situ ND and investigated the structural changes and Li migration in Li-rich layered oxides during battery operation. They calculated the evolution of the lattice parameter, O position, and two different Li sites within the Li and transition-metal layers. From the results as depicted in Figure 17C, the Li originating from the transition-metal layer showed predominantly irreversible behavior, whereas only Li reinsertion occurred within the Li layers during the initial cycle. Similarly, Dolotko et al.³⁰⁶ used in situ ND to study the structural changes occurring in commercial cells based on a $\text{Li}_x(\text{Ni}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2})\text{O}_2$ cathode. By analyzing the diffraction patterns for both electrodes, shown in Figure 17D, they found that the anode material transformed into higher-ordered Li intercalated carbons, while the cathode material showed distinct volume reduction and increase with reversible cation mixing.

In situ ND can observe the average structural changes, including phase transitions, lattice parameter changes, and atomic position variations. However, similar to XRD, ND primarily applies to crystalline materials, except for amorphous or liquid phases. Furthermore, ND requires a substantial amount of samples to obtain reliable results. Additionally, it is imperative to generate a large number of neutrons, which involves consumption of significant amounts of energy. While powerful neutron sources have been developed, there still remains a demand for improved devices capable of producing high neutron flux to analyze more complex and sophisticated samples. In this way, in situ ND offers the advantage of providing crystallographic information about light elements like Li, but it is crucial to be aware of various considerations related to the analysis of noncrystalline phases, sample quantity, and the energy-intensive nature of the technique.

6.2 | In situ SANS

SANS reveals nanoscale structural information of nanoparticles, pores, and other features in battery components.^{316–321} While the working principle is similar to SAXS, the difference in radiation source leads to distinct scattering patterns and offers complementary information. As a beam of neutrons is directed onto the sample, a fraction of the neutrons scatter at a small angle. These

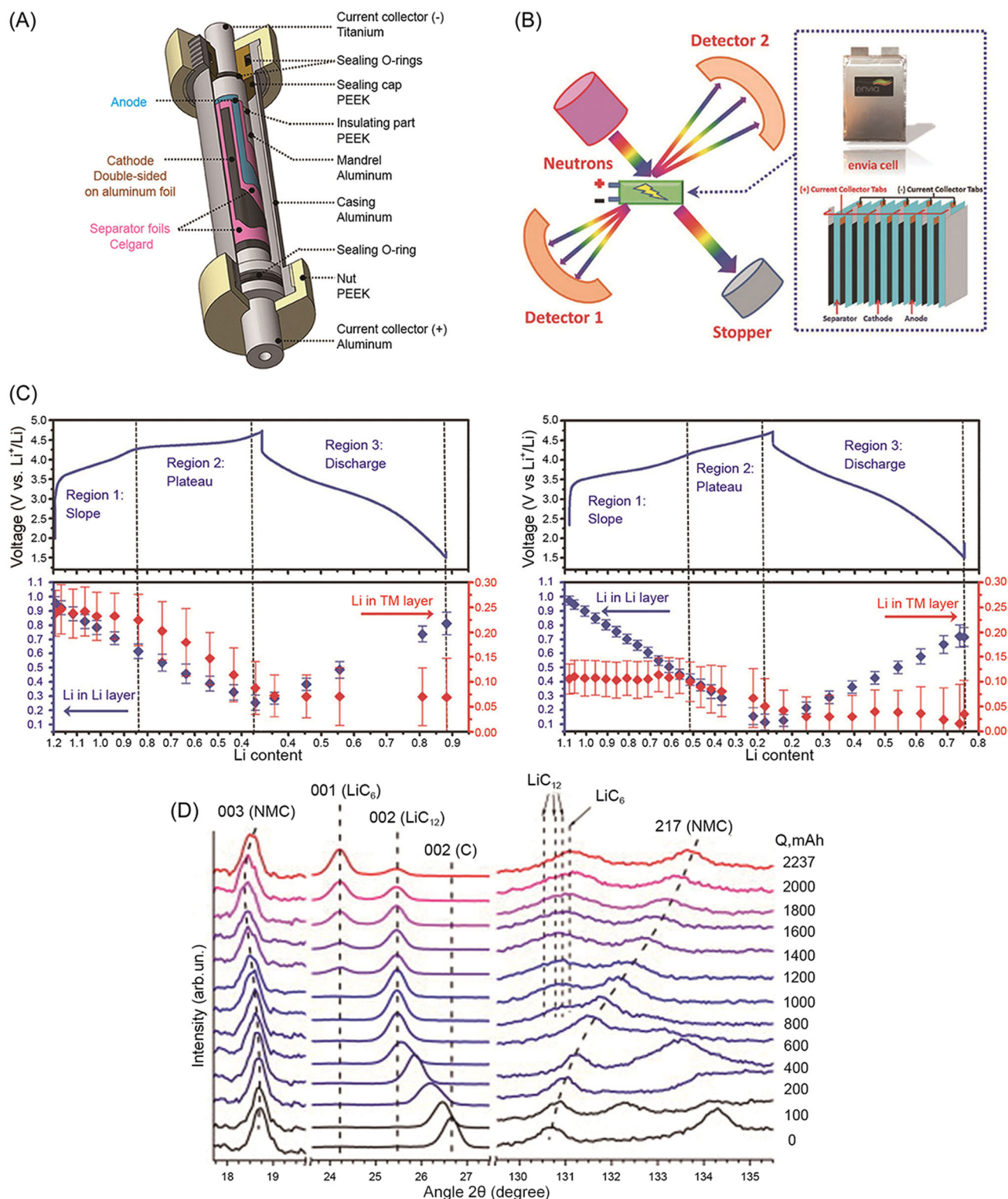


FIGURE 17 (A) Cylindrical cell for in situ ND. Reproduced under terms of the CC-BY license (<http://creativecommons.org/licenses/by/4.0/>). Copyright 2018, The Authors, published by Frontiers Media S.A.³¹² (B) Schematic illustration of an experimental setup of in situ ND for a pouch-type cell. (C) Voltage profiles and the corresponding lithium occupancy during a charge/discharge cycle for high Li-rich Li ($\text{Li}_{x/3}\text{Ni}_{(3/8-3x/8)}\text{Co}_{(1/4-x/4)}\text{Mn}_{(3/8+7x/24)}\text{O}_2$ ($x = 0.6$)) (right) and low Li-rich Li ($\text{Li}_{x/3}\text{Ni}_{(3/8-3x/8)}\text{Co}_{(1/4-x/4)}\text{Mn}_{(3/8+7x/24)}\text{O}_2$ ($x = 0.6$)) materials (left). Reproduced with permission: Copyright 2016, Wiley.³⁰⁷ (D) ND patterns of a commercial 18650 cylindrical $\text{LiNi}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2}\text{O}_2$ cell at different charged states. Reproduced with permission: Copyright 2014, Elsevier.³⁰⁶

scattered neutrons are captured by a detector, and the resulting data are analyzed to determine the scattering intensity as a function of the scattering angle.

When setting up in situ SANS experiments, similar cell designs to those used in XRD can be used such as coin cells with windows or single-layer pouch cells for neutron

penetration.^{320,321} For example, Sacchi et al.³¹⁸ utilized in situ SANS to examine the structural and compositional features of the SEI layer formed on lithiated carbon electrodes as shown in Figure 18A. Through the comparison of protiated and deuterated (D_2O) samples, they observed differences in scattering intensity, and

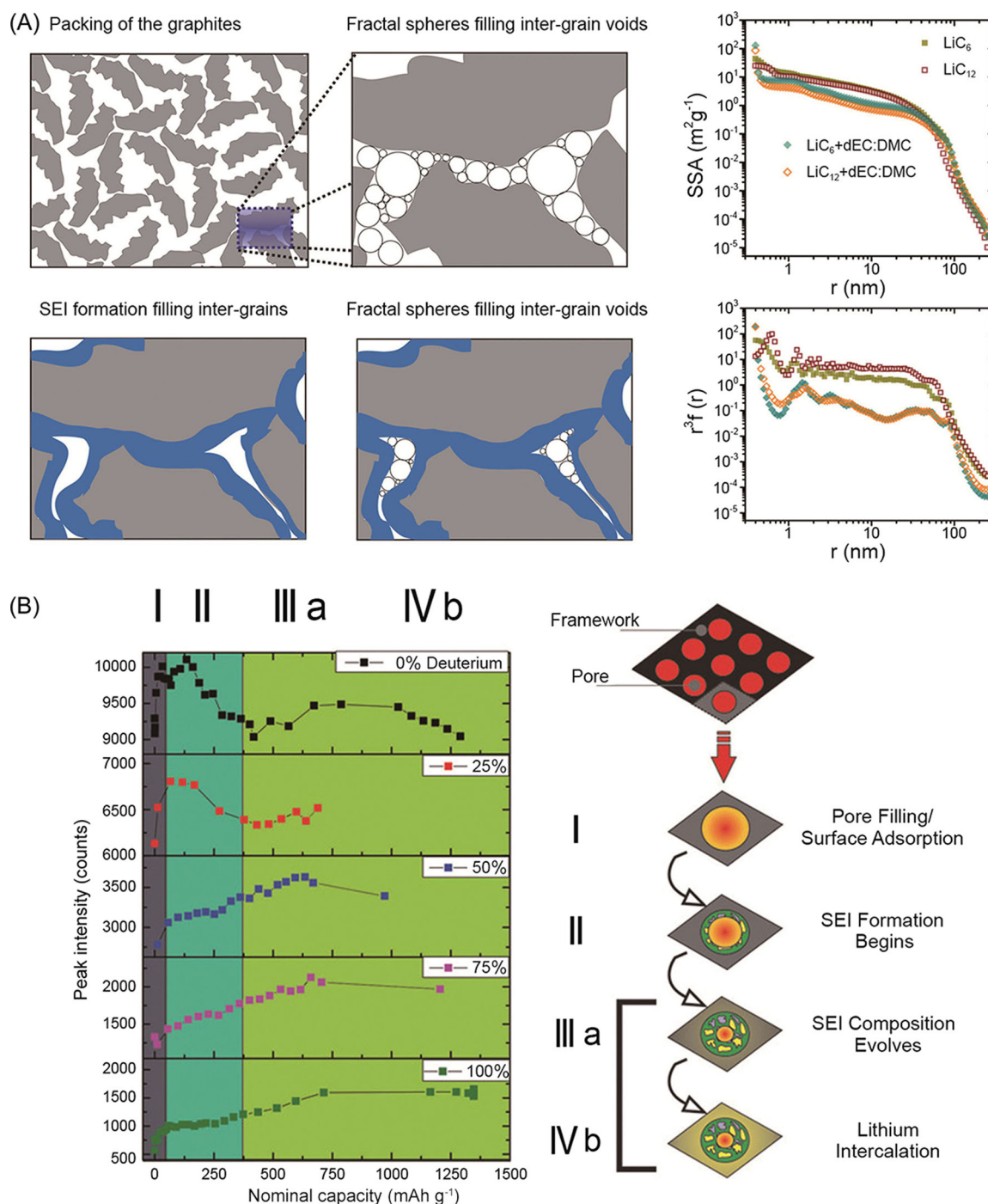


FIGURE 18 (A) Analysis regions of SANS and calculated specific surface area (SSA, top) and pore volume distributions (bottom) of LiC_6 and LiC_{12} . Reproduced with permission: Copyright 2015, American Chemical Society.³¹⁸ (B) Peak intensity according to the nominal capacity obtained from SANS (left) and the schematic illustration of different stages during the first charge/discharge cycle of the nondeuterated cell (right). Reproduced with permission: Copyright 2012, American Chemical Society.³¹⁹

revealed surface fractal scattering on the carbon surface. Also, Bridges et al.³¹⁹ explored the formation of the SEI layer and lithiation processes in mesoporous hard carbon electrodes by using in situ SANS. Their study, shown in Figure 18B, uncovered the presence of an SEI layer within the carbon framework, with its composition evolving during discharge. However, by combining SANS with single-component measurements and scattering length density (SLD) calculations, Seidlmayer et al.³²⁰ utilized in situ SANS to track the compositional changes of active materials in the pouch cell system under realistic operating conditions. Through scattering contributions from a defined volume, they successfully tracked the gradual lithiation of active material particles and monitored core-shell lithiation processes.

In situ SANS is useful for studying the structural evolution of macromolecules and nanostructures with the characteristic of light element sensitivity due to the neutron sources. However, because of a similar working principle as SAXS, SANS encounters challenges in distinguishing different components due to low scattering contrast. This limitation can make it challenging to differentiate materials with similar properties, potentially leading to ambiguities in data interpretation. Additionally, SANS has a narrow resolution range, typically from 1 to 100 nm, limiting the application to nanoscale structures.³²² For larger structures, alternative techniques such as light scattering or microscopy would be more appropriate. Furthermore, multiple scattering effects can complicate data analysis and interpretation in SANS measurements.³²³ Thus, although in situ SANS is a tool that can reveal structural and compositional features in the SEI layer and active materials, it is important to bear in mind that its application comes with several constraints.

6.3 | In situ NR

The history of NR traces back to 1946, when Fermi et al.³²⁴ first reported the phenomenon of total reflection of neutrons. Through the precise direction of a focused neutron beam onto a surface or interface, NR provides information of the structural and compositional characteristics of interfaces and thin films, including layer thickness, roughness, and SLD.³²⁵ The superior penetration depth of neutrons also allows them to probe interfaces hidden behind thick solid materials such as Al, Si, and stainless-steel windows, as well as interfaces immersed in liquids.³²⁶ When highly collimated neutron beams interact with the material, they bounce back like how light reflects off a mirror, specifically called specular reflection. The working principle of NR is based on

measuring the specular reflection from a surface, characterized by the wave vector transfer.^{8,327} The reflectivity variation is influenced by the SLD, which calculates the composition density of the surface layer. Therefore, the accurate determination of SLD values is crucial, and various approaches have been developed to extract this information from reflectivity data.³²⁸

In situ NR provides high spatial and temporal resolution, enabling observation and quantification of (de)lithiation kinetics,^{329–332} and the variations in the composition or thickness of electrode-electrolyte interphase.^{333–338} Wagemaker et al.³²⁴ utilized in situ NR for the first time in the battery field to investigate the interfacial layer formation on a TiO₂ thin-film battery made of quartz, allowing the passage of a neutron beam. Currently, custom-made electrochemical cells, including three- or two-electrode cells, are commonly used for in situ NR studies.^{324,332} For instance, Kawaura et al.³³⁵ conducted in situ NR investigations on the SEI layer during Li insertion and extraction with the addition of Li bis(oxalate)borate (LiBOB). As shown in Figure 19A, they used a two-electrode system with an amorphous carbon film electrode on a Si substrate, with Li metal as the counter and reference electrodes. As shown in Figure 19B, they analyzed the thickness and SLD of the electrode and the electrode-electrolyte interface layer and revealed an increase in the thickness of the carbon electrode during charging and a decrease in SLD correspondingly. The addition of LiBOB reduced the charge-transfer resistance, increased the charging capacity, and facilitated the formation of a chemically stable SEI layer composed of Li₂CO₃. Moreover, Wagemaker et al.³²⁹ used a three-electrode cell to investigate the (de)lithiation mechanism in a TiO₂ anatase electrode. In the in situ NR data shown in Figure 19C,D determined the spatial distribution of Li ions perpendicular to the interface of the TiO₂ anatase layer during the initial cycle. These results indicated that a phase boundary between anatase and Li titanate moved progressively inside the TiO₂ material in a direction parallel to its surface.

In situ NR is used to study interfacial phenomena and characterize the surface structure and chemical composition of battery components. However, one limitation of in situ NR is the requirement for smooth surfaces/interfaces to ensure accurate measurements. Smooth surfaces yield better data quality and a higher signal-to-noise ratio. Additionally, access to neutron scattering facilities equipped for in situ NR is not as widespread compared to other characterization techniques. Therefore, it is essential to be aware of the limitations of this technique, such as surface smoothness and facility accessibility, to obtain accurate data.

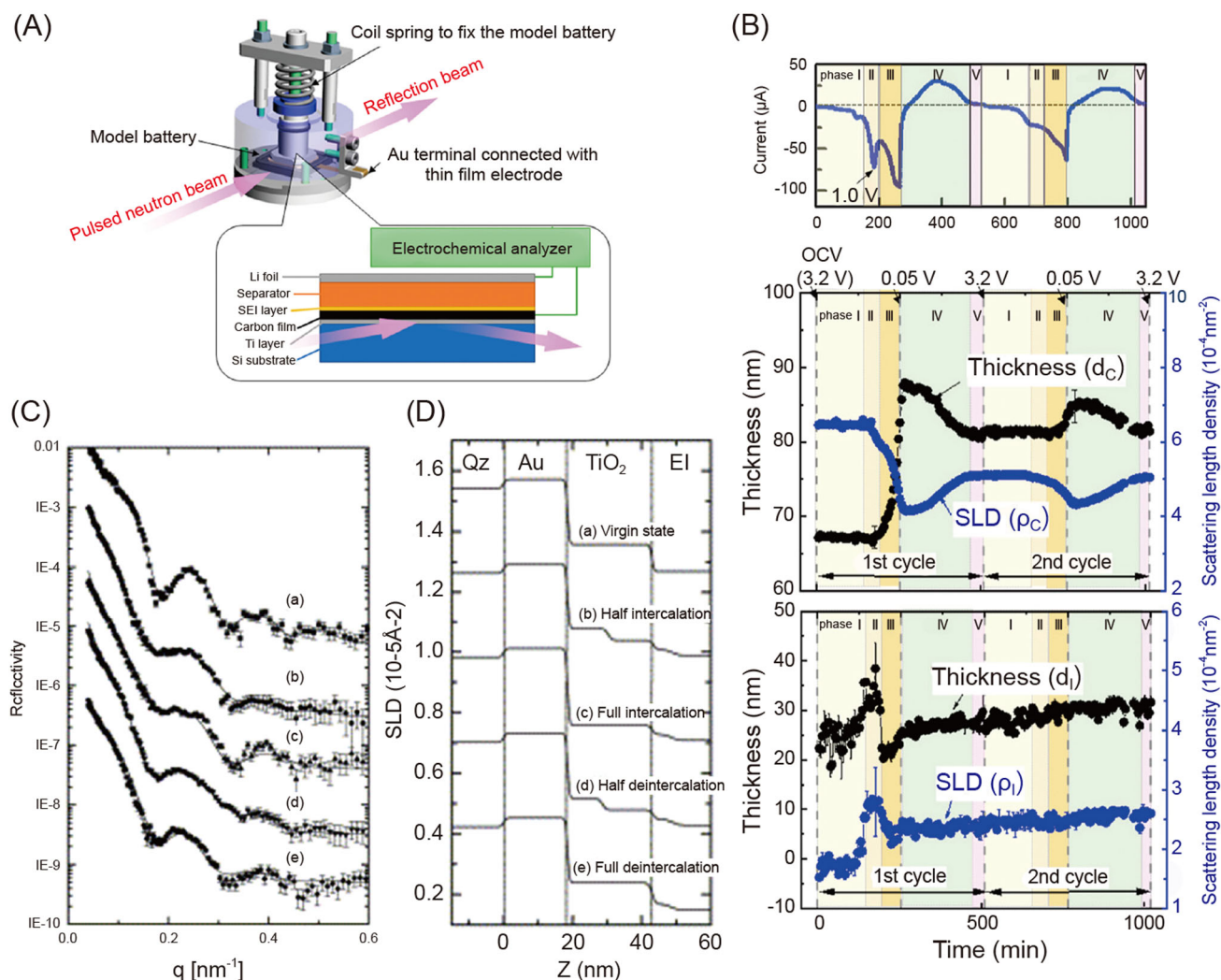


FIGURE 19 (A) Electrochemical cell for in situ NR. (B) Current profiles of amorphous carbon electrodes (top), and thickness and SLD of the carbon electrode (middle) and electrode–electrolyte interface layer (bottom) during charge–discharge reactions. Reproduced with permission: Copyright 2022, American Chemical Society.³³⁵ (C) In situ neutron reflectometry data measured at various points during the intercalation cycle and (D) the corresponding SLD results with the best individual fit. Reproduced with permission: Copyright 2003, Elsevier.³²⁹

6.4 | In situ NI

NI enables visualization of internal phenomena, which is not easily discernible by conventional X-ray methods, using a collimated neutron beam directed through the battery cell. Within NI, two specific applications are commonly used: neutron radiography imaging (NRI) and neutron tomography (NT). NRI generates 2D images by measuring the absorption gradients of the neutron beam, while NT reconstructs 3D images by acquiring a series of 2D radiographic images at different rotation angles and processing them accordingly. NI relies on the attenuation of the neutron beam. The mass-specific attenuation coefficient of neutrons, which represents the attenuation per unit mass, does not show a consistent trend with

atomic number and shows a higher propensity for interaction with H and Li compared to most heavy elements.³³⁹ NI accordingly enables the simultaneous observation of the Li-ion distribution and volume changes within the electrode, facilitating the tracking of Li dendrite growth,³⁴⁰ gas evolution,^{341,342} and electrolyte consumption.³⁴³ Therefore, the first NI experiment in battery research was applied to visualize the Li-ion diffusion in a spinel-type $\text{Li}_{1.33}\text{Ti}_{1.67}\text{O}_4$ electrode.³⁴⁴

In the 2010s, in situ NI experiments began to be conducted. For example, Siegel et al.³⁴⁵ first observed the real-time Li concentration across the electrodes in LFP/Graphite pouch-type cells. Recently, Song et al.³⁴⁰ utilized in situ NRI and NT to investigate the Li distribution after shorting. Figure 20A illustrates the

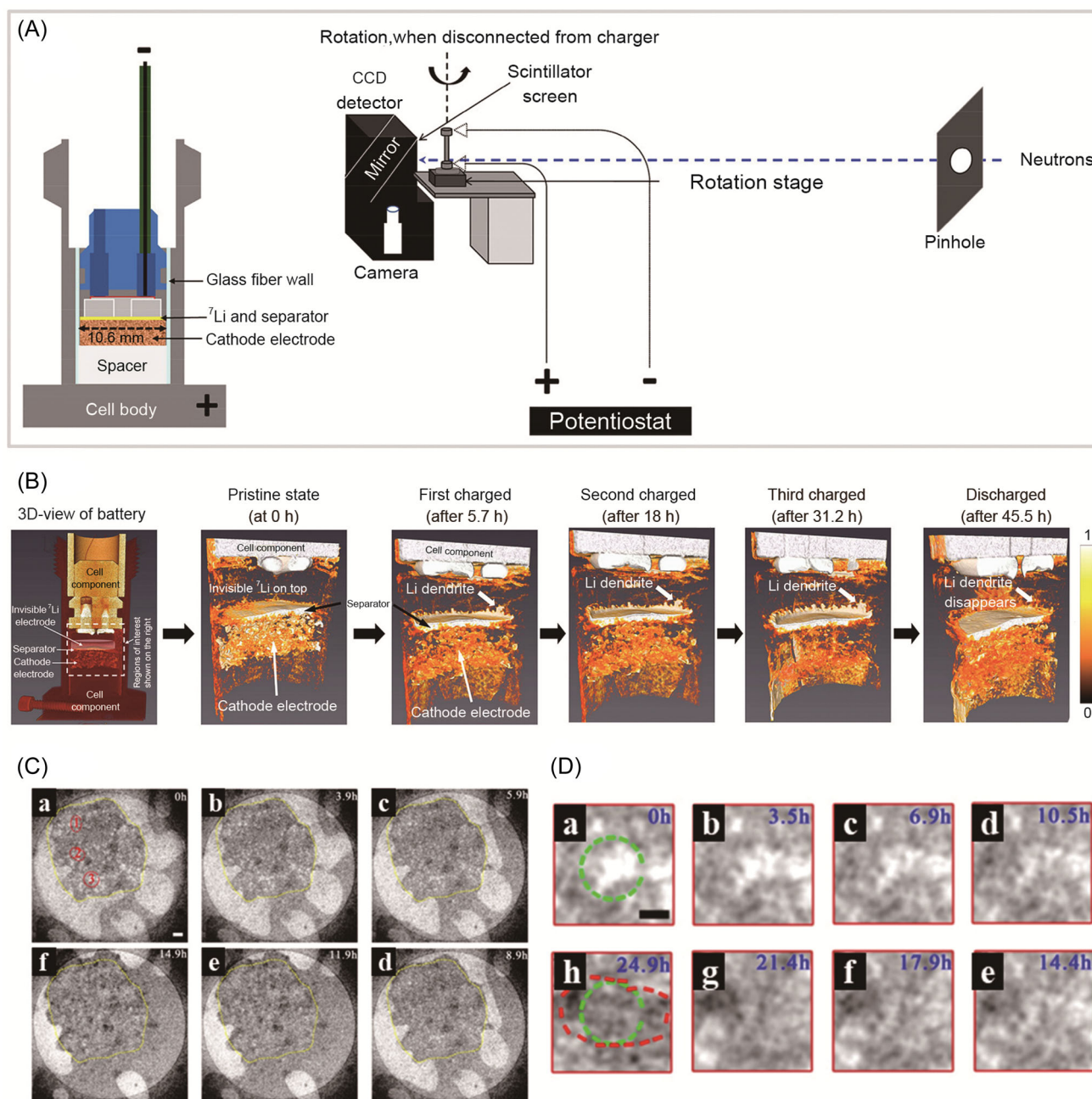


FIGURE 20 (A) Electrochemical cell for in situ NI and (B) tomography data for the Li distribution in a Li/LiMn₂O₄ battery at different charge–discharge states. Reproduced with permission: Copyright 2019, American Chemical Society.³⁴⁰ (C) Snapshots of in situ NRI during the first lithiation of a Si anode (a–f, the sequence of letters represents the passage of time). (D) NRI data for different states of charge of the Si during the first lithiation (a–f, the sequence of letters represents the passage of time) Reproduced with permission: Copyright 2017, Elsevier.³⁴⁶

electrochemical cell and the instrument setup for in situ NI. The core part was made of Ti_{2.08}Zr alloy, which shows a coherent scattering length of about 0. A high-thickness LiMn₂O₄ electrode was used in the half-cell, resulting in the formation of visible Li dendrites. During multiple charging cycles, the Li dendrites gradually grew and approached the cathode electrode as shown in Figure 20B. Furthermore, Sun et al.³⁴⁶ conducted in situ

NRI to investigate the lithiation behavior in Si anode. As can be seen in Figure 20C,D, NRI visualized a significant volume expansion of Si particles during initial lithiation, simultaneously capturing the alloying behavior of Li ions within Si electrodes.³⁴⁷

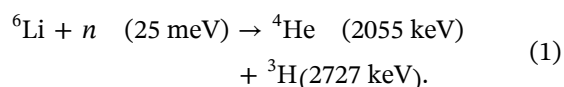
In situ NI can be used to study the internal structure and morphology changes of a Li-rechargeable battery during cell operations. However, one limitation of NI lies

in achieving high spatial resolution and temporal resolution compared to other imaging techniques, such as X-ray imaging. Especially, the spatial resolution of neutron-based imaging ranges from 10 to 25 μm , which may restrict the detailed characterization of fine structures within the battery.³⁴⁸ Additionally, the strong attenuation of neutrons by light elements like H and Li can occasionally obscure important details in the neutron images, posing challenges in differentiating between various battery components. Therefore, researchers must be mindful of potential limitations in NI capabilities when interpreting data.

6.5 | In situ NDP

NDP offers a nondestructive and depth-resolved analysis of light elements, particularly focused on Li in this paper. Moreover, NDP enables isotope-specific measurements of concentration versus depth profiles in the near-surface region of solids.³⁴⁹ This depth profile allows for the selective characterization of various Li-related phenomena, including Li diffusion and³⁵⁰ electrode-electrolyte interactions.³⁵¹ The low energy of thermal neutrons also benefits these analyses in that they cause little neutron-induced Li diffusion. As one example, Goldner et al.³⁵² first utilized NDP to acquire Li concentration distribution for experimental evidence of a rocking-chair structure in Li-rechargeable battery cells.

The probing depth of NDP varies depending on the stopping power of the material and is typically around 30–40 μm for battery materials, with a resolution in the range of 10–60 nm. During NDP analysis, a low-energy neutron is directed toward the surface of the targeted material.³⁵³ Also, the neutron absorption leads to the emission of alpha (α) particles (^4He) and triton particles (^3H) with well-defined energies as shown below:



By measuring the energy loss of these particles, the depth at which they were initially formed can be determined.

The first instance of in situ NDP being used in battery research can be found in the study carried out by Lamaze et al.,³⁵⁴ who investigated the distribution and quantification of Li, N, and Co in LiCoO_2 lithium-ion battery materials. To conduct in situ NDP, a vacuum or inert gas environment is required to avoid parasitic energy absorption of reaction products, which restricts the use of liquid electrolytes.³⁵⁵ Furthermore, the reaction

products have limited penetration lengths; thus, the thickness of the current collector and metal casings should be minimized. To solve these problems, various cell designs were suggested, such as a windowless cell and a cell sealed with a Kapton film.^{356,357} In a recent study, Linsenmann et al.³⁵⁸ presented a modified coin cell design that provides cycling performance comparable to that of standard coin cells. The electrochemical cell is shown in Figure 21A, which incorporate a thin Kapton window, selectively permeable to ^3H particles, and a coin cell cap with 500- μm hole patterns.

As an example of a real application, Li et al.³⁵⁹ conducted an in situ NDP investigation to analyze the Li plating behavior in a $\text{Li/Li}_{6.4}\text{La}_3\text{Zr}_{1.4}\text{Ta}_{0.6}\text{O}_{12}$ (LLZTO)/Ti solid-state battery, which featured a 3D Ti electrode with holes. Figure 21B demonstrates a reliable correspondence between the time counts and the amount of Li, even when the discharge capacity surpassed the capacity of the void space. These findings suggested that the presence of a 3D Ti electrode facilitates the Li plating within their voids, potentially mitigating volumetric changes at the electrode-electrolyte interface. Similarly, Ping et al.³⁶⁰ investigated the reversible nature of short-circuit behavior in garnet-based batteries using in situ NDP. As shown in Figure 21C, they monitored the transport and distribution of Li ions within Li/garnet/Li all-solid-state batteries by counting the ^3H particles, which served as an indicator of Li-ion movement. As a result of in situ NDP, a sudden voltage drop occurred during the plating stage at the higher current densities, suggesting the onset of a short-circuit.

In situ NDP can be used to analyze the depth-resolved details of battery composition concentration, especially for ^6Li isotopes. However, NDP has limitations related to the construction of experimental setups. The need for a vacuum or inert gas environment can present challenges and restrict the applicability of NDP to certain battery systems and configurations. On overcoming the design of the experimental setup and accessibility to intense neutron facilities, NDP will be used more widely as a useful method for surface-interface analysis.

6.6 | Recent studies

In this section, we have introduced a range of neutron-based in situ techniques, including ND, SANS, NR, NI, and NDP. Neutron beams offer distinct advantages for probing specific elements such as hydrogen and lithium, which present challenges for X-ray measurements due to their divergent attenuation coefficients. However, it is worth noting that most neutron-based techniques involve extended measurement times and necessitate large sample quantities due to the relatively low neutron beam

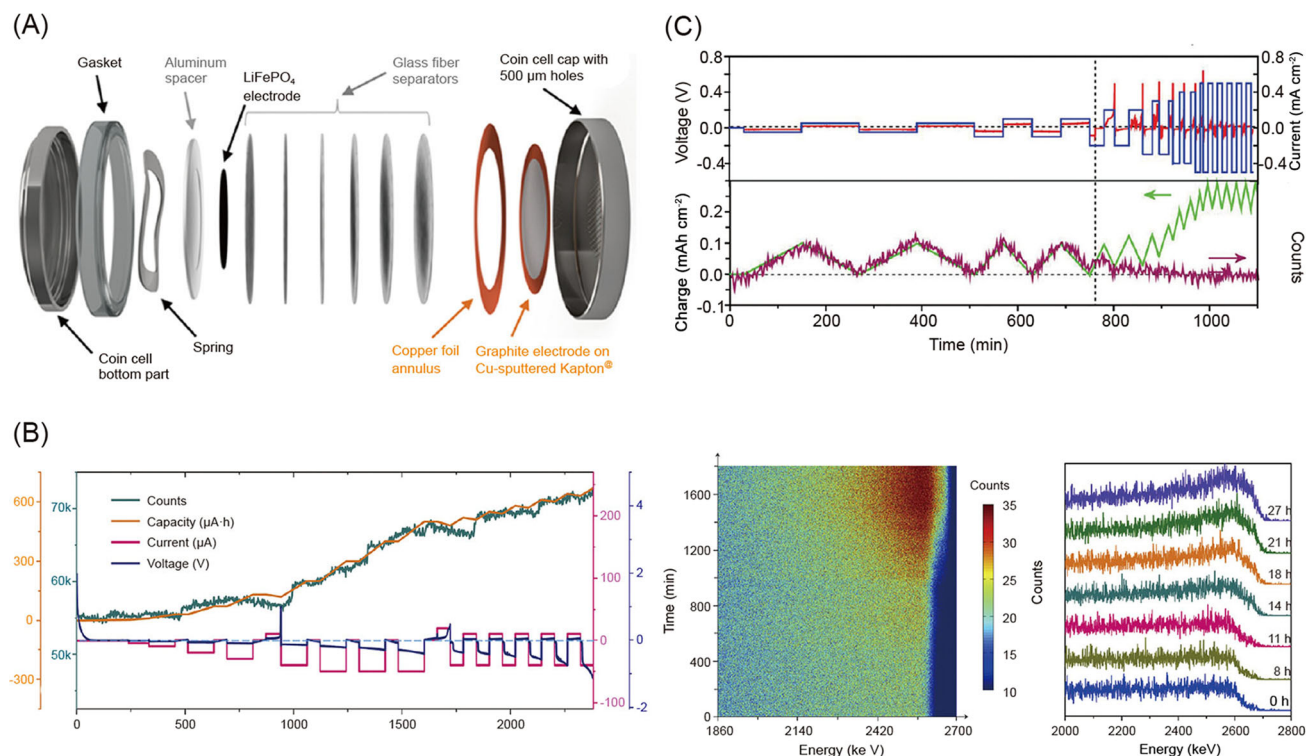


FIGURE 21 (A) Electrochemical cell with a Kapton window for in situ NDP. Reproduced under terms of the CC-BY license (<http://creativecommons.org/licenses/by/4.0/>). Copyright 2020, The Authors, published by IOP publishing.³⁵⁸ (B) Graphs of voltage, current, capacity, and total NDP signal counts (left), in situ NDP contour plot for a Li/LLZTO/Ti cell during cycling (middle), and the integrated curve data of NDP obtained every 1 h (right). Reproduced with permission: Copyright 2019, Elsevier.³⁵⁹ (C) Voltage and current profiles of a Li/garnet/Li cell (top) and the corresponding charge curve and in situ NDP counts (bottom). Reproduced with permission: Copyright 2020, Wiley.³⁶⁰

flux. To mitigate these limitations, recent developments in neutron-based techniques have focused on the exploration of advanced neutron sources to obtain high-flux neutron sources. These state-of-the-art sources facilitate time-resolved experiments, thereby allowing neutron-based techniques to expedite data acquisition.^{361,362} Comprehensive literature on the latest neutron-based in situ techniques, including time-resolved methods, can be found in Table 5. As demonstrated in the table, modern neutron-based techniques are applied to observe Li deposition and Li-diffusion pathways across diverse battery systems.

7 | OTHERS

In addition to radiation-based techniques, several in situ tools are extensively used in battery research. Analytical tools utilizing scanning probes provide topographic information about the surface of the electrodes. On the other hand, analytical tools capable of detecting the mass of molecular species offer insights into the generated gas phases within the system. In this section, two in situ techniques are introduced: AFM and DEMS.

7.1 | In situ AFM

The inception of AFM can be traced back to when Binnig and Rohrer discovered a novel form of electron tunneling microscopy using a scanning needle tip for atomic-level surface observation.³⁷⁵ Therefore, AFM is referred to as a force-based technique for investigating morphological and interface-related properties with exceptional spatial resolution at the nanometer scale. The mechanisms and processes occurring at the electrode–electrolyte interface can be explored by analyzing the electrode surface film layer,^{376,377} Li deposition,³⁷⁸ and electrode material integrity.^{379,380} In these studies, AFM has a significant advantage as it generated 3D images through the combination of lateral scanning movement with vertical deflection, allowing the creation of detailed topographic maps. Furthermore, AFM is not limited by sample conductivity, making it suitable for studying a wide range of components, including insulators and electrolytes. AFM uses a fine tip attached to a flexible cantilever, detecting the interaction forces between the tip and the sample surface. As the tip approaches the surface, the resulting interaction forces cause the cantilever to deflect. This deflection is

TABLE 5 Recent studies of neutron-based in situ techniques.

Techniques	Material @ cell	Main description/results
ND	LiNi _{0.5} Co _{0.2} Mn _{0.3} O ₂ -LiNi _{0.9} Co _{0.05} Al _{0.05} O ₂ /graphite @ cylindrical cell	Zhu et al. ³⁶³ examined the structural degradation mechanisms of 18650-type batteries that cycled within various SoC ranges.
	LiNi _{0.6} Co _{0.2} Mn _{0.2} O ₂ /graphite @ cylindrical cell	Wu et al. ³⁶⁴ studied the structural change in both electrodes at different charging rates from 0.27 to 4.4 C.
	LiNi _{0.5} Co _{0.2} Mn _{0.3} O ₂ /graphite @ pouch cell	Pender et al. ³⁶⁵ discovered that the Li/Ni antisite concentration varies throughout cycling.
NR	Ni @ in-house in situ cell	Charlton et al. ³⁶⁶ studied the SEI formation on a Ni electrode and observed the sequential formation of the Li-rich layer and the organic layer during lithiation.
	WO ₃ @ in-house in situ cell	Rus et al. ³⁶⁷ investigated the interphase composition and electrode volume change between the WO ₃ electrode and the LiPF ₆ -based electrolyte.
	LIPON/Li @ in-house in situ cell	Browning et al. ³⁶⁸ analyzed the interphase formation between Li metal and the solid electrolyte LIPON.
NI	NMC/graphite @ modified coin cell	Yusuf characterized the deposited Li at rates of 1C and 6C using neutron microcomputed tomography. ³⁶⁹
	NMC, LiCoO ₂ , LiMn ₂ O ₄ /graphite @ cylindrical cell	Nazer et al. ³⁷⁰ used in situ NR to trace the Li redistribution and electrolyte circulation at various SoC and current densities.
	In/InLi Li ₆ PS ₅ Cl S/C/Li ₆ PS ₅ Cl @ in-house in situ cell	Bradbury et al. ³⁷¹ limited the pixel size to enhance the time resolution of NT and traced the reaction front propagation from the separator to the current collector.
	In/InLi Li ₆ PS ₅ Cl S/C/Li ₆ PS ₅ Cl @ in-house in situ cell	Bradbury et al. ³⁷² visualized the distribution of mobile or trapped Li in an all-solid-state Li-S battery.
NDP	Li Li _{6.4} La ₃ Zr _{1.4} Ta _{0.6} O ₁₂ Ti @ open cell	Li et al. ³⁵⁹ investigated the Li plating process, which primarily occurs in the voids of the three-dimensional electrodes.
	Li ₇ La ₃ Zr ₂ O ₁₂ , LiPON, Li ₃ PS ₄ @ open cell	Han et al. ³⁷³ compared the growth patterns of Li dendrite on three representative solid electrolytes with different electronic conductivities.
	Li garnet Li @ symmetric cell with Kapton window	Ping et al. ³⁷⁴ suggested the reversible nature of the short-circuit process in garnet-based all-solid-state batteries.

Abbreviations: ND, neutron diffraction; NDP, neutron depth profiling; NI, neutron imaging; NR, neutron reflectometry; NT, neutron tomography; SEI, solid electrolyte interface.

monitored using a laser beam and a position-sensitive photodetector, providing feedback to maintain a constant force or deflection during scanning. Based on the movement of the cantilever, AFM offers various imaging modes, including contact mode, tapping mode, and noncontact mode.^{377,381–383}

To perform in situ AFM studies, the cell construction must provide unobstructed access from the top, allowing the AFM cantilever to make direct contact with the working electrode as shown in Figure 22A–C.^{382–384} These experiments are typically conducted inside a glovebox to maintain an inert atmosphere. Additionally, the materials used to construct the cell must be nonreactive with the battery components to avoid unwanted interactions that could affect the accuracy of

the measurements. In situ AFM studies, taking these considerations into account, can be primarily categorized into two approaches: investigations using a three-electrode system and those utilizing a two-electrode system, which are depicted in Figure 22A,B, respectively.

For example, Park et al.³⁸³ utilized a two-electrode fluid cell, shown in Figure 22C, to investigate the deformation process of LiCoO₂ during lithiation and delithiation. By monitoring dimensional changes, the AFM experiment yielded a measurement range of 1.28%–1.3% for the height increase as the Li content changed from 1 to 0.35. This result matched well with the measurements using XRD techniques, which indicated an average height increase of approximately 1.34%. On the other hand, Doi et al.³⁸⁵ used a three-electrode

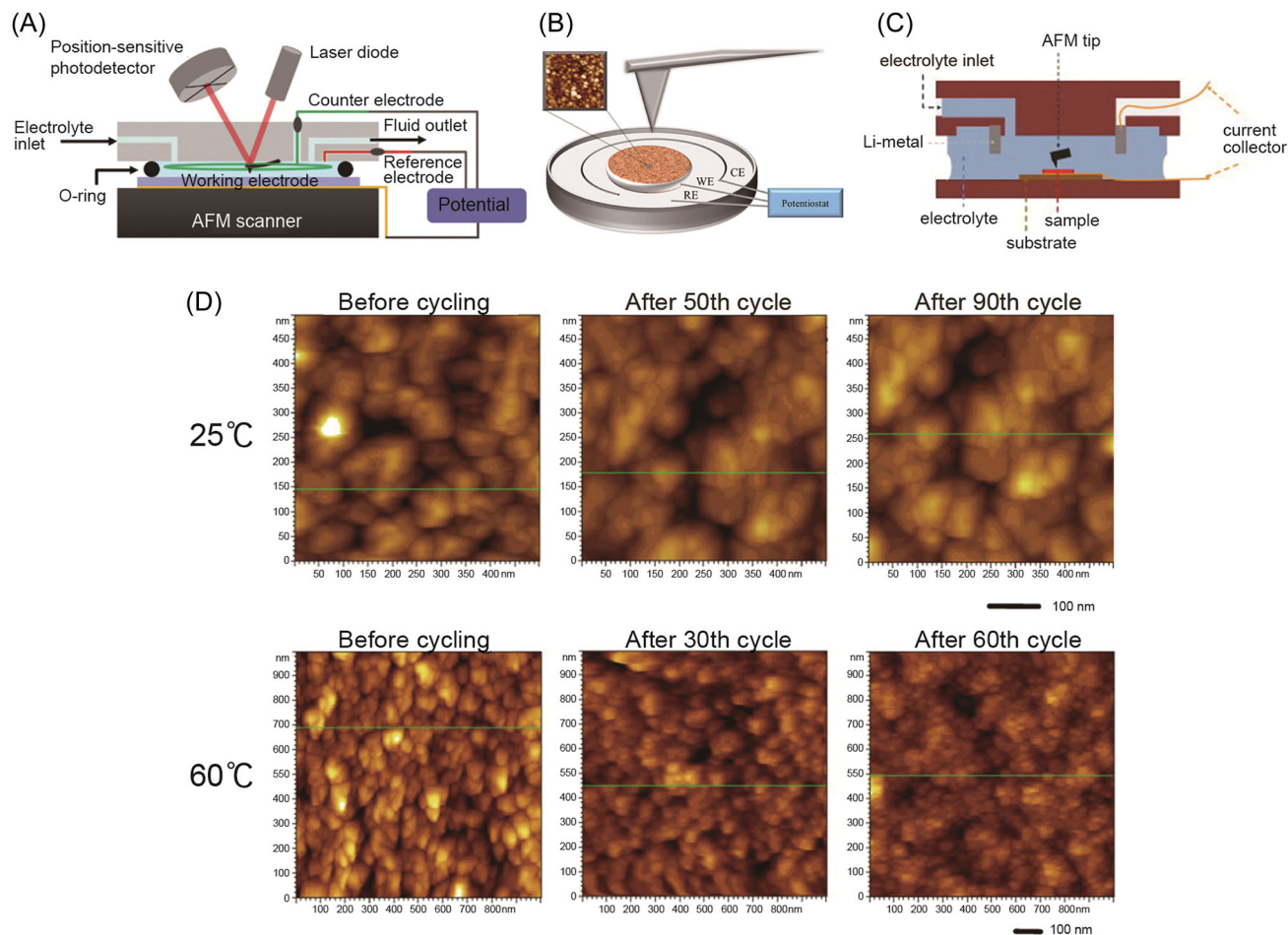


FIGURE 22 (A) Schematic illustration of a three-electrode electrochemical cell for in situ AFM. Reproduced with permission: Copyright 2015, Elsevier.³⁸² (B) In situ AFM setup using a three-electrode system. Reproduced with permission: Copyright 2019, American Chemical Society.³⁸⁴ (C) Two-electrode electrochemical cell for in situ AFM. Reproduced with permission: Copyright 2013, Elsevier.³⁸³ (D) AFM images of a LiMn_2O_4 electrode obtained at 25°C (top) and 60°C (bottom) for different numbers of cycles. Reproduced with permission: Copyright 2008, Elsevier.³⁸⁵

system for the observation of morphological changes of the LiMn_2O_4 thin film during cycling or storage at elevated temperatures. They observed morphological changes of the degraded LiMn_2O_4 thin film as shown in Figure 22D. They found the formation of small particles, loss of crystallinity, and accelerated capacity fading during cycling or high-temperature storage.

In situ AFM uses fine tips to study the surface morphology and nanoscale structural changes of battery components, providing high-resolution 3D images with interface-related properties. Nevertheless, the construction of the operating condition presents challenges as it requires the design of an open system cell, deviating from the conventional encapsulated system. Moreover, AFM imaging can be time-consuming, especially when scanning large areas or acquiring high-resolution images. It is also essential to take longer data acquisition times when dealing with extensive sample areas. External vibrations

and noise can affect the accuracy and quality of the acquired data, necessitating the implementation of vibration isolation measures and stable experimental conditions. The choice of the probe tip and its parameters can also impact imaging results, requiring careful calibration to prevent sample damage or artifacts. While the experimental designs for in situ AFM may be challenging and intricate, the abundance of available probe tips makes this technique a versatile tool for the determination of interfacial properties in Li-rechargeable batteries.

7.2 | In situ DEMS

DEMS has high sensitivity for identifying and quantifying gaseous and volatile species generated during electrochemical reactions. After Bruckenstein et al.³⁸⁶

developed combined mass spectrometry (MS) techniques for electrochemical reactions and Wolter et al.³⁸⁷ introduced the term “DEMS,” this method is mostly used as an in situ method and is suitable for analyzing the composition or generation rates of gas species. DEMS combines gas chromatography (GC) and MS. Within the mass spectrometer, the gaseous molecules undergo collisions with high-voltage electrons, resulting in the formation of charged fragments. These fragments are guided to the detector by a magnetic field, reaching the detector at different flight times based on their mass-to-charge ratio (m/z). By analyzing the flight times, the mass spectrometer can identify the specific vaporizable species present in the gas. By integrating the mass spectral data with the retention time obtained from GC analysis, the relative concentrations or partial pressures of the targeted gaseous species can be identified.

Figure 23A,B shows a schematic example of the electrochemical cell for the DEMS experiment and its inner part, respectively.³⁸⁸ It is noteworthy that the current collectors are designed with holes that allow effective gas flow. The carrier gas is introduced into the cell from the upper part, by which any evolved gases during the electrochemical process exit the cell and flow to the mass spectrometer.

In a study by Zhao et al.,³⁸⁹ DEMS was used to analyze the gas released from half-cells with different Ni-rich layered oxides, $\text{LiNi}_x\text{Mn}_y\text{Co}_{1-x-y}\text{O}_2$, during voltammetric scans and storage depicted in Figure 23C. Through DEMS analysis, it was found that the O species released reacted with carbonate solvents, leading to the simultaneous production of CO_2 and protons at the cathode interface. In another study, Peng et al.³⁹⁰ suggested using a dimethyl sulfoxide electrolyte and a porous gold electrode to achieve reversible formation and decomposition of Li peroxide (Li_2O_2) in a Li- O_2 battery. As shown in Figure 23D, they used DEMS to analyze the gas species generated during the discharge and charge processes and discovered that the discharge process was primarily dominated by the formation of Li_2O_2 . The only gas detected during discharge was O_2 , indicating the absence of unintended CO_2 , SO_2 , or SO_3 and suggesting no evidence of electrolyte decomposition.

DEMS can be used to investigate the mass-resolved characteristics to study the relationship between electrochemical processes and gas evolution in Li-rechargeable battery systems. However, one primary limitation of DEMS is its inability to accurately detect liquid or nonvolatile species, which poses challenges for their analysis. Overcoming this limitation necessitates the use of advanced electrochemical cell setups to facilitate the detection and quantification of these species. Furthermore, efficient transfer of gas species from the

electrode–electrolyte interfaces to the MS system is crucial for obtaining reliable results. The high viscosity of common electrolytes, for instance, presents obstacles to the mass transfer of gas species, and the delayed gas transfer can lead to erroneous interpretations.³⁹¹ To address this challenge, various flow cell configurations, such as impinging jets and rotating disc electrodes, have been developed to enhance the transport of gas species for timely and accurate detection.^{392–395}

7.3 | Recent studies

In this section, we have introduced in situ AFM and DEMS. AFM, which uses a cantilever, is advantageous for probing fine surface structures. Nevertheless, ongoing research and development efforts are dedicated to enhancing its capabilities. Recent initiatives include the development of high-speed AFM to address the inherent limitations of conventional AFM with a relatively slow scan speed.³⁹⁶ Furthermore, there have been systematic attempts to combine AFM with other techniques to acquire comprehensive analytical information.²⁹⁵ Similarly, for DEMS, a technique capable of quantitatively analyzing gaseous and volatile species, there have been efforts to integrate it with other analytical methodologies.³⁹⁷ Additional recent applications of in situ AFM and DEMS are summarized in Table 6. There seems to be no doubt that the continuous advancement of these techniques will significantly impact the growth of battery research.

8 | SUMMARY AND PERSPECTIVES

In situ analysis elucidates underlying patterns by correlating discrete data points and dispels ambiguity in interpreting the phenomena that transpired before and after the event. By enabling real-time observation of the dynamic physical and chemical changes, in situ techniques can provide a clear understanding of a particular system by correcting misconceptions that can arise in post-mortem analysis of ex situ techniques. In this comprehensive review, a series of widely used in situ techniques used in the field of Li-rechargeable batteries were investigated and systematically organized depending on radiation sources. Distinctive factors including the beam type, material interaction principle, and the signal detection mode, dictate each technique's capability to scrutinize the target material, spanning various types of information from electronic and atomic scales to morphological scales, and across a range of probing

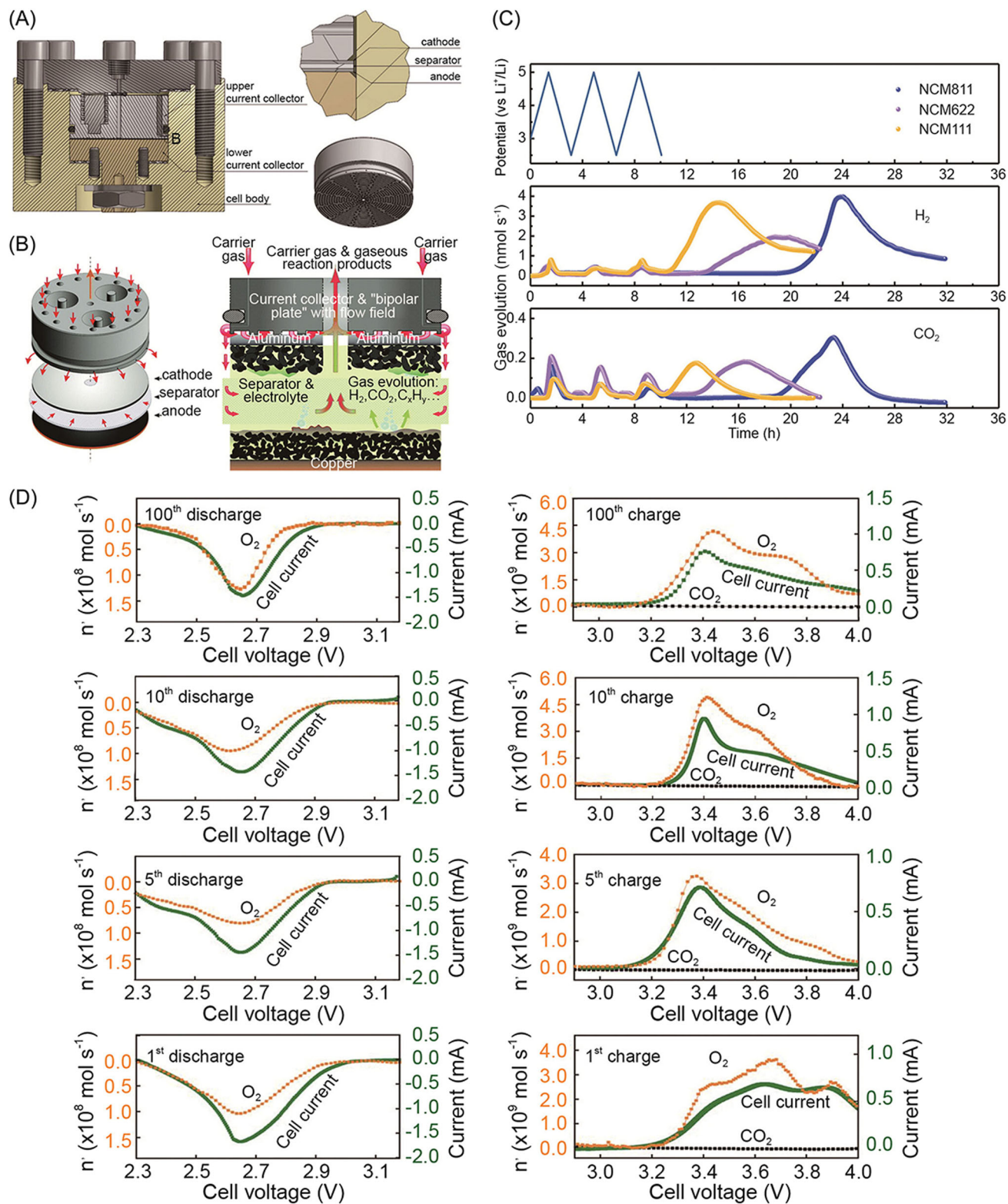


FIGURE 23 (A) Schematic illustration of the electrochemical cell for DEMS and (B) its inner part. Reproduced with permission: Copyright 2015, American Chemical Society.³⁸⁸ (C) Potential curves for LiNi_xMn_yCo_{1-x-y}O₂-Li cells (top) and the evolution rates of H₂ (middle) and CO₂ (bottom) gases. Reproduced with permission: Copyright 2021, Wiley.³⁸⁹ (D) DEMS of the discharge (left) and charge (right) for the porous gold electrode in different cycles. Reproduced with permission: Copyright 2012, American Association for the Advancement of Science.³⁹⁰

TABLE 6 Recent studies of other in situ techniques.

Techniques	Material @ cell	Main description/results
AFM	Li @ exposed electrode in Ar atmosphere	Zhang et al. ²⁹⁵ combined in situ AFM with environmental TEM to observe the growth of Li whiskers and measured the resulting stress.
	Li @ homemade 3-electrode cell	Zhang et al. ³⁹⁸ investigated the formation process and mechanical property of the SEI layer on highly oriented pyrolytic graphite in trioxane-modulated electrolytes.
	LiMn ₂ O ₄ @ homemade inverted AFM cell	Yang et al. ³⁹⁶ observed the morphological change of LiMn ₂ O ₄ during the overdischarging process with high-speed AFM (~0.85 fps).
DEMS	LiCoO ₂ , LiNiO ₂ , Li ₂ MnO ₃ @ customized Swagelok cell	Papp et al. ³⁹⁹ compared the gas evolution phenomena from representative cathode materials during high-voltage charging.
	Li ₂ CO ₃ @ customized Swagelok cell	Cao et al. ⁴⁰⁰ utilized DEMS in combination with gas chromatography to measure the gas that evolved during the oxidation of Li ₂ CO ₃ on a carbon substrate.
	Ag, Bi ₂ MoO ₆ @ customized Swagelok cell	Li et al. ⁴⁰¹ verified the reversible formation and decomposition of Li ₂ O ₂ in a photoassisted Li–O ₂ battery.

Abbreviations: AFM, atomic force microscopy; DEMS, differential electrochemical mass spectrometry; SEI, solid electrolyte interface; TEM, transmission electron microscopy.

depth, from surface to bulk, which is schematically summarized in Figure 24.

Table 7 summarizes the purposes, advantages, and challenges of each technique, along with several research cases. Multiple techniques can be used to analyze one phenomenon, and it is worth reiterating the importance of understanding the different facets of information that each methodology can offer. For example, techniques such as XRD, ND, and TEM can all be used to investigate electrode material phase transitions. However, there are differences in the information that can be gained from each experiment. Both XRD and ND provide crystal structural information based on diffraction patterns. However, XRD, with its high beam flux, is suitable for structural analysis with smaller sample volumes and at shorter time intervals. ND, on the other hand, leverages the high sensitivity of neutrons to lighter elements and its extended penetration depth, making it suitable for determining the position of Li or O, or for observing both the cathode and the anode at once. TEM, meanwhile, has the advantage of capturing atomic-resolved images regardless of crystallinity as long as an appropriate cell design is provided. Therefore, it is essential to choose the proper technique based on the material to be analyzed and the level of detail to be obtained.

It is recommended to use multiple techniques in combination to eliminate the blind spots that exist when analyzing with a single technique and to gain a multifaceted understanding of a phenomenon in Li-rechargeable battery systems. Combining analytical methods with different radiation sources can provide a more robust verification of information. For example, X-rays and neutrons can be used together to complement the respective constraints. Ziesche

et al.³⁴⁷ analyzed changes in the 3D distribution of Li ions in a Li/MnO₂ battery cell using XCT and NT. X-rays in tomography allow for better temporal and spatial resolution to effectively trace the mechanical degradation. The process of Li-ion diffusion, which was indirectly observed through XCT, has been more directly visualized using NT. Recently, neutron and X-ray tomography techniques have also been applied to battery materials,⁵³⁴ which allows simultaneous analysis of both XCT and NT, further increasing reliability. On the other hand, even when utilizing a similar radiation source, a phenomenon can be investigated differently depending on the methodologies implemented. One classic example is the combinatorial research conducted by Nam et al.⁹ to analyze the thermal degradation behavior of charged Ni-rich cathode materials. The overall crystal structural changes were analyzed by time-resolved XRD, while element-specific local structure changes were explored by hard XAS. In this way, by observing several correlated phenomena, or by observing a single phenomenon in multiple ways, researchers can gain an inclusive understanding of the underlying processes within the battery.

Recent research has primarily concentrated on fast-charging technologies aimed at reducing charging times to just a few minutes. High current densities can lead to issues like nonuniform reactions and dendrite formation, necessitating the development of real-time diagnostic techniques. However, some in situ methods require extended time-frames, ranging from tens of minutes to hours, for obtaining single data. These extended measurements, during fast charging, yield limited data due to significant SoC change within a single measurement, rendering the analysis less reliable. For example, in hard XAS analyses, detailed structural assessments usually take nearly 30 min when

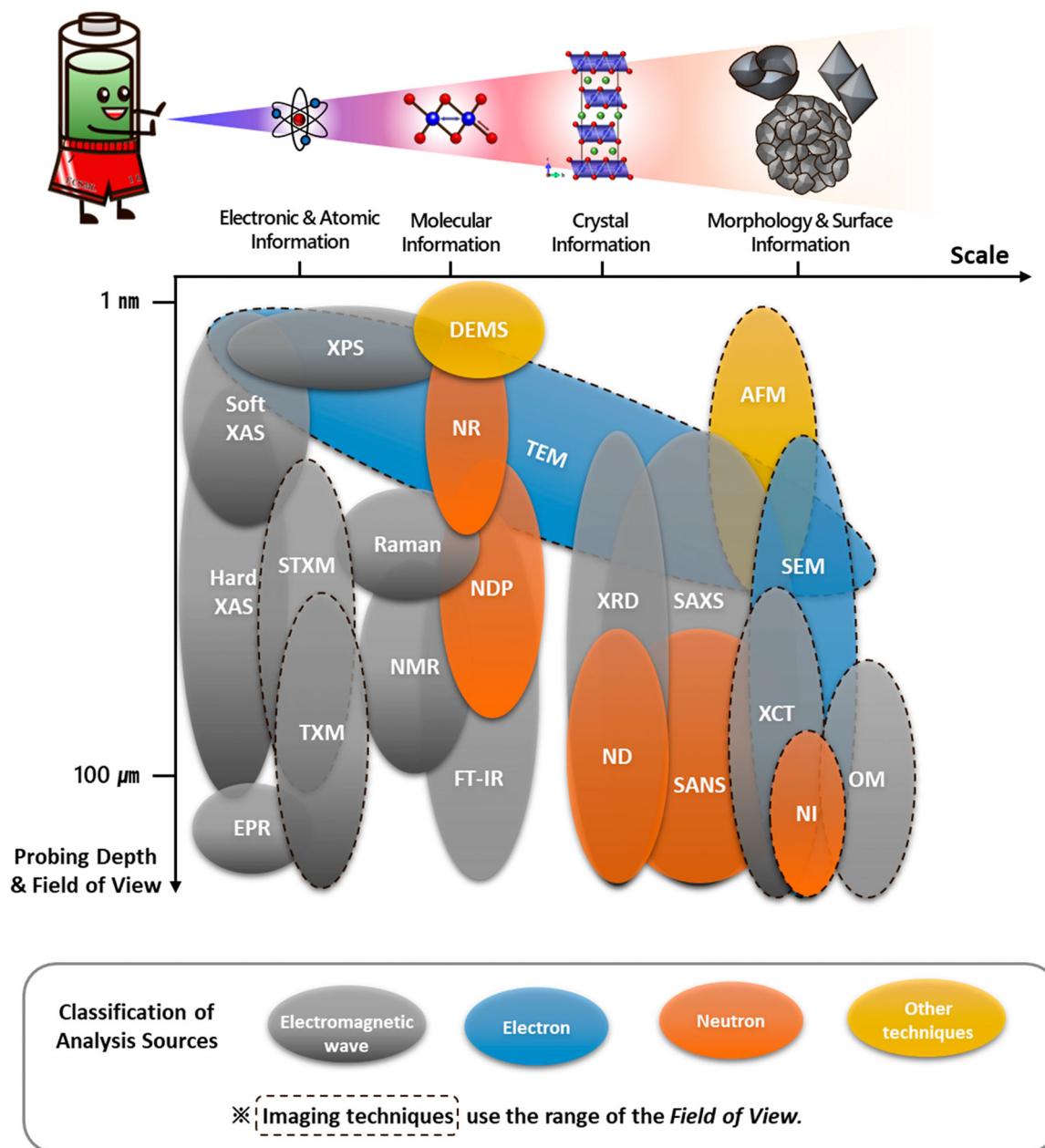


FIGURE 24 Categorization of in situ techniques for lithium rechargeable battery research based on types of information and general probing depth or field of view. The imaging techniques are presented by dotted lines and categorized according to the scale of field of view.

scanning over long energy ranges. To reduce the timeframe, enhancements in mechanical design are imperative. Some synchrotron facilities offer XAS beamlines designed for quick energy scanning,⁴⁷ enabling the acquisition of a single spectrum in 1 min or less via a custom monochromator. On the other hand, traditional AFM typically operates under a low-current cycling environment, acquiring pixel data sequentially to construct 2D images. A recent innovation, high-speed AFM,^{396,535} achieves speeds of about 1 frame s^{-1} by using an inverted setup and an objective lens optimized for efficient cantilever deflection detection, reducing the tip-surface interaction time significantly. However, NT

remains a challenging technique, requiring several hours to capture an entire structure, making real-time analysis problematic. Nevertheless, the temporal resolution can be decreased by adapting to cell design and spatial resolution. Furthermore, ongoing research in fast NT^{371,536} with enhanced neutron flux augments the potential for NT to be used during battery operation. In this way, temporal resolution in formerly time-consuming experiments can be advanced by altering experimental requisites. Ongoing advancements may enable distortion-free observation of phenomena within the cell, even under high current densities.

TABLE 7 Summary of in situ characterization techniques; electromagnetic wave-, electron-, and neutron-based approaches and other techniques.

Techniques	Purposes and advantages	Challenges	Details
<i>X-ray-based</i>			
XRD	Understanding information about the average crystal structure.	Inaccurate data of amorphous samples, Interpretation of overlapped diffraction peaks.	Phase transition ^{9,15–18,20,21,45,46,402–416} Lattice parameter change ^{14–16,18,20,167,411–414,417,418} Phase fraction ^{14,415,416,419}
SAXS	Understanding the structural evolution of macromolecules and nanostructure.	Low scattering contrast, Interpretation of overlapped diffraction peaks.	Microstructure and morphology change ^{36,37,420,421}
Hard XAS	Understanding electronic and local structures, element-specific technique, high sensitivity and resolution.	Control of sample thickness to achieve an optimal edge step.	Valence state change ^{9,18,53,408,415,422–433} Local structure change ^{9,53,428–434}
Soft XAS	Understanding electronic structure, element-specific technique, high sensitivity and resolution, surface sensitivity.	Need for vacuum conditions, achievement of real operating conditions.	Valence state change ^{65,66,435}
XPS	Understanding electronic states and chemical species, surface sensitivity.	Need for vacuum conditions, interpretation of overlapped or shifted peaks, achievement of real operating conditions.	Electrode–electrolyte interface composition ^{70–75,78,79,436}
XCT	Understanding the internal structure and morphology, wide range of spatial resolutions.	Trade-offs between spatial and temporal resolution, limited penetration depth, distinction between elements with similar X-ray attenuation coefficients.	Volume change ^{88,105} Phase transition ^{88,95,96} Crack formation ^{89,437} Li deposition ¹¹² Electrode layer degradation ^{90,438}
TXM	Understanding the changes in electronic structure with corresponding images.	Trade-offs between spatial and temporal resolution, limited penetration depth and field of view, achievement of real operating conditions.	Phase propagation ^{106–109,111,113,114,439–442}
STXM	Understanding the changes in electronic structure with corresponding images.	Low temporal resolution, need for vacuum conditions, achievement of real operating conditions, limited penetration depth and field of view.	Phase propagation ^{119,120}
<i>Ultraviolet and visible light and infrared-based</i>			
OM	Understanding morphological changes, large field of view, high temporal resolution.	Low spatial resolution, overcoming low contrast and signal-to-noise ratio.	Phase propagation ^{156,157,159,160} Li dendrite and deposition ^{149–153,443–451}
Raman	Understanding chemical bonding and molecular composition, high temporal resolution.	Application limited to Raman-active samples.	Ion concentration change in electrolyte ¹⁷¹ Phase transition ^{165–168,170,172,176–178,420,452–454}
FT-IR	Understanding chemical bonding and molecular composition, high temporal resolution.	Application limited to IR-active samples, achievement of real operating conditions.	Electrode–electrolyte interface reaction ^{183,184,187,188,191,455–459} Electrolyte reaction, ^{185,189,192,460} Phase distribution ¹⁸⁴ Phase transition ⁴⁶¹

(Continues)

TABLE 7 (Continued)

Techniques	Purposes and advantages	Challenges	Details
<i>Microwave and radiowave-based</i>			
NMR	Understanding chemical and molecular environments by magnetic response of nuclei.	Magnetic susceptible effect, low spatial/temporal resolution.	Bulk magnetic susceptibility ^{237,462,463} Li concentration ^{226,227,464,465} Li deposition ^{221,228–230} Phase transition ^{222–225,236,466} Li site ⁴⁶⁷
EPR	Understanding electronic properties of intermediates, radicals, and other paramagnetic species.	Magnetic susceptible effect, difficulty in measuring samples with a nonradiative absorbent, achievement of real operating conditions.	Li deposition ^{245–248} Anionic redox ^{249–251}
<i>Electron-based</i>			
SEM	Understanding microstructural surface changes and elemental distributions, high temporal and spatial resolution, modifiable magnification.	Need for vacuum conditions, achievement of real operating conditions.	SEI formation ²⁶⁴ Li deposition ^{265,266,468–472} Morphology changes ^{268,269,271,473–476}
TEM	Understanding nanomorphology, crystal structure, and atomic composition, high spatial resolution.	Need for vacuum conditions, achievement of real operating conditions, need for highly thin samples.	SEI formation ^{279,280,282} Li deposition ²⁸¹ Atomic image ^{278,477} Morphology changes ^{290,478–484} Phase transition ^{277,278,282–285,287,288,485–487}
<i>Neutron-based</i>			
ND	Understanding information about the average crystal structure, light element sensitivity.	Need for a large amount of sample, inaccurate data about amorphous samples, interpretation of overlapped diffraction peaks.	Phase transition ^{306,308,488–495} Lattice parameter change ^{306–309,493–495} Phase fraction ^{305,496} Atomic position of light element ³⁰⁷
SANS	Understanding the structural evolution of macromolecules and nanostructure, light element sensitivity.	Need for a large amount of sample, low scattering contrast, interpretation of overlapped diffraction peaks.	Compositional change ^{37,319,320} Microstructure and morphology change ^{37,318,319,321}
NR	Understanding surface structure and chemical composition, high spatial and temporal resolution.	Requirement for smooth surface, achievement of real operating conditions.	SEI formation ^{335,336,338,497–500} (De)lithiation mechanism ^{329–332}
NI	Understanding the internal structure and morphology, light element sensitivity, high penetration capability.	Low spatial and temporal resolution.	Phase propagation ^{501–503} Li concentration ³⁴⁵ Li deposition ³⁴⁰ Volume change ^{343,504–506}
NDP	Understanding light element concentration over depth, high sensitivity to ⁶ Li isotopes, high depth resolution.	Need for vacuum or inert gas environment.	Li distribution ^{357,359,360,507–511}

TABLE 7 (Continued)

Techniques	Purposes and advantages	Challenges	Details
<i>Other techniques</i>			
AFM	Understanding nanostructure and interface-related properties, high spatial resolution with 3D images, unlimited by sample conductivity.	Slow scanning speed, impact of external vibrations and noise, achievement of real operating condition.	Chemical and morphological structures of SEI ^{512–522} Phase transition ^{523,524} Li deposition ^{383,525,526}
DEMS	Understanding mass-resolved characteristics of gaseous species, high sensitivity, excellent temporal resolution.	Inability to detect nonvolatile or liquid species, mass transfer disturbance by electrolytes with high viscosity.	Gas evolution ^{389,390,397,527–533}

Abbreviations: AFM, atomic force microscopy; DEMS, differential electrochemical mass spectrometry; EPR, electron paramagnetic resonance; FT-IR, Fourier-transform infrared; IR, infrared; ND, neutron diffraction; NDP, neutron depth profiling; NI, neutron imaging; NMR, nuclear magnetic resonance; NR, neutron reflectometry; OM, optical microscopy; SANS, small-angle neutron scattering; SAXS, small-angle X-ray scattering; SEI, solid electrolyte interface; SEM, scanning electron microscopy; STXM, scanning transmission X-ray microscopy; TEM, transmission electron microscopy; TXM, transmission X-ray microscopy; XAS, X-ray absorption spectroscopy; XCT, X-ray computed tomography; XPS, X-ray photoelectron spectroscopy; XRD, X-ray diffraction.

Despite its great potential, the utilization of in situ analysis still faces challenges such as designing of experimental setups, compatibility among cell components, and data interpretation issues related to resolution, sensitivity, and the complex nature of battery systems. Continuing advances in experimental techniques, instrumentation, and data processing methodologies are further refining the capabilities of in situ analysis, enabling researchers to easily explore complex phenomena occurring in battery systems and to obtain more precise information. The stereoscopic approach based on information type and probing depth through in situ techniques can provide a key solution that can overcome the limitations of current Li-rechargeable batteries and provide new perspectives on the design of innovative energy storage systems.

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CONFLICT OF INTEREST STATEMENT

The authors declare that there are no conflicts of interests.

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